▶ Notebook

Lecture 2 at a glance: six parts, twenty-nine sections

1 · Why generate?
1.1 Things that do not exist
1.2 Text, pictures and sound
1.3 Beyond pictures
1.4 What they all share

2 · Fit a distribution, then sample
2.1 The data: Old Faithful
2.2 One Gaussian: fit, score, sample
2.3 Two bumps: a hidden choice
2.4 Fit the mixture, then sample it
2.5 How many bumps?
2.6 Two numbers per eruption

3 · Images: PCA and Gaussians
3.1 An image is 784 numbers
3.2 Gaussians on raw pixels
3.3 PCA: a few numbers per image
3.4 Generate from PCA numbers

4 · Autoencoders
4.1 From PCA to an autoencoder
4.2 Train one
4.3 How many numbers in the code?
4.4 What codes are good for

5 · Can an autoencoder generate?
5.1 Decode a random code
5.2 Where the codes are
5.3 Fit a density to the codes

6 · The variational autoencoder
6.1 The generative story
6.2 Which code made this 3?
6.3 Variational inference for one image
6.4 An encoder does it for every image
6.5 The loss, term by term
6.6 What the VAE learned
6.7 The KL weight β
6.8 Wrap-up

Click a section to jump to it. We start with why generative models matter, and end by training a VAE.

▶ Notebook 1.1

Who are these people?



> Look closely: ages, hair, glasses, backgrounds. Who are these people?

Faces generated by StyleGAN: Karras, Laine and Aila, "A Style-Based Generator Architecture for GANs", CVPR 2019. NVIDIA, CC BY-NC 4.0.

▶ Notebook 1.1

Who are these people?



None of them exist. A network turned random numbers into each face.

> Nobody. Each face is a sample from a generative model (StyleGAN, 2019): random numbers in, a face out.

Faces generated by StyleGAN: Karras, Laine and Aila, "A Style-Based Generator Architecture for GANs", CVPR 2019. NVIDIA, CC BY-NC 4.0.

▶ Notebook 1.1

Real or generated? Six of these twelve digits are fake



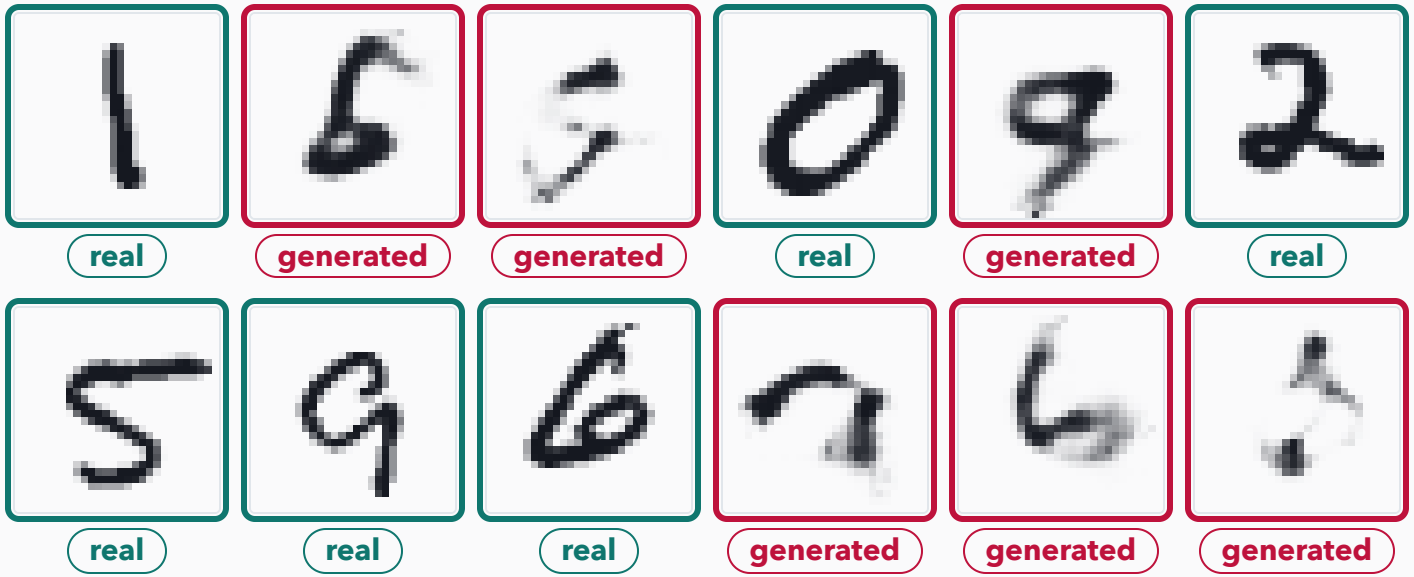
Six are handwritten MNIST test digits. Six come from a network.

> Six are real, six are generated. Write down which six you think are fake.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 1.1

Real or generated? Six of these twelve digits are fake



Six are handwritten MNIST test digits. Six come from a network.

The fakes: numbers 2, 3, 5, 10, 11, 12 (counting left to right, top to bottom).

By the end of today you will have trained the network that made them: a VAE with 16 random numbers per digit.

> Answer: the six outlined in red and labelled “generated”. Some are easy (broken strokes); some fool most people.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 1.2

A chatbot is a generative model: one prompt, three different answers

Prompt: Write one sentence that opens a short story about the first monsoon rain in Gandhinagar.

Sample 1. As the first heavy drops of the monsoon struck the sunbaked avenues of Sector 21, the parched earth of Gandhinagar exhaled a sweet, dusty petrichor that signaled the end of the long summer.

We sent the same prompt to a language model three times. Here is the first reply.

Generated for this lecture on 21 October 2026 with Gemini 3.5 Flash, temperature 1 (`assets/examples/provenance.json`).

▶ Notebook 1.2

A chatbot is a generative model: one prompt, three different answers

Prompt: Write one sentence that opens a short story about the first monsoon rain in Gandhinagar.

Sample 1. As the first heavy drops of the monsoon struck the sunbaked avenues of Sector 21, the parched earth of Gandhinagar exhaled a sweet, dusty petrichor that signaled the end of the long summer.

Sample 2. As the first heavy drops struck the sun-baked asphalt of Sector 17, Gandhinagar exhaled its scorching summer heat in a sweet, petrichor-laced sigh that rippled down the quiet, tree-lined avenues.

Sample 3. The brutal summer heat surrendered in a single afternoon when heavy, bruised clouds cracked open over Gandhinagar, washing its dusty neem trees and quiet, grid-lined sectors in the intoxicating scent of the season's first rain.

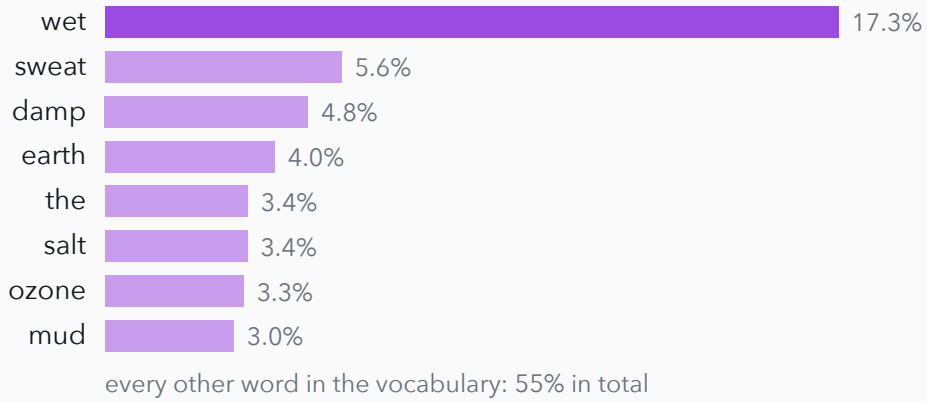
Same prompt, three different replies. Each is a sample from a distribution over sentences.

Generated for this lecture on 21 October 2026 with Gemini 3.5 Flash, temperature 1 (`assets/examples/provenance.json`).

▶ Notebook 1.2

Under the hood: a probability for every possible next word

“After months of heat, the first monsoon rain smelled of …”



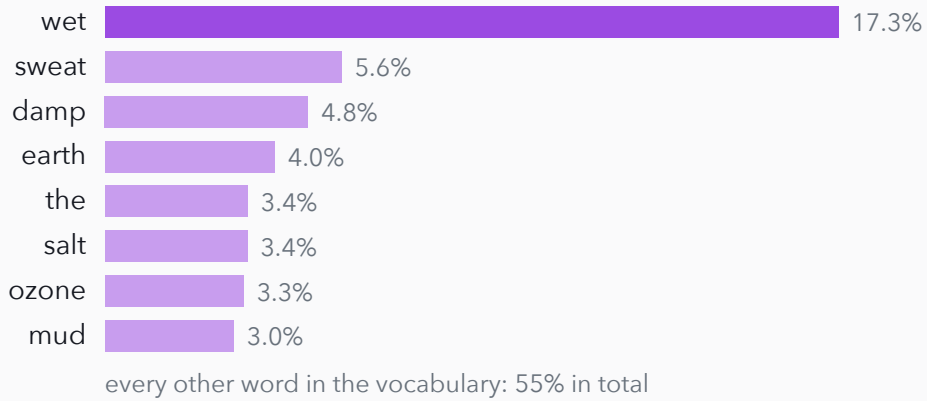
> A small language model (SmolLM2, 360 million weights) scores every word it knows as the next word. “wet” gets 17%.

SmolLM2-360M-Instruct (Hugging Face), next-token probabilities and three samples at temperature 1, computed for this lecture.

▶ Notebook 1.2

Under the hood: a probability for every possible next word

"After months of heat, the first monsoon rain smelled of …"



Sampling = pick a word with these probabilities, append it, repeat.

The same model's three continuations: "melons and oranges.", "sweat and damp ruins. It was moist, it was s", "desperation. What had occurred, and what transpired that its".

> Sample a word, append it, repeat. That is all a chatbot does when it writes, at a much larger scale.

SmolLM2-360M-Instruct (Hugging Face), next-token probabilities and three samples at temperature 1, computed for this lecture.

▶ Notebook 1.2

Text to image: one prompt, four samples

Prompt: A watercolour painting of a camel walking across the white salt desert of the Rann of Kutch at sunrise, soft light, wide view.



sample 1

Same idea for pictures. One prompt, and the first picture the model returned.

Generated for this lecture on 21 October 2026 with Gemini 3 Pro Image (`assets/examples/provenance.json`).

▶ Notebook 1.2

Text to image: one prompt, four samples

Prompt: A watercolour painting of a camel walking across the white salt desert of the Rann of Kutch at sunrise, soft light, wide view.



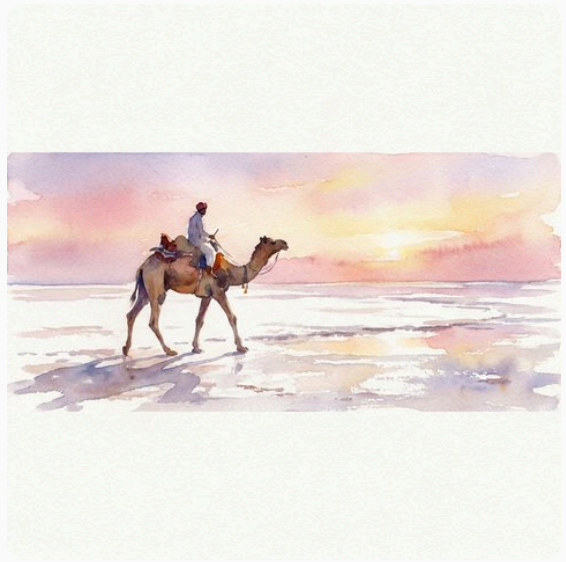
sample 1



sample 2



sample 3



sample 4

Three more runs, same prompt: a rider appears, a walker, a different horizon. Four samples.

Generated for this lecture on 21 October 2026 with Gemini 3 Pro Image (`assets/examples/provenance.json`).

▶ Notebook 1.2

Image in, image out


edges of a shoe


pix2pix: a photo


a label map


SPADE: a landscape


a horse


CycleGAN: a zebra

> The input can be an image: a sketch becomes a photo, a coloured map becomes a landscape, a horse becomes a zebra.

Isola et al., pix2pix (CVPR 2017); Park et al., SPADE (CVPR 2019); Zhu et al., CycleGAN (ICCV 2017). Attributed teaching excerpts; copyright with the authors.

▶ Notebook 1.2

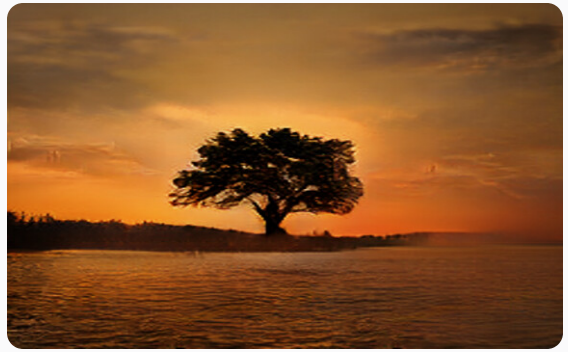
Image in, image out



one label map



sample 1



sample 2



sample 3

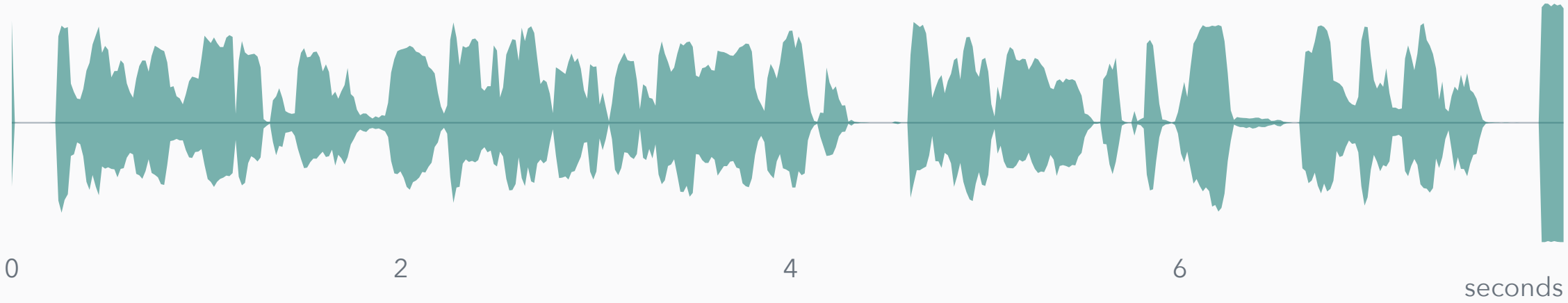
> The same label map, sampled three times: three different photographs. Still a distribution, now given an input.

Isola et al., pix2pix (CVPR 2017); Park et al., SPADE (CVPR 2019); Zhu et al., CycleGAN (ICCV 2017). Attributed teaching excerpts; copyright with the authors.

▶ Notebook 1.2

Sound: text in, 24,000 numbers per second out

Text: “A generative model turns a few random numbers into a new example: a sentence, a picture, or this voice.”



0:00 / 0:00

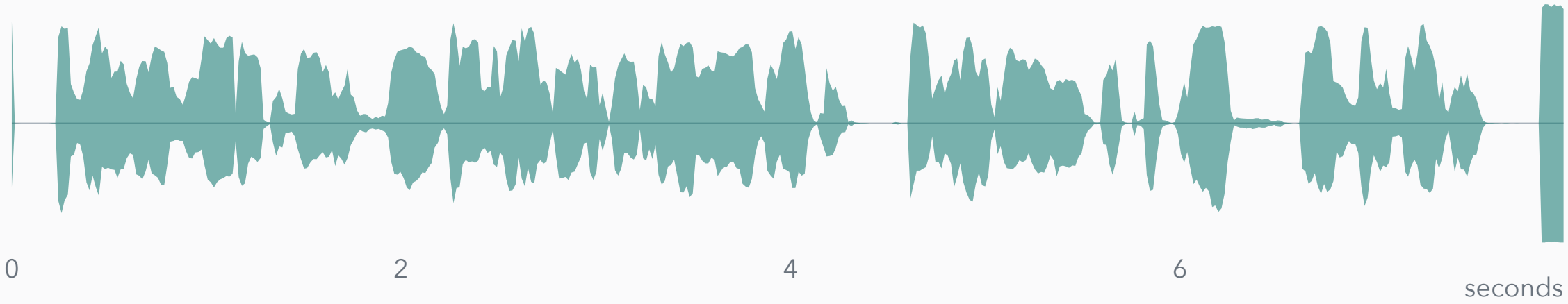
A text-to-speech model read this sentence aloud. Press play.

Generated for this lecture on 21 October 2026 with Gemini 3.8 Flash TTS, voice “Kore”.

▶ Notebook 1.2

Sound: text in, 24,000 numbers per second out

Text: “A generative model turns a few random numbers into a new example: a sentence, a picture, or this voice.”



0:00 / 0:00

191,219 numbers for 8.0 seconds. A few of them: 0.102, 0.089, 0.083, 0.072, 0.062, 0.051, …

The output is just a long list of numbers: the air pressure, 24,000 times a second. Generating sound = generating that list.

Generated for this lecture on 21 October 2026 with Gemini 3.8 Flash TTS, voice “Kore”.

▶ Notebook 1.3

Beyond pictures: generative models in science

Weather

GenCast samples 50 possible futures for the next 15 days. The spread of the samples says how sure the forecast is.

Price et al., Nature 2025

Proteins

RFdiffusion generates new protein shapes, for example ones that bind a chosen target.

Watson et al., Nature 2023

> Weather and proteins: the output is a sample, and many samples together describe what could happen.

Price et al., "Probabilistic weather forecasting with machine learning" (GenCast), Nature 2025; Watson et al., "De novo design of protein structure and function with RFdiffusion", Nature 2023; Zeni et al., "A generative model for inorganic materials design" (MatterGen), Nature 2025.

▶ Notebook 1.3

Beyond pictures: generative models in science

Weather

GenCast samples 50 possible futures for the next 15 days. The spread of the samples says how sure the forecast is.

Price et al., Nature 2025

Proteins

RFdiffusion generates new protein shapes, for example ones that bind a chosen target.

Watson et al., Nature 2023

Materials

MatterGen proposes new crystal structures with properties you ask for.

Zeni et al., Nature 2025

Missing data

Sensors drop readings. A model of p(x) fills the gap with plausible values, not a flat average, and says how unsure it is.

for example, air-quality and smart-meter records

> Materials, and the everyday case: filling in missing sensor readings with plausible values.

Price et al., "Probabilistic weather forecasting with machine learning" (GenCast), Nature 2025; Watson et al., "De novo design of protein structure and function with RFdiffusion", Nature 2023; Zeni et al., "A generative model for inorganic materials design" (MatterGen), Nature 2025.

▶ Notebook 1.3

What a model of p(x) is good for

Create
new text, images, sound, designs

Fill in
missing pixels, missing readings, the rest of a sentence

Simulate
many plausible futures, not one average

> Three uses that need samples: create, fill in, simulate.

▶ Notebook 1.3

What a model of p(x) is good for

Create
new text, images, sound, designs

Fill in
missing pixels, missing readings, the rest of a sentence

Simulate
many plausible futures, not one average

Spot the odd one out
a reading with very low p(x) is suspicious: a faulty sensor, a fraud

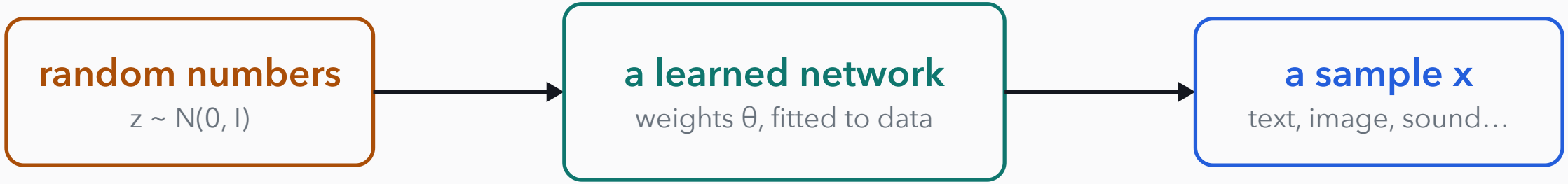
Compress
likely data needs fewer bits: Lecture 1's codes

Make more data
extra training examples, or shareable synthetic records

> Three that need scores: spot anomalies, compress, and generate extra data when real data is scarce or private.

▶ Notebook 1.4

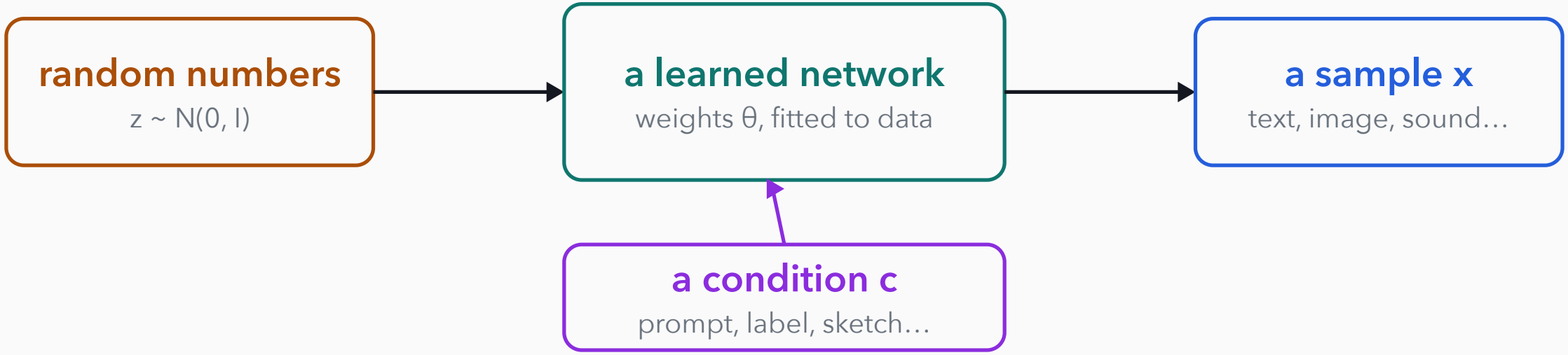
What they share: random numbers in, a learned network, a sample out



> Every model on the tour: random numbers z go in, a network with learned weights θ turns them into a sample x.

▶ Notebook 1.4

What they share: random numbers in, a learned network, a sample out



> Most also take a condition c that steers the sample: a prompt, a class label, a sketch. Then the model is p(x | c).

▶ Notebook 1.4

Conditions steer the sample; today we learn p(x) without one

	condition c	sample x
chatbot	your prompt	a reply
text to image	a prompt	a picture
pix2pix, SPADE, CycleGAN	a sketch, a map, a photo	a photo
text to speech	a sentence	a waveform
today	none	a 28 × 28 digit

> Each example on the tour fills in c and x differently. Same recipe: random numbers in, learned network, sample out.

▶ Notebook 1.4

Conditions steer the sample; today we learn p(x) without one

	condition c	sample x
chatbot	your prompt	a reply
text to image	a prompt	a picture
pix2pix, SPADE, CycleGAN	a sketch, a map, a photo	a photo
text to speech	a sentence	a waveform
today	none	a 28 × 28 digit

Learning a distribution over images is the hard part, with or without a condition.

> Today we drop the condition and learn p(x) for handwritten digits: the core problem underneath all of these.

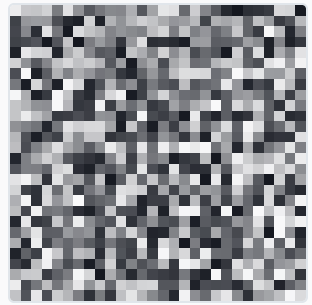
▶ Notebook 1.4

A model of p(x) can do two jobs: score and sample

Score: how plausible is this x?

$$p(\text{a clean 3}) \text{ high}, \qquad p(\text{static}) \text{ tiny}$$





Job one, score: a number that is high for a real-looking 3 and tiny for random static.

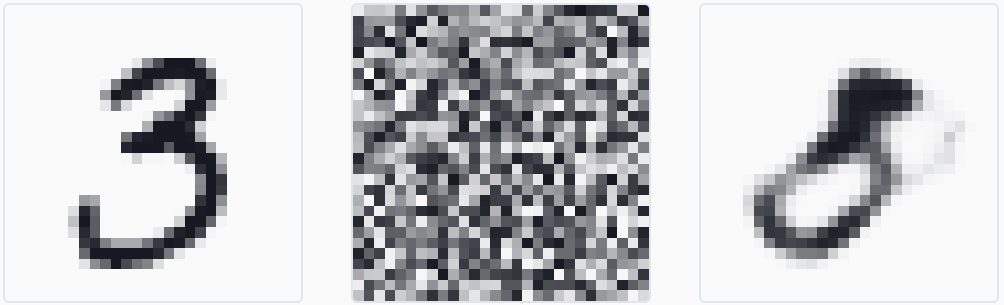
▶ Notebook 1.4

A model of p(x) can do two jobs: score and sample

Score: how plausible is this x?

$$p(\text{a clean 3}) \text{ high}, \qquad p(\text{static}) \text{ tiny}$$

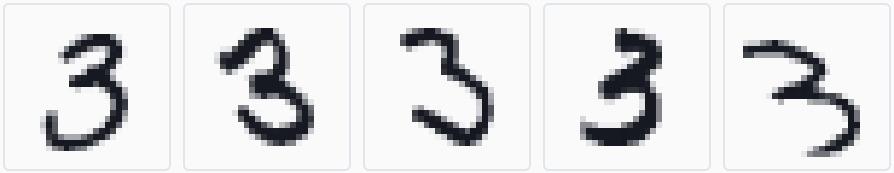
Sample: make a new x that looks like the data.



> Job two, sample: produce a new x. Both jobs come from the same fitted distribution.

▶ Notebook 1.4

Why not just output the average?



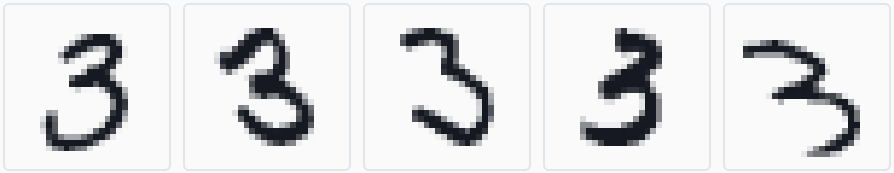
average of 100 threes:



Five real 3s, and the pixel-by-pixel average of 100 of them.

▶ Notebook 1.4

Why not just output the average?



average of 100 threes:



Nobody writes this 3. The average is the blur in the middle of many real 3s.

> The average looks like no real 3. We need the whole distribution, so we can draw one sharp 3 at a time.

▶ Notebook 1.4

Today's route: six parts

1 · Why generate | 2 · Fit, then sample | 3 · PCA and Gaussians | 4 · Autoencoders | 5 · Can it generate? | 6 · The VAE

1 · Why generate?
what generative models do, and the shared recipe

2 · Fit, then sample
one number per example: a Gaussian, then two bumps

3 · PCA and Gaussians
784 numbers per digit: compress first, then fit

4 · Autoencoders
a learned, bent version of PCA

5 · Can it generate?
random codes, and why they fail

6 · The VAE
a recipe we can sample, trained with Lecture 1's ELBO

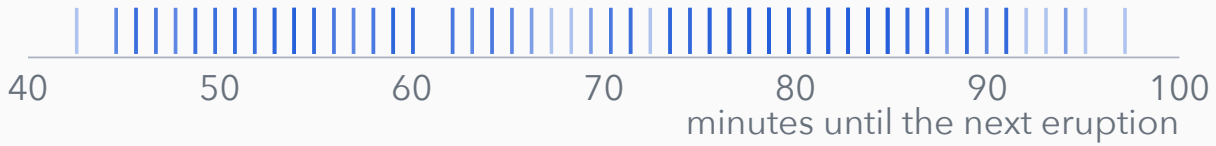
> From one number per example to 784, from fitting by formula to training networks, ending with the variational autoencoder.

▶ Notebook 2.1

Old Faithful: 272 eruptions, and the wait before each one

A geyser in Yellowstone. Rangers record how long visitors wait between eruptions.

The first five waits, in minutes: **79, 54, 74, 62, 85**, …



> One number per example: the wait until the next eruption. Each tick is one of 272 eruptions.

Old Faithful geyser data (Azzalini and Bowman, 1990), as distributed with R (`datasets::faithful`): 272 eruptions.

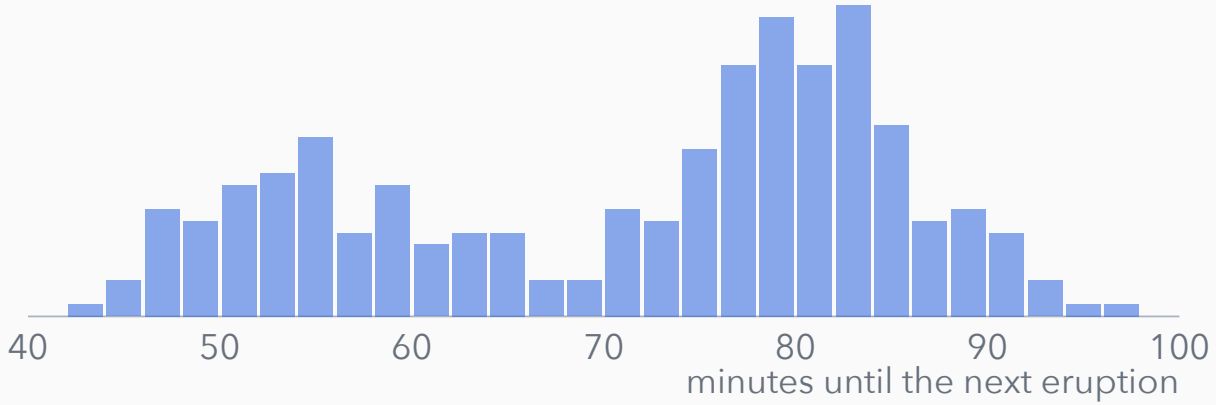
▶ Notebook 2.1

Old Faithful: 272 eruptions, and the wait before each one

A geyser in Yellowstone. Rangers record how long visitors wait between eruptions.

The first five waits, in minutes: **79, 54, 74, 62, 85**, …

All 272 waits as a histogram: two clumps, around 55 and 80 minutes.



> The histogram shows two clumps: short waits around 55 minutes, long ones around 80. Few waits near 70.

Old Faithful geyser data (Azzalini and Bowman, 1990), as distributed with R (`datasets::faithful`): 272 eruptions.

▶ Notebook 2.1

The task: examples in, a distribution out, new examples from it

examples

272 waiting times

> We start with examples: 272 waiting times.

▶ Notebook 2.1

The task: examples in, a distribution out, new examples from it



> Fit: choose a distribution p(x) from a family by maximum likelihood. Lecture 1: maximum likelihood minimises KL(data ‖ model).

▶ Notebook 2.1

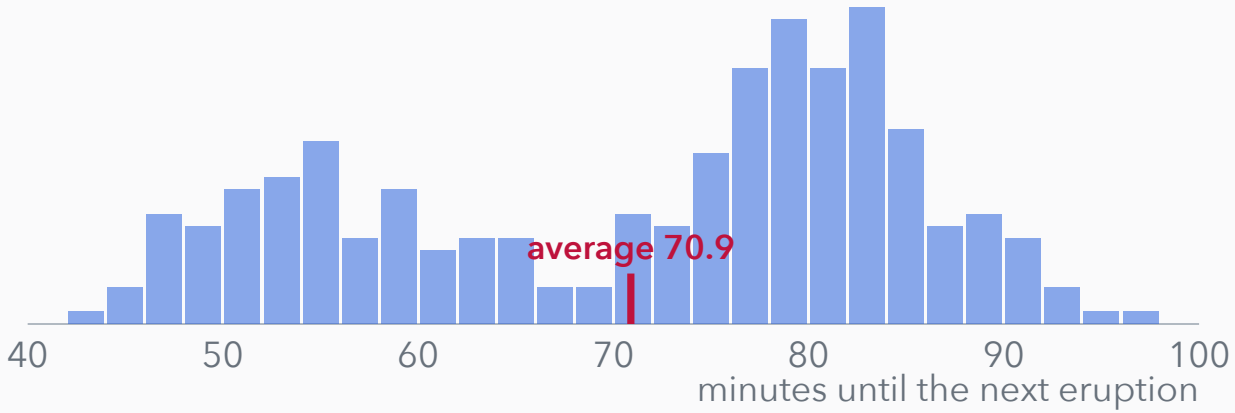
The task: examples in, a distribution out, new examples from it



> Sample: run the fitted distribution forwards to get new waits that nobody measured.

▶ Notebook 2.1

The average wait, 70.9 minutes, is a poor sample



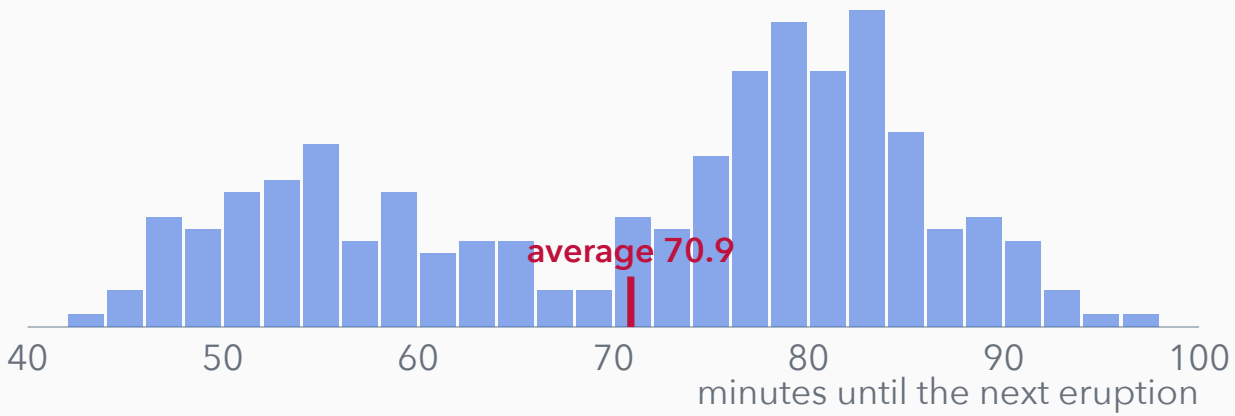
Only **46** of 272 waits lie within 5 minutes of the average.

> The average is 70.9 minutes, in the dip between the clumps. Few real waits are near it.

Old Faithful geyser data (Azzalini and Bowman, 1990), as distributed with R (`datasets::faithful`): 272 eruptions.

▶ Notebook 2.1

The average wait, 70.9 minutes, is a poor sample



Only **46** of 272 waits lie within 5 minutes of the average.

The average sits in the dip. To make realistic waits we need the shape of the data, not one number.

> A realistic new wait should come from one of the clumps, as often as the clumps occur. That needs a distribution.

Old Faithful geyser data (Azzalini and Bowman, 1990), as distributed with R (`datasets::faithful`): 272 eruptions.

▶ Notebook 2.2

One Gaussian: the best μ and σ have a formula

$$p(x) = \mathcal{N}(x;\, \mu, \sigma^2)$$

> Family: one Gaussian, two numbers μ and σ.

Old Faithful geyser data (Azzalini and Bowman, 1990), as distributed with R (`datasets::faithful`): 272 eruptions.

▶ Notebook 2.2

One Gaussian: the best μ and σ have a formula

$$p(x) = \mathcal{N}(x;\, \mu, \sigma^2)$$

$$\text{maximise} \quad \frac{1}{n} \sum_{i=1}^{n} \log \mathcal{N}(x_i;\, \mu, \sigma^2)$$

> Fit: choose μ and σ to maximise the average log-likelihood of the 272 waits.

Old Faithful geyser data (Azzalini and Bowman, 1990), as distributed with R (`datasets::faithful`): 272 eruptions.

▶ Notebook 2.2

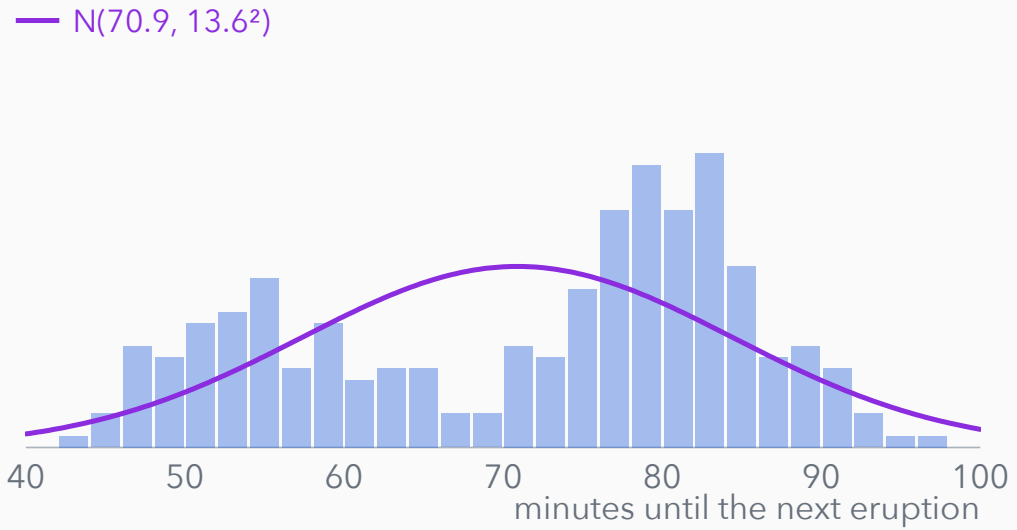
One Gaussian: the best μ and σ have a formula

$$p(x) = \mathcal{N}(x;\, \mu, \sigma^2)$$

$$\text{maximise} \quad \frac{1}{n}\sum_{i=1}^{n} \log \mathcal{N}(x_i;\, \mu, \sigma^2)$$

$$\hat{\mu} = \frac{1}{n}\sum_{i} x_i = 70.9$$

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i} (x_i - \hat{\mu})^2 = 13.6^2$$

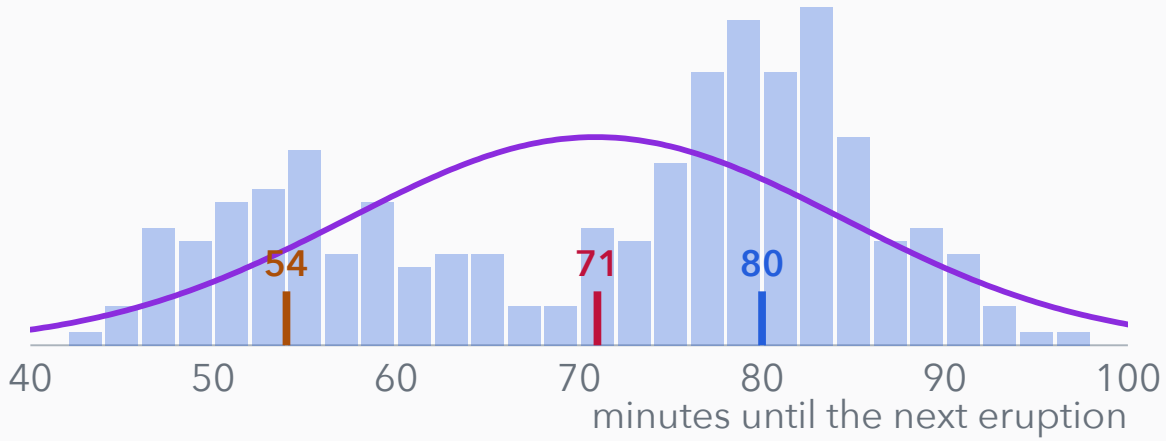


> Set the derivatives to zero: μ̂ is the average and σ̂² the average squared deviation. One bump, centred in the dip.

Old Faithful geyser data (Azzalini and Bowman, 1990), as distributed with R (`datasets::faithful`): 272 eruptions.

▶ Notebook 2.2

Score: how probable is each wait under the bell?



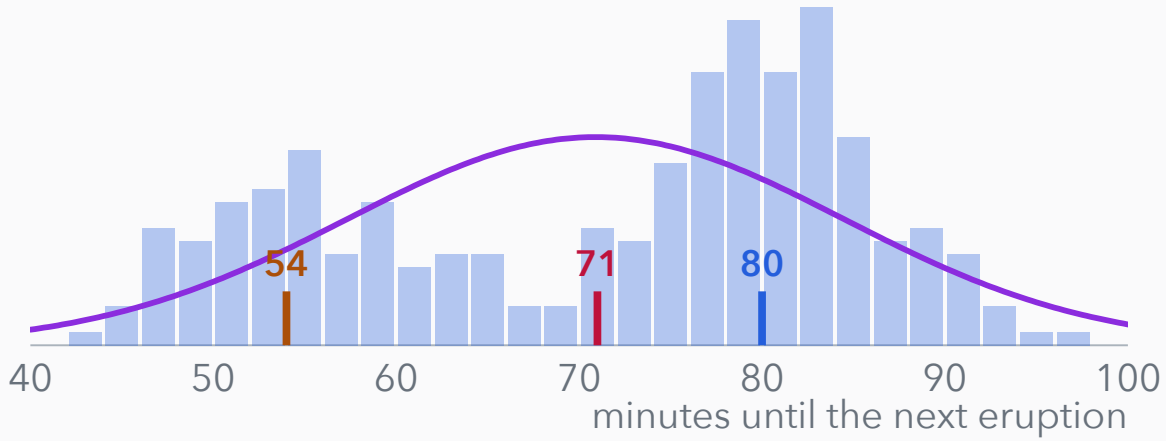
wait x	p(x) under the bell
54 minutes	0.0135
71 minutes	0.0294
80 minutes	0.0235

> The bell gives a density for every wait. It rates 71 minutes, in the empty dip, as the most likely wait of all.

Old Faithful geyser data (Azzalini and Bowman, 1990), as distributed with R (`datasets::faithful`): 272 eruptions.

▶ Notebook 2.2

Score: how probable is each wait under the bell?



wait x	p(x) under the bell
54 minutes	0.0135
71 minutes	0.0294
80 minutes	0.0235

Average log-likelihood of all 272:
−4.027

> One number scores the whole model: the average log p(x) over the data. We will try to beat it.

Old Faithful geyser data (Azzalini and Bowman, 1990), as distributed with R (`datasets::faithful`): 272 eruptions.

▶ Notebook 2.2

Sample from the bell: the draws fill the empty dip

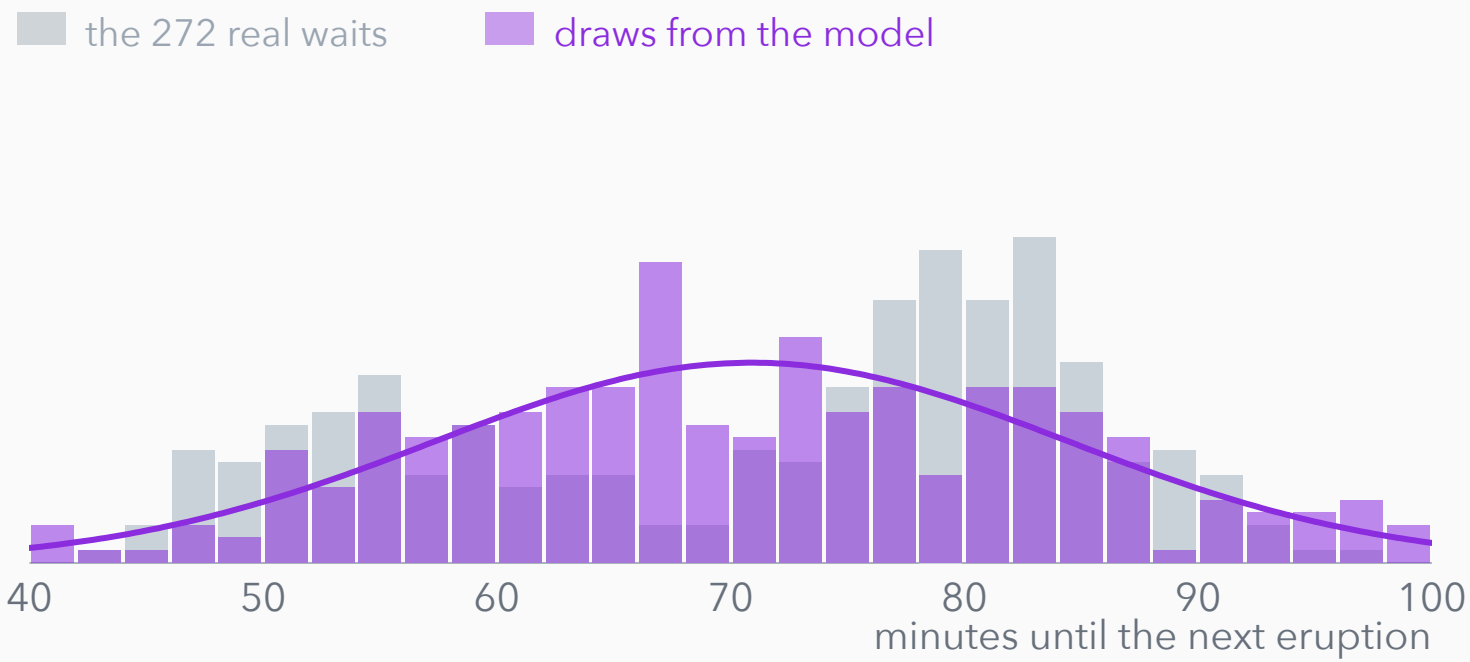
one step: draw a time

$N(70.9, 13.6^2)$
the one fitted Gaussian

latest: 81.4 minutes

73 of 272 draws land in 62–72 min
(the data: 30 of 272)

272 draws so far



Press +1 a few times, then +272: draws from the fitted Gaussian, over the real waits.

Old Faithful geyser data (Azzalini and Bowman, 1990), as distributed with R (`datasets::faithful`): 272 eruptions.

▶ Notebook 2.2

Sample from the bell: the draws fill the empty dip

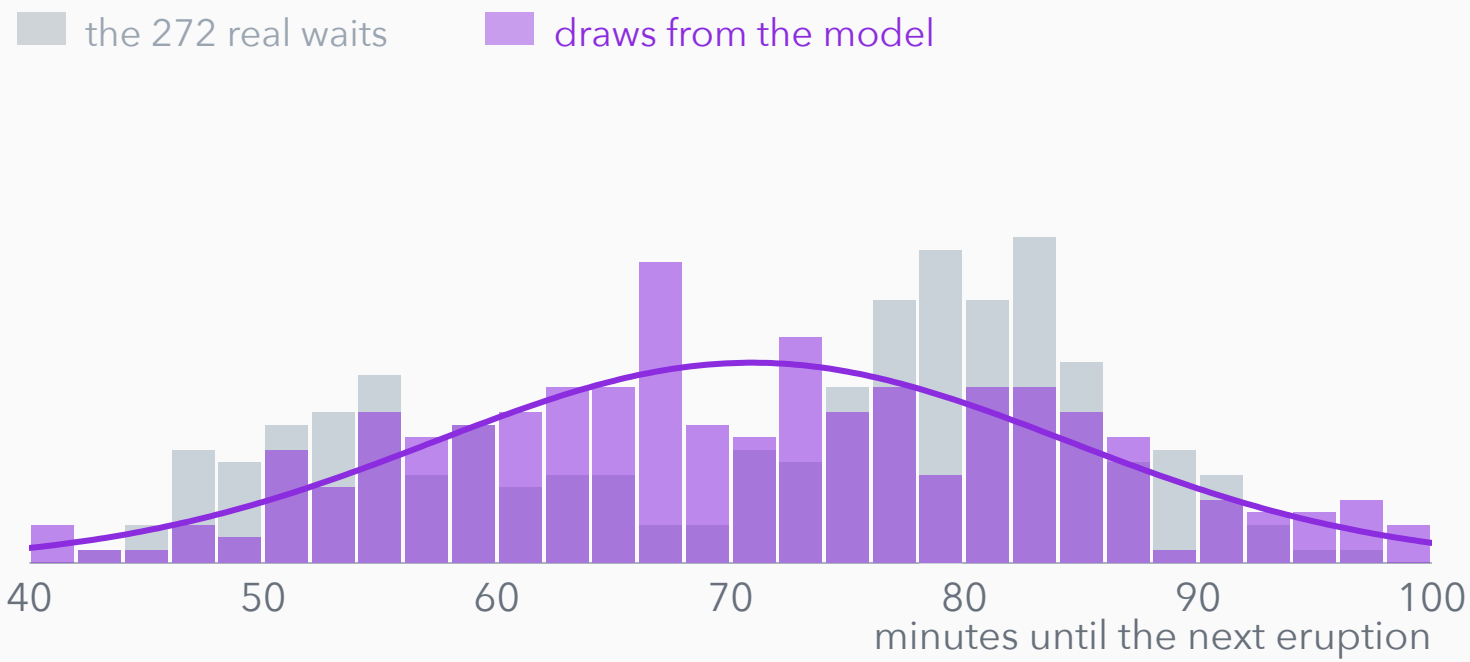
one step: draw a time

$N(70.9, 13.6^2)$
the one fitted Gaussian

latest: 81.4 minutes

73 of 272 draws land in 62–72 min
(the data: 30 of 272)

272 draws so far

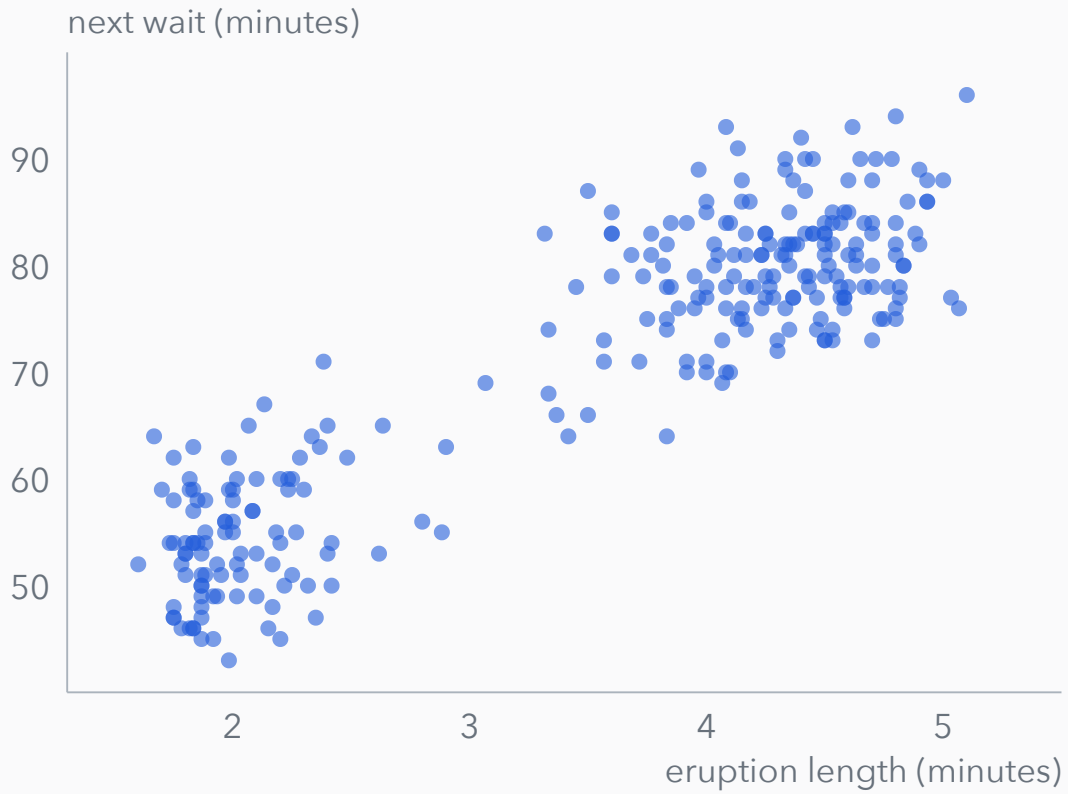


The draws pile up in the dip where real eruptions rarely are, and miss the two clumps.

Old Faithful geyser data (Azzalini and Bowman, 1990), as distributed with R (`datasets::faithful`): 272 eruptions.

▶ Notebook 2.3

The geyser has two kinds of eruption



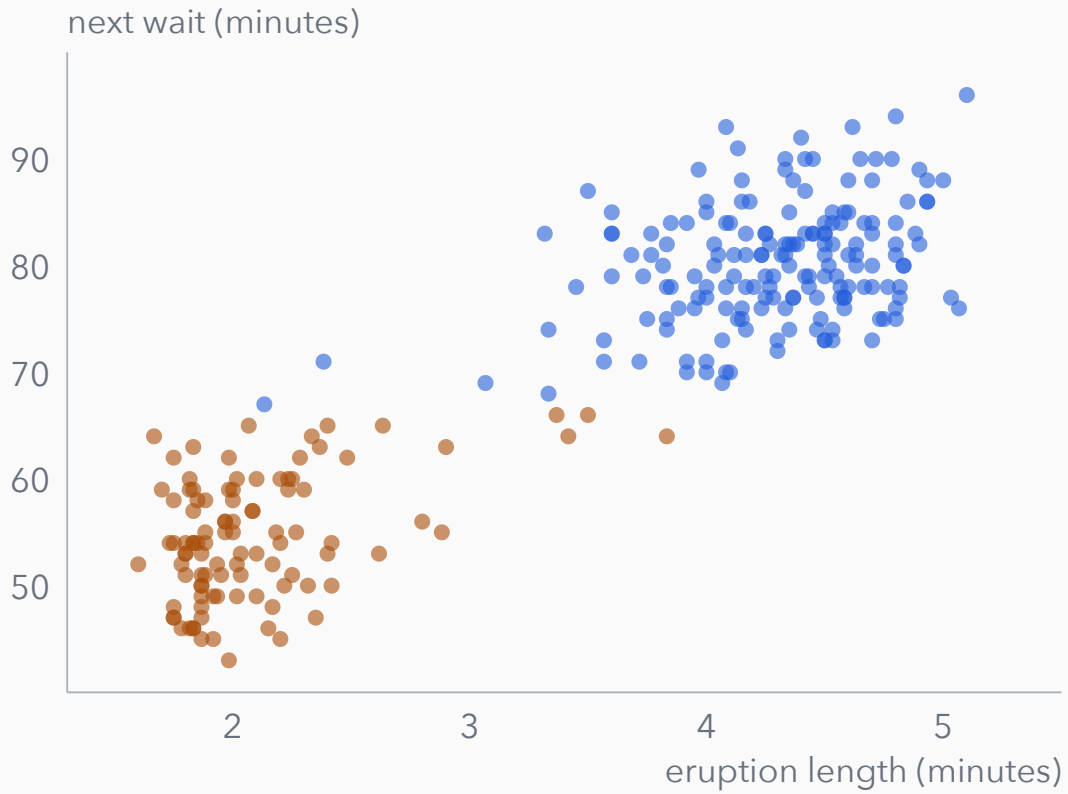
Add a second measurement: how long each eruption lasted.

Plot each eruption's length against the wait that follows it. Two groups.

Old Faithful geyser data (Azzalini and Bowman, 1990), as distributed with R (`datasets::faithful`): 272 eruptions.

▶ Notebook 2.3

The geyser has two kinds of eruption



Add a second measurement: how long each eruption lasted.

Short eruptions (about 2.1 min) come before short waits; long ones (about 4.3 min) before long waits.

The kind of eruption is a hidden choice. If we only see the wait, we never see the kind directly.

> Nature first "chooses" a short or a long eruption; the wait then depends on that choice.

Old Faithful geyser data (Azzalini and Bowman, 1990), as distributed with R (`datasets::faithful`): 272 eruptions.

▶ Notebook 2.3

A recipe with a hidden step: pick a bump, then draw

$$k \sim \text{Categorical}(\pi_1, \pi_2) \qquad \text{(short or long)}$$

$$x \mid k \sim \mathcal{N}(\mu_k, \sigma_k^2) \qquad \text{(the wait, given the kind)}$$

Two steps: choose a kind k with probabilities π_1, π_2; then draw the wait from that kind's Gaussian.

▶ Notebook 2.3

A recipe with a hidden step: pick a bump, then draw

$$k \sim \text{Categorical}(\pi_1, \pi_2) \qquad (\text{short or long})$$

$$x \mid k \sim \mathcal{N}(\mu_k, \sigma_k^2) \qquad (\text{the wait, given the kind})$$

k is a **latent** (hidden) variable: it shapes the data, but we never observe it.

> k is a latent variable: hidden, but part of the recipe. This idea returns in every later part.

▶ Notebook 2.3

A recipe with a hidden step: pick a bump, then draw

$$k \sim \text{Categorical}(\pi_1, \pi_2) \qquad \text{(short or long)}$$

$$x \mid k \sim \mathcal{N}(\mu_k, \sigma_k^2) \qquad \text{(the wait, given the kind)}$$

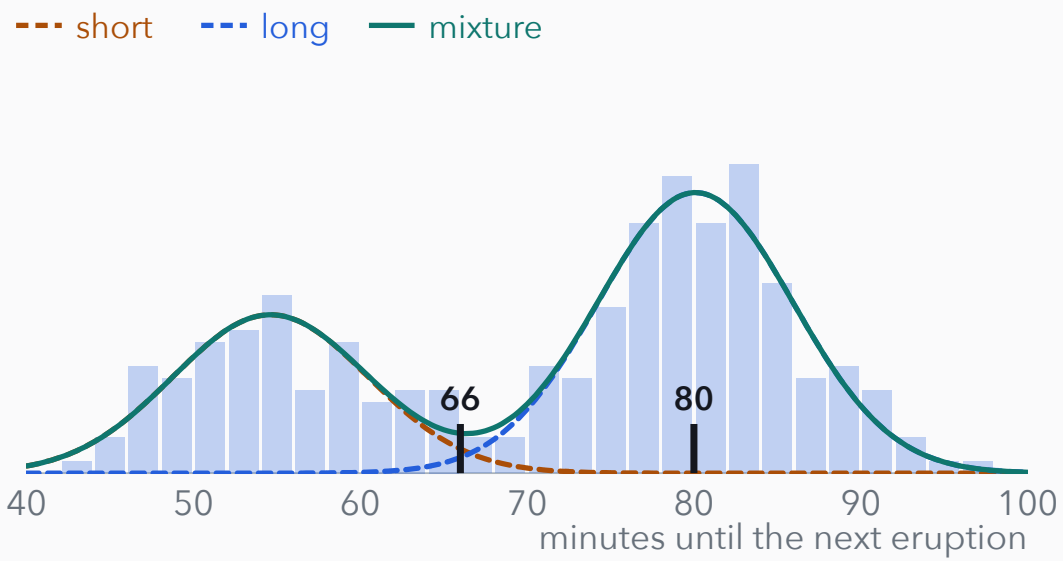
k is a **latent** (hidden) variable: it shapes the data, but we never observe it.

$$p(x) = \sum_{k=1}^{2} \pi_k \, \mathcal{N}(x;\, \mu_k, \sigma_k^2) \qquad \text{a mixture of Gaussians}$$

> The distribution of x alone adds up both ways x could have been made: a mixture of Gaussians.

▶ Notebook 2.3

To score a wait, add over both kinds



$$p(66) = \pi_1 \mathcal{N}(66;\ \text{short}) + \pi_2 \mathcal{N}(66;\ \text{long})$$

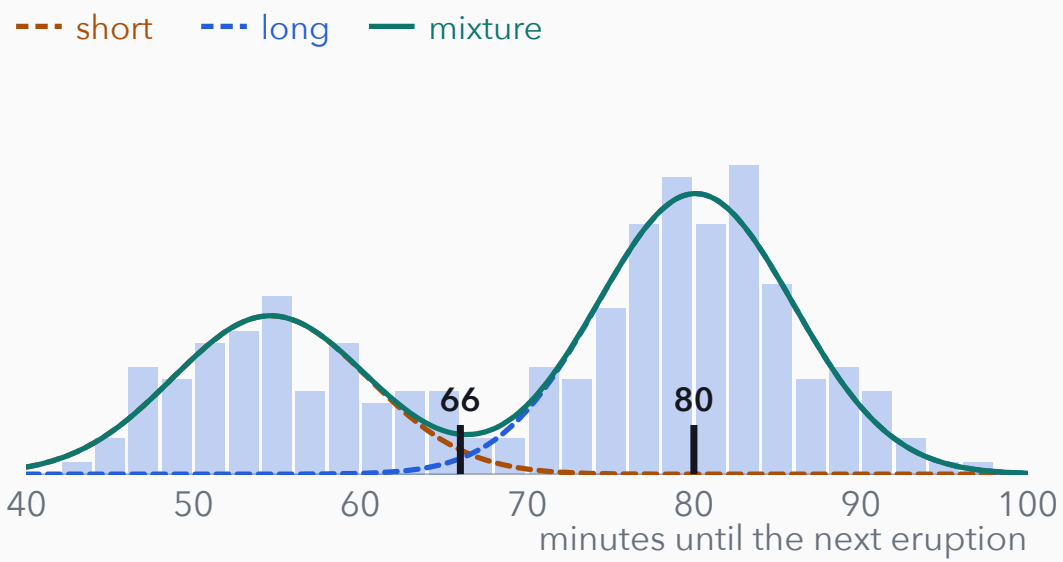
$$= 0.0037 + 0.0024 = 0.0062$$

> At 66 minutes, between the clumps, both kinds contribute. Add them.

Old Faithful geyser data (Azzalini and Bowman, 1990), as distributed with R (`datasets::faithful`): 272 eruptions.

▶ Notebook 2.3

To score a wait, add over both kinds



$$p(66) = \pi_1 \mathcal{N}(66;\ \text{short}) + \pi_2 \mathcal{N}(66;\ \text{long})$$

$$= 0.0037 + 0.0024 = 0.0062$$

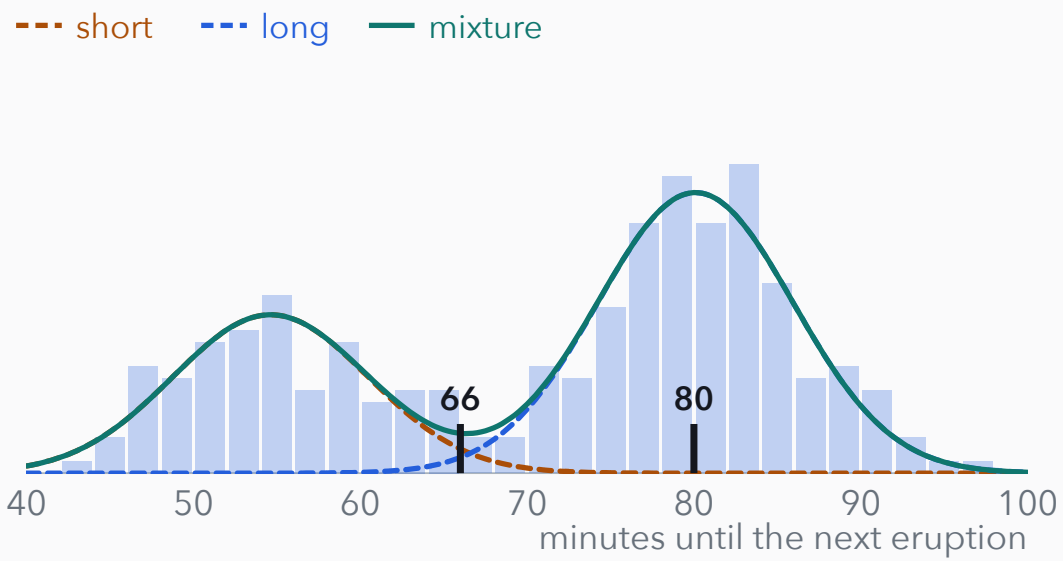
$$p(80) = 0.000002 + 0.0435 = 0.0435$$

> At 80 minutes, almost all the probability comes from the long kind.

Old Faithful geyser data (Azzalini and Bowman, 1990), as distributed with R (`datasets::faithful`): 272 eruptions.

▶ Notebook 2.3

To score a wait, add over both kinds



$$p(66) = \pi_1 \mathcal{N}(66;\ \text{short}) + \pi_2 \mathcal{N}(66;\ \text{long})$$

$$= 0.0037 + 0.0024 = 0.0062$$

$$p(80) = 0.000002 + 0.0435 = 0.0435$$

Average log-likelihood: **−3.801**, up from −4.027 for one bump.

> Over all 272 waits the mixture scores −3.801 against −4.027: the data are e^61 times more probable under it.

Old Faithful geyser data (Azzalini and Bowman, 1990), as distributed with R (`datasets::faithful`): 272 eruptions.

▶ Notebook 2.4

Fit it by gradient descent, as Lecture 1 fitted q


```
# x: the 272 waits, standardised
logits = torch.zeros(2, requires_grad=True)
mu = torch.tensor([-.25, .25], requires_grad=True)
log_sd = torch.zeros(2, requires_grad=True)
opt = Adam([logits, mu, log_sd], lr=0.03)
for step in range(600):
    log_pi = torch.log_softmax(logits, 0)
    comp = Normal(mu, log_sd.exp()).log_prob(x[:, None])
    loss = -torch.logsumexp(log_pi + comp, 1).mean()
    opt.zero_grad(); loss.backward(); opt.step()
```

> Same loop as Lecture 1: parameters, a loss (the negative average log-likelihood), Adam.

▶ Notebook 2.4

Fit it by gradient descent, as Lecture 1 fitted q


```
# x: the 272 waits, standardised
logits = torch.zeros(2, requires_grad=True)
mu = torch.tensor([-.25, .25], requires_grad=True)
log_sd = torch.zeros(2, requires_grad=True)
opt = Adam([logits, mu, log_sd], lr=0.03)
for step in range(600):
    log_pi = torch.log_softmax(logits, 0)
    comp = Normal(mu, log_sd.exp()).log_prob(x[:, None])
    loss = -torch.logsumexp(log_pi + comp, 1).mean()
    opt.zero_grad(); loss.backward(); opt.step()
```

`log_softmax`: π_1, π_2 stay positive and add to 1.

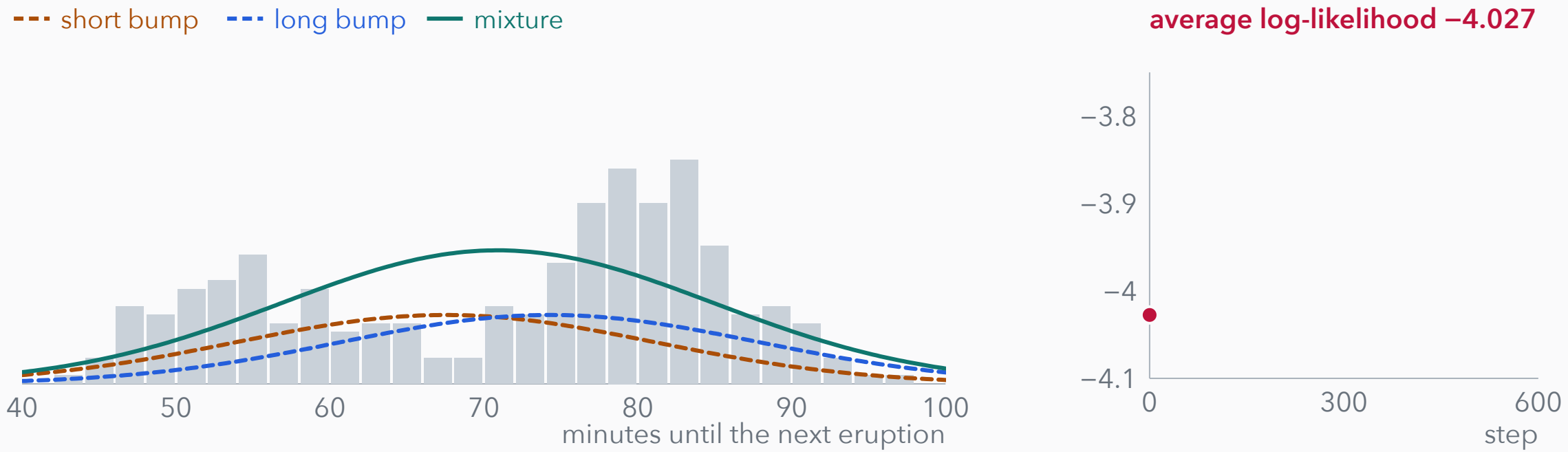
`log_sd.exp()`: σ stays positive.

`logsumexp`: log of the sum over kinds, computed safely.

> Three details keep the parameters valid: softmax for π, exp for σ, and logsumexp for the log of a sum.

▶ Notebook 2.4

Press play: the bumps separate onto the two clumps

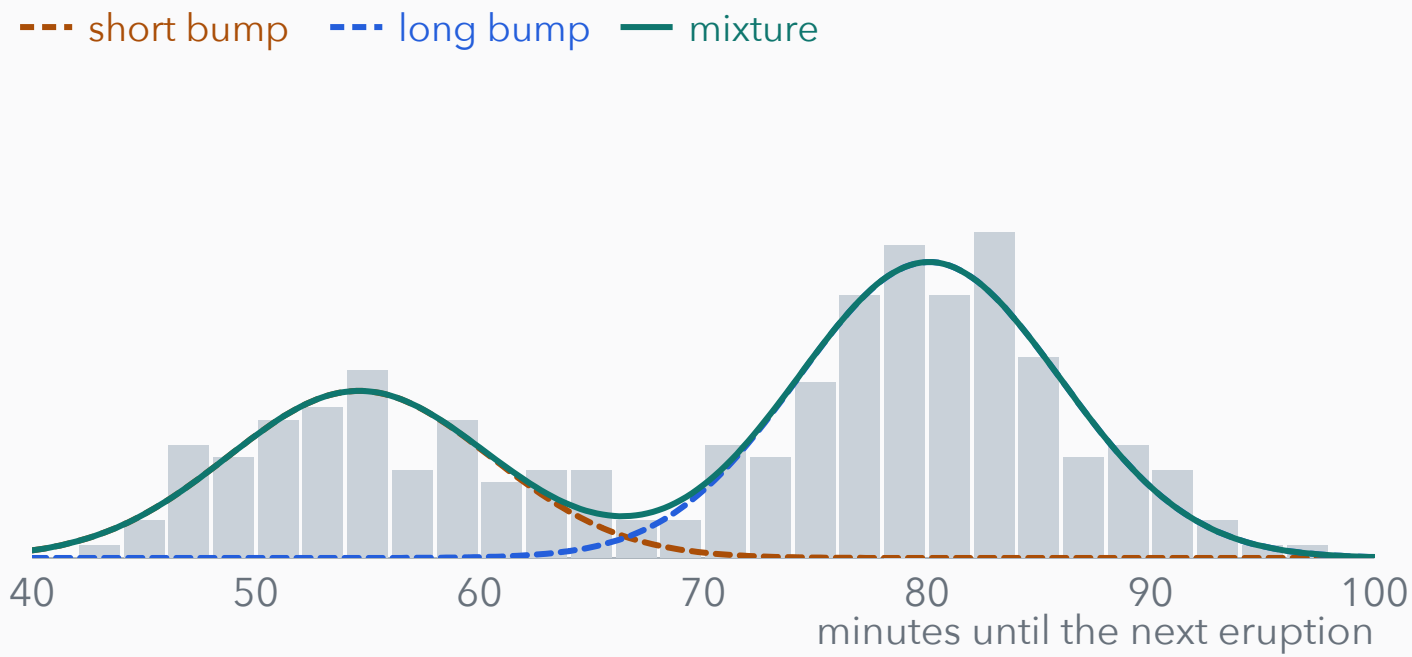


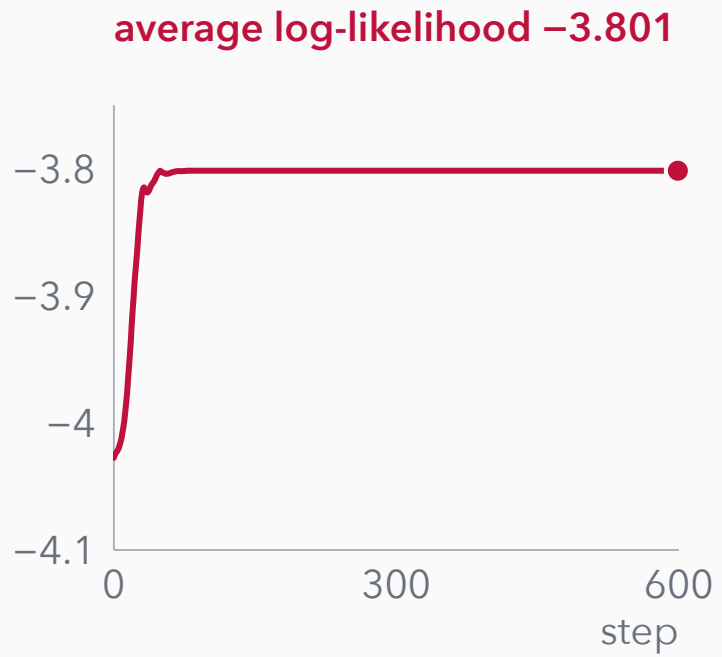
Both bumps start wide and near the middle.

Old Faithful geyser data (Azzalini and Bowman, 1990), as distributed with R (`datasets::faithful`): 272 eruptions.

▶ Notebook 2.4

Press play: the bumps separate onto the two clumps





They pull apart onto the clumps; the score rises from −4.03 to −3.80.

Old Faithful geyser data (Azzalini and Bowman, 1990), as distributed with R (`datasets::faithful`): 272 eruptions.

▶ Notebook 2.4

It found the two kinds of eruption without being told

	short kind	long kind
share π	0.36	0.64
centre μ (min)	54.6	80.1
width σ (min)	5.9	5.9

> About 36% short waits near 55 minutes, 64% long waits near 80.

Old Faithful geyser data (Azzalini and Bowman, 1990), as distributed with R (`datasets::faithful`): 272 eruptions.

▶ Notebook 2.4

It found the two kinds of eruption without being told

	short kind	long kind
share π	0.36	0.64
centre μ (min)	54.6	80.1
width σ (min)	5.9	5.9

Waits it puts in the short bump follow eruptions of **2.09** min on average; long bump: **4.29** min.

It never saw eruption lengths. The hidden variable it invented matches a real one.

> Check against the eruption lengths, which the fit never saw: the two bumps line up with short and long eruptions.

Old Faithful geyser data (Azzalini and Bowman, 1990), as distributed with R (`datasets::faithful`): 272 eruptions.

▶ Notebook 2.4

Sample: pick a bump, then a time

step 1: pick a bump

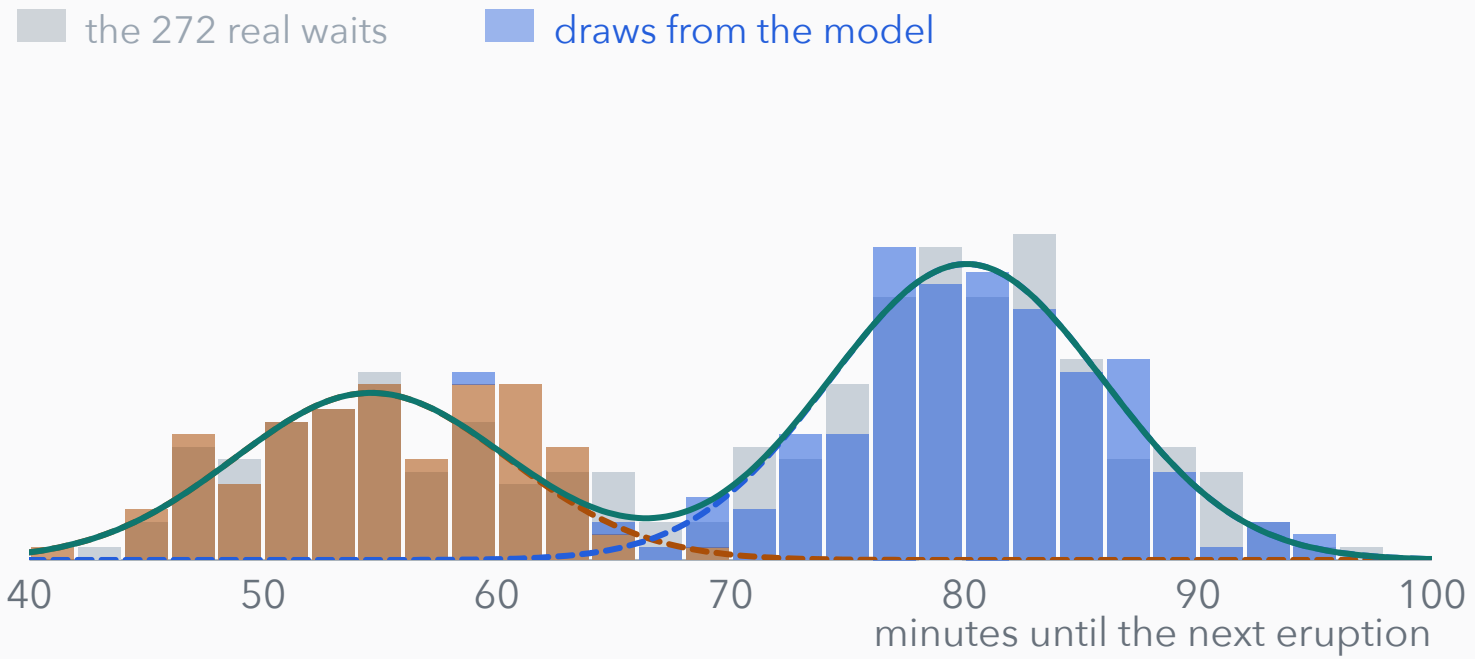
short wait: probability 0.36
N(54.6, 5.9²) · picked 107×

long wait: probability 0.64
N(80.1, 5.9²) · picked 165×

step 2: draw a time from that bump

latest: long bump, 78.9 minutes

272 draws so far



> Sampling runs the recipe: pick short or long with probabilities 0.36 and 0.64, then draw from that bump.

Old Faithful geyser data (Azzalini and Bowman, 1990), as distributed with R (`datasets::faithful`): 272 eruptions.

▶ Notebook 2.4

Sample: pick a bump, then a time

step 1: pick a bump

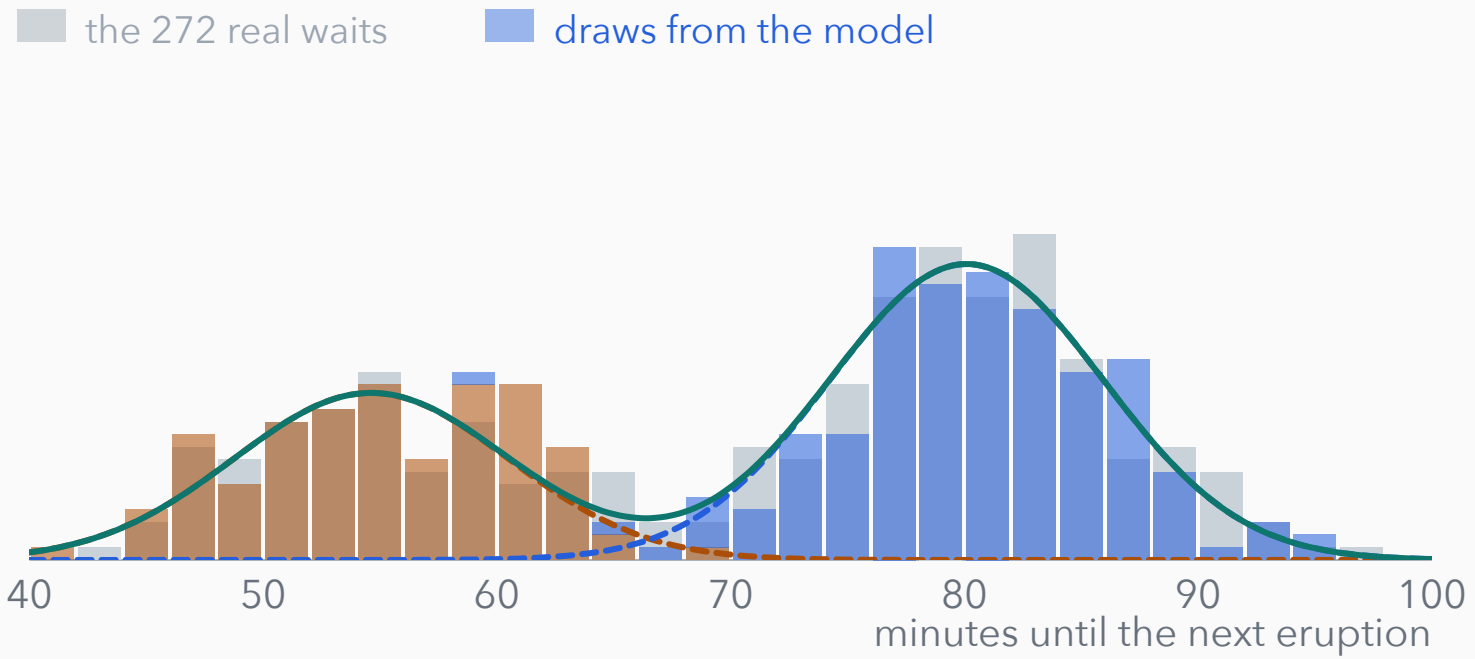
short wait: probability 0.36
N(54.6, 5.9²) · picked 107×

long wait: probability 0.64
N(80.1, 5.9²) · picked 165×

step 2: draw a time from that bump

latest: long bump, 78.9 minutes

272 draws so far

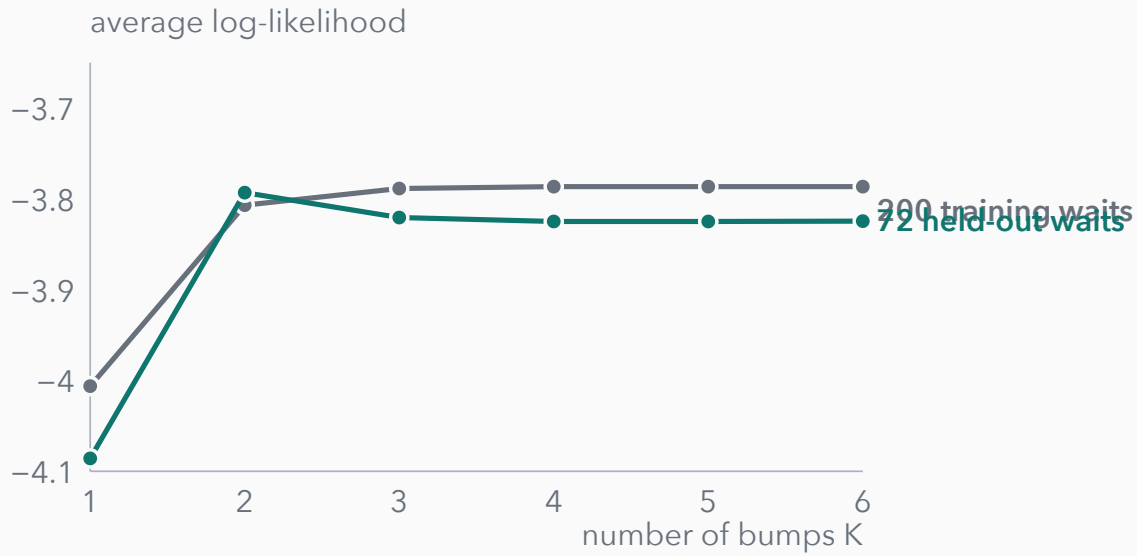


After 272 draws the histogram of samples matches the data: two clumps, few in the dip.

Old Faithful geyser data (Azzalini and Bowman, 1990), as distributed with R (`datasets::faithful`): 272 eruptions.

▶ Notebook 2.5

More bumps always fit the training waits better; new waits decide



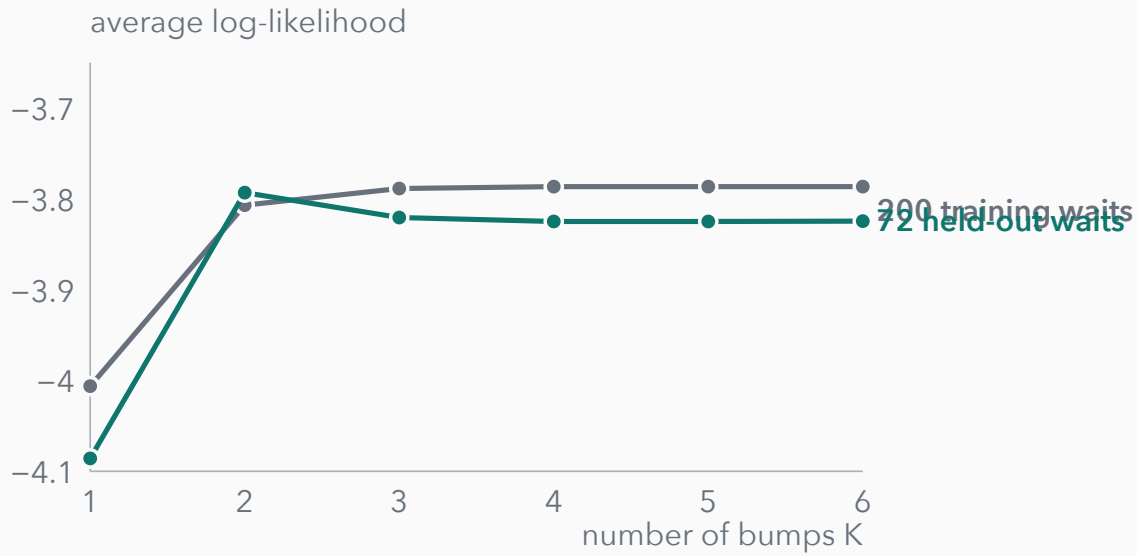
Fit on 200 waits, score on the other 72.

Fit K = 1 to 6 bumps on 200 waits. The training score only goes up, or stays level.

Old Faithful geyser data (Azzalini and Bowman, 1990), as distributed with R (`datasets::faithful`): 272 eruptions.

▶ Notebook 2.5

More bumps always fit the training waits better; new waits decide



Fit on 200 waits, score on the other 72.

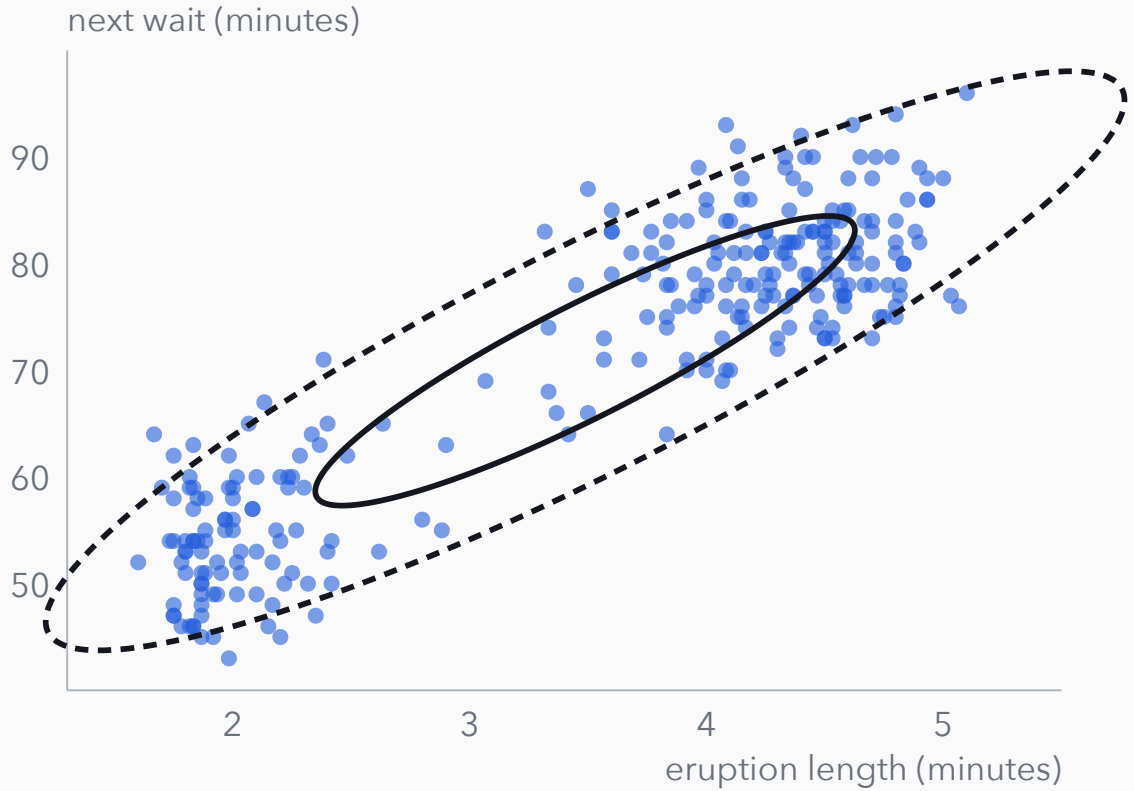
Held-out score is best at **K = 2**, then drops slightly.

> On the 72 held-out waits the score peaks at two bumps. Extra bumps fit noise in the training waits.

Old Faithful geyser data (Azzalini and Bowman, 1990), as distributed with R (`datasets::faithful`): 272 eruptions.

▶ Notebook 2.6

Two numbers per eruption: Gaussians need a covariance



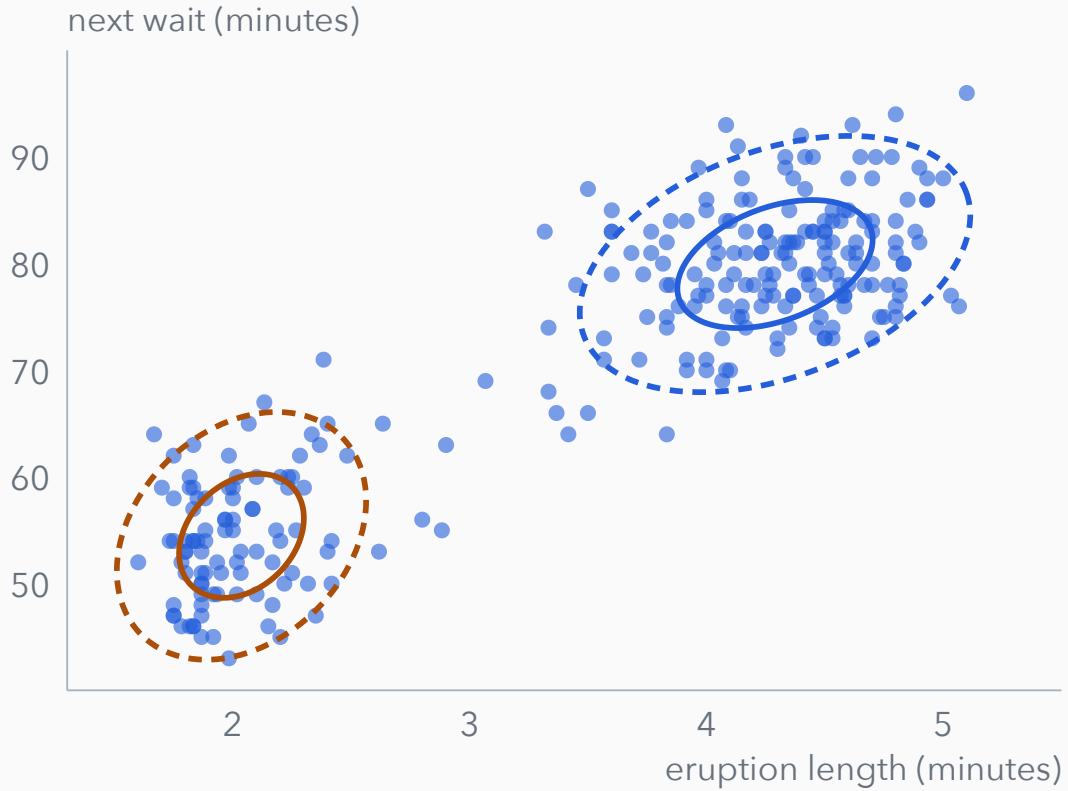
One 2-D Gaussian: a mean and a 2 × 2 covariance. Correlation 0.90.

> With two numbers per example, a Gaussian needs a 2 × 2 covariance: how length and wait vary together (Lecture 1, section 3.3).

Old Faithful geyser data (Azzalini and Bowman, 1990), as distributed with R (`datasets::faithful`): 272 eruptions.

▶ Notebook 2.6

Two numbers per eruption: Gaussians need a covariance



One 2-D Gaussian: a mean and a 2 × 2 covariance. Correlation 0.90.

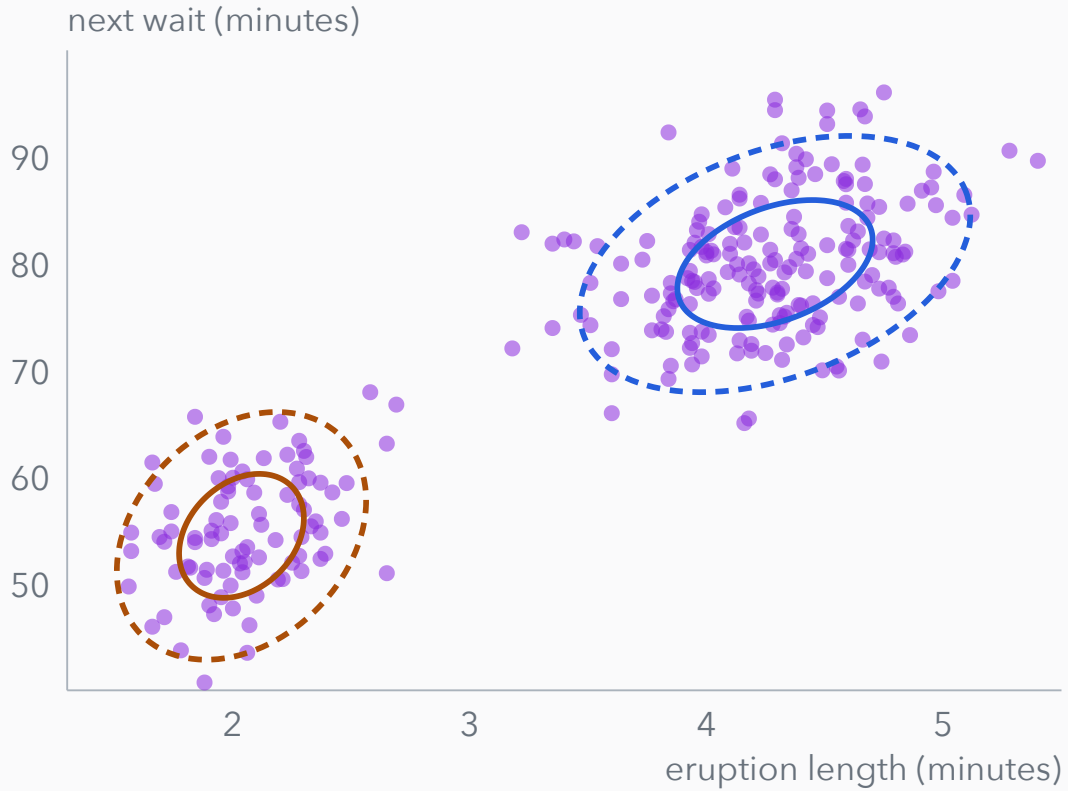
Two 2-D bumps, each with its own covariance: average log-likelihood −4.16 against −4.74.

A mixture of two 2-D Gaussians: one tilted ellipse per kind of eruption.

Old Faithful geyser data (Azzalini and Bowman, 1990), as distributed with R (`datasets::faithful`): 272 eruptions.

▶ Notebook 2.6

Two numbers per eruption: Gaussians need a covariance



One 2-D Gaussian: a mean and a 2 × 2 covariance. Correlation 0.90.

Two 2-D bumps, each with its own covariance: average log-likelihood −4.16 against −4.74.

Samples from the 2-D mixture look like the eruptions.

> Sampling works exactly as before: pick a kind, then draw a pair from its 2-D Gaussian.

Old Faithful geyser data (Azzalini and Bowman, 1990), as distributed with R (`datasets::faithful`): 272 eruptions.

▶ Notebook 2.6

Part 2 in one slide

step	one number per example	what we learned
fit	choose parameters to maximise the average log p(x)	same loop as Lecture 1
score	evaluate log p(x) of any wait	held-out score picks the model
sample	run the recipe: pick a bump, then draw	hidden variables make samples realistic

> Fit, score and sample, with a hidden choice that made the samples realistic.

▶ Notebook 2.6

Part 2 in one slide

step	one number per example	what we learned
fit	choose parameters to maximise the average log p(x)	same loop as Lecture 1
score	evaluate log p(x) of any wait	held-out score picks the model
sample	run the recipe: pick a bump, then draw	hidden variables make samples realistic

Next: each example is a 28 × 28 image, 784 numbers. Same three steps, much harder.

> Next, images: the same three steps with 784 numbers per example.

▶ Notebook 3.1

MNIST: 70,000 handwritten digits, 28 × 28 pixels each



60,000 for training, 10,000 for testing.

The first 16 training digits. Each is a 28 × 28 grey image of one handwritten digit.

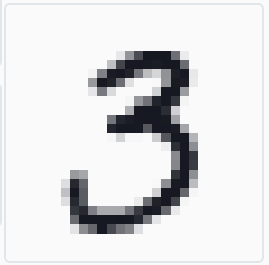
MNIST (LeCun, Cortes and Burges), grey levels scaled to [0, 1].

▶ Notebook 3.1

MNIST: 70,000 handwritten digits, 28 × 28 pixels each



60,000 for training, 10,000 for testing.



Our running example: test digit #90, a 3.

We follow one digit all lecture: test digit #90, a 3.

MNIST (LeCun, Cortes and Burges), grey levels scaled to [0, 1].

▶ Notebook 3.1

Our 3 is 784 numbers between 0 and 1



0	0	0	0.3	0.8	1.0	1.0	1.0	1.0	1.0	1.0
0	0.0	0.5	1.0	1.0	0.9	0.6	0.3	0.2	0.5	1.0
0	0.4	1.0	0.9	0.6	0.1	0	0	0.0	0.5	1.0
0	0.1	0.6	0.1	0	0	0	0	0.2	1.0	1.0
0	0	0	0	0	0.0	0.4	0.6	0.9	1.0	0.9
0	0	0	0	0	0.3	1.0	1.0	1.0	0.9	0.2
0	0	0	0.6	0.9	1.0	1.0	1.0	1.0	1.0	0.4
0	0	0	0.9	1.0	1.0	1.0	0.7	1.0	1.0	1.0
0	0	0	0.0	0.2	0.0	0.0	0	0.2	0.7	1.0
0	0	0	0	0	0	0	0	0	0.1	0.9

Each pixel is a number: 0 is paper, 1 is full ink.

The pixels of a patch of our 3 (rows 6–15, columns 8–18), as numbers.

▶ Notebook 3.1

Our 3 is 784 numbers between 0 and 1



0	0	0	0.3	0.8	1.0	1.0	1.0	1.0	1.0	1.0
0	0.0	0.5	1.0	1.0	0.9	0.6	0.3	0.2	0.5	1.0
0	0.4	1.0	0.9	0.6	0.1	0	0	0.0	0.5	1.0
0	0.1	0.6	0.1	0	0	0	0	0.2	1.0	1.0
0	0	0	0	0	0.0	0.4	0.6	0.9	1.0	0.9
0	0	0	0	0	0.3	1.0	1.0	1.0	0.9	0.2
0	0	0	0.6	0.9	1.0	1.0	1.0	1.0	1.0	0.4
0	0	0	0.9	1.0	1.0	1.0	0.7	1.0	1.0	1.0
0	0	0	0.0	0.2	0.0	0.0	0	0.2	0.7	1.0
0	0	0	0	0	0	0	0	0	0.1	0.9

Each pixel is a number: 0 is paper, 1 is full ink.

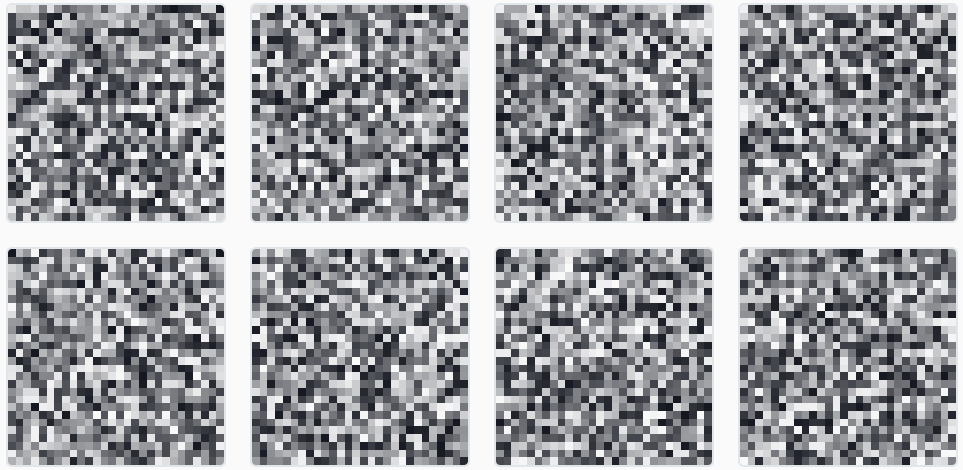
$$x \in [0, 1]^{784}$$

One image = one point in a 784-dimensional space.

> Flatten the 28 × 28 grid into a list: x is one point in a 784-dimensional space. A model of digits is a distribution over that space.

▶ Notebook 3.1

A random point in that space looks like static

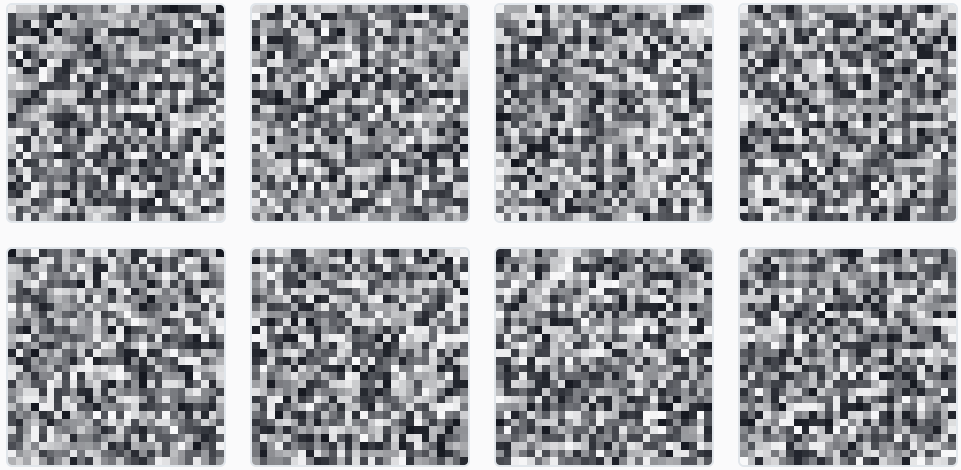


Each pixel drawn uniformly between 0 and 1.

> Eight random points in the 784-dimensional space: every pixel uniform on [0, 1].

▶ Notebook 3.1

A random point in that space looks like static



Each pixel drawn uniformly between 0 and 1.

Almost every point in the space is static. Digits live in a tiny region; p(x) must find it.

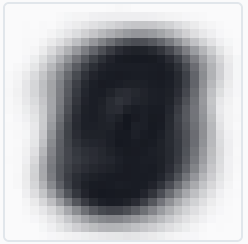
> You would have to draw for longer than the age of the universe to hit a digit. A good p(x) concentrates on the tiny region where digits live.

▶ Notebook 3.2

First try: one Gaussian per pixel, all independent



mean of each pixel



spread of each pixel

> Fit a Gaussian to each pixel separately, like Part 2: its mean and its spread over the 60,000 training digits.

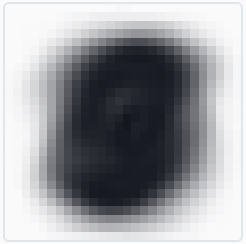
MNIST training set. Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 3.2

First try: one Gaussian per pixel, all independent



mean of each pixel



spread of each pixel

$$p(x) = \prod_{j=1}^{784} \mathcal{N}(x_j;\, m_j, s_j^2)$$

2 × 784 = 1,568 numbers.

> The model multiplies 784 independent bells: 1,568 numbers.

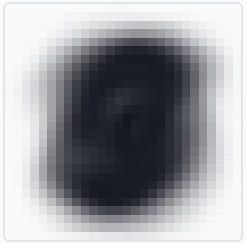
MNIST training set. Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 3.2

First try: one Gaussian per pixel, all independent



mean of each pixel



spread of each pixel

$$p(x) = \prod_{j=1}^{784} \mathcal{N}(x_j;\, m_j, s_j^2)$$

8 samples



2 × 784 = 1,568 numbers.

> Samples are speckled noise around the average digit: every pixel decides on its own.

MNIST training set. Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 3.2

Neighbouring pixels move together



correlation of the centre pixel (14, 14) with every pixel: teal positive, red negative

pixel pair	correlation
centre and its right neighbour	0.78
two to the right	0.29
five to the right	−0.21
a far corner pixel	−0.01

> If the centre pixel is inked, its neighbours usually are too: correlation 0.78 with the pixel to its right.

Correlations over the 60,000 MNIST training digits. Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 3.2

Neighbouring pixels move together



correlation of the centre pixel (14, 14) with every pixel: teal positive, red negative

pixel pair	correlation
centre and its right neighbour	0.78
two to the right	0.29
five to the right	−0.21
a far corner pixel	−0.01

An independent model throws all of this away.

> Strokes are made of neighbouring pixels that move together. A model of digits must capture that.

Correlations over the 60,000 MNIST training digits. Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 3.2

Second try: one 784-dimensional Gaussian with a full covariance

$$p(x) = \mathcal{N}(x;\, m, \Sigma), \qquad \Sigma \in \mathbb{R}^{784\times 784}$$

308,504 numbers: 784 for the mean,
307,720 for the covariance.

8 samples



> One big Gaussian whose covariance links every pair of pixels: over 300,000 numbers.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 3.2

Second try: one 784-dimensional Gaussian with a full covariance

$$p(x) = \mathcal{N}(x;\, m, \Sigma), \qquad \Sigma \in \mathbb{R}^{784 \times 784}$$

308,504 numbers: 784 for the mean, 307,720 for the covariance.

Samples have smoother, stroke-like blobs, but they are ghosts, not digits.

8 samples

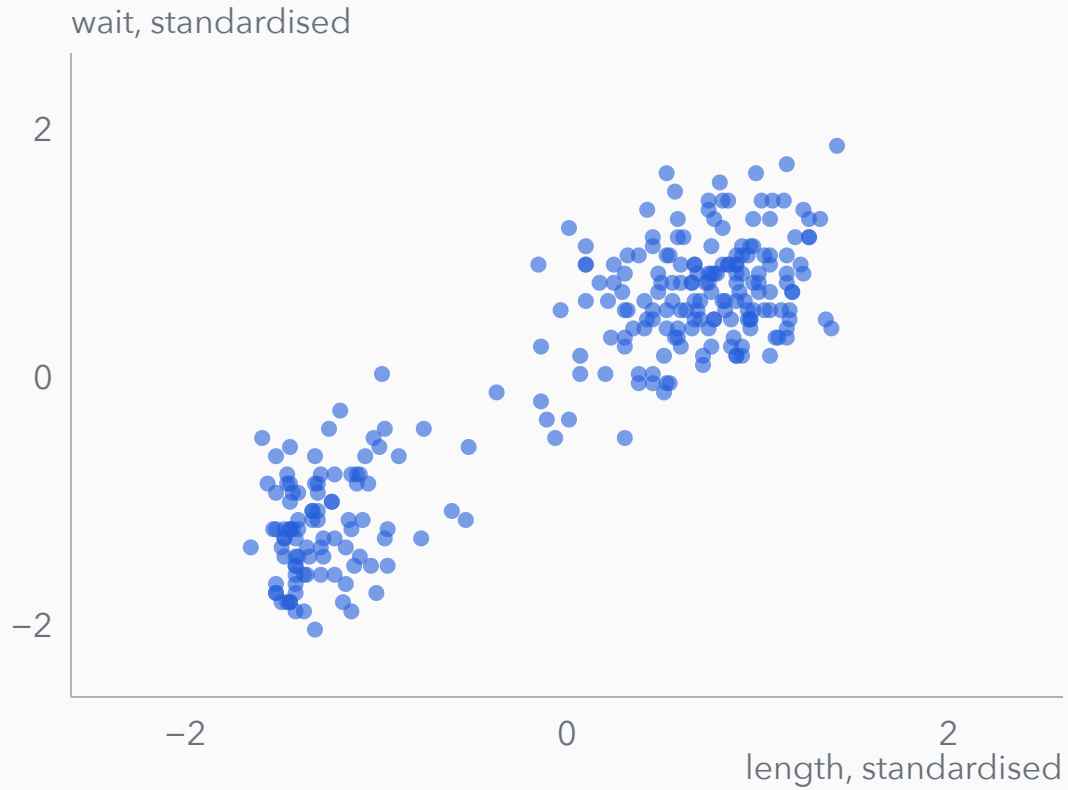


> Samples are smoother, because neighbouring pixels now move together. Still not digits: one bump for ten kinds of digit, and far too many numbers.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 3.3

Idea: describe each example with fewer numbers. PCA in 2-D first

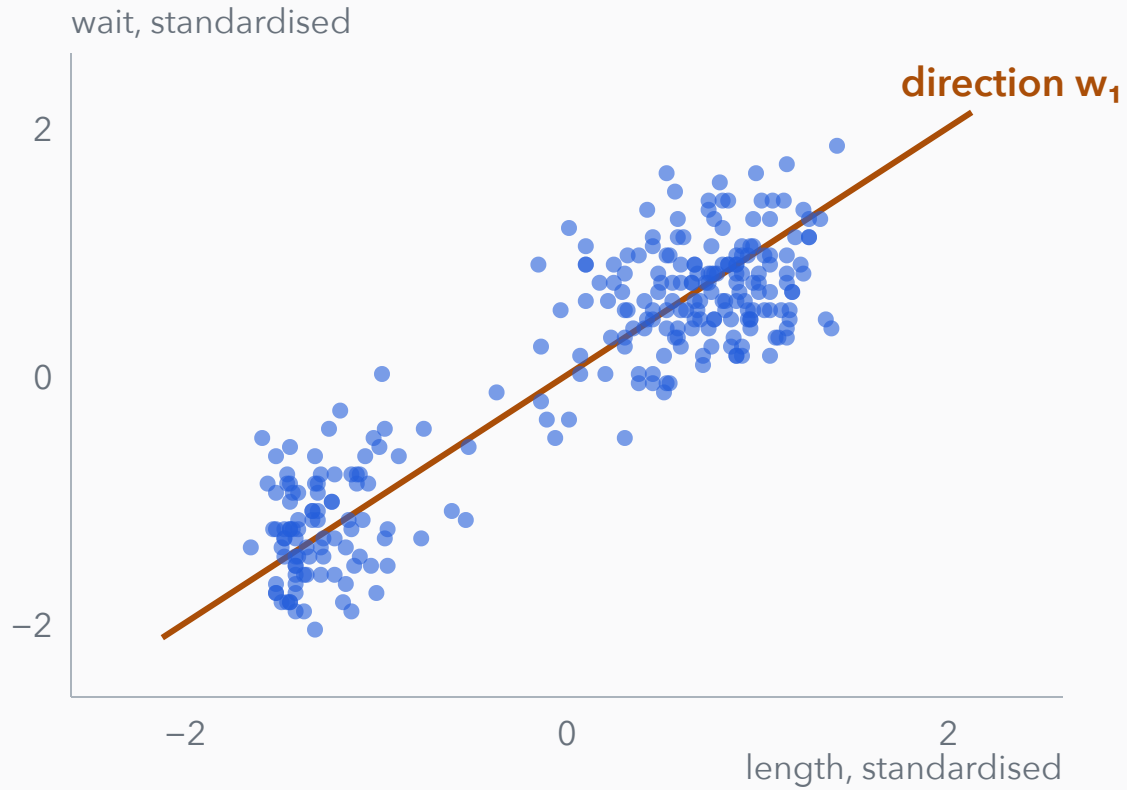


Back to the eruptions: two numbers each (standardised), strongly correlated.

Old Faithful geyser data (Azzalini and Bowman, 1990), as distributed with R (`datasets::faithful`): 272 eruptions.

▶ Notebook 3.3

Idea: describe each example with fewer numbers. PCA in 2-D first



share of the spread along w_1: 95%

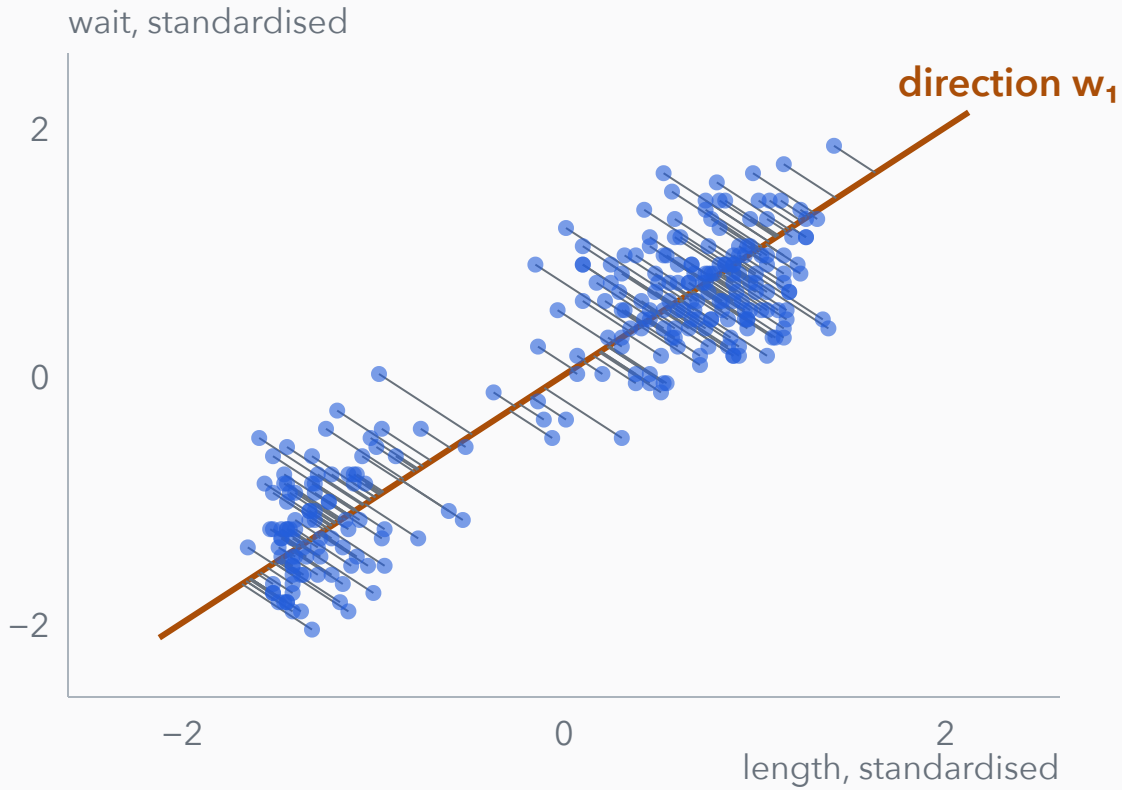
the other direction keeps 5%

> Find the direction along which the points spread most: here 95% of the spread.

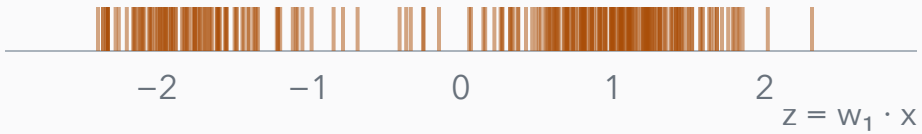
Old Faithful geyser data (Azzalini and Bowman, 1990), as distributed with R (`datasets::faithful`): 272 eruptions.

▶ Notebook 3.3

Idea: describe each example with fewer numbers. PCA in 2-D first



one number per eruption



share of the spread along w_1: 95%

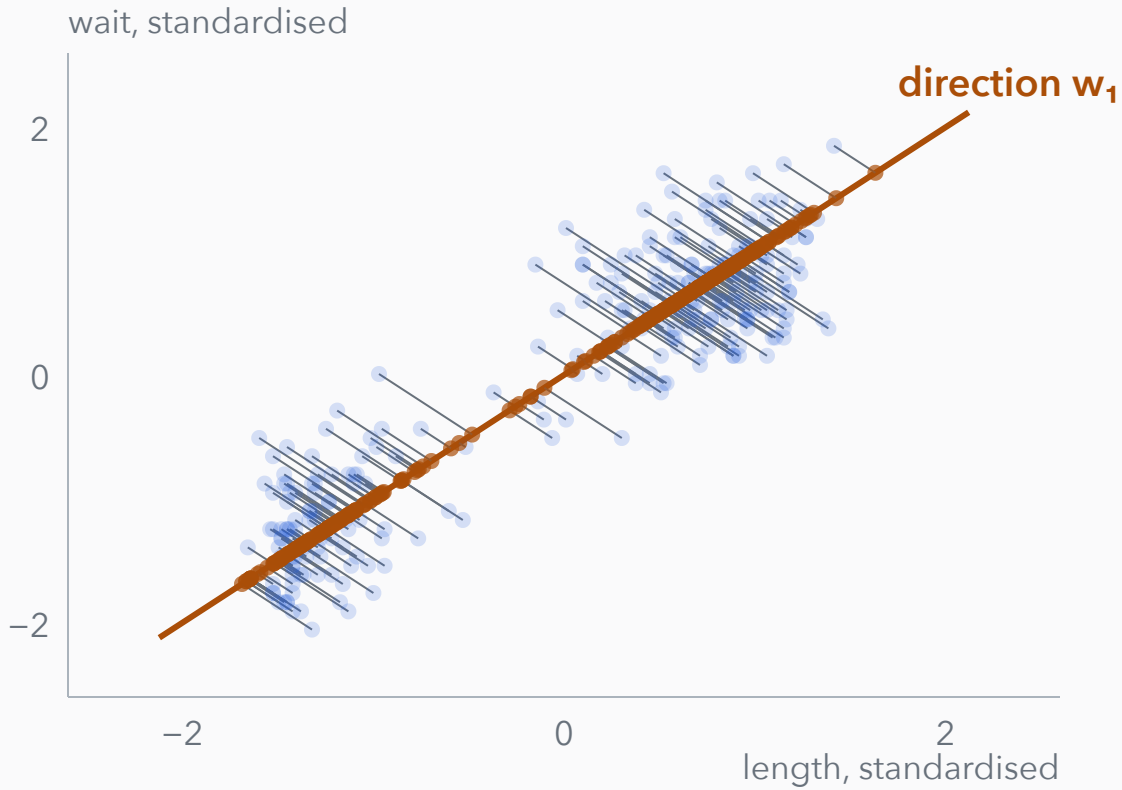
the other direction keeps 5%

> Project each point onto that line: one number z per eruption instead of two.

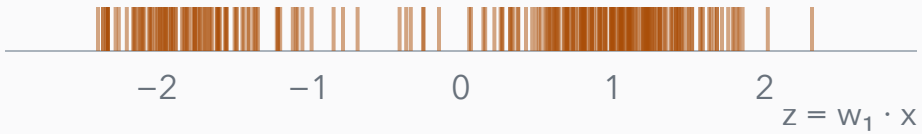
Old Faithful geyser data (Azzalini and Bowman, 1990), as distributed with R (`datasets::faithful`): 272 eruptions.

▶ Notebook 3.3

Idea: describe each example with fewer numbers. PCA in 2-D first



one number per eruption



share of the spread along w_1: 95%

the other direction keeps 5%

rebuild: $\hat{x} = z \cdot w_1$ (orange)

two numbers per eruption became one

> Rebuild each point from its one number: it lands on the line, close to where it started. That is PCA.

Old Faithful geyser data (Azzalini and Bowman, 1990), as distributed with R (`datasets::faithful`): 272 eruptions.

▶ Notebook 3.3

PCA on 60,000 digits: a mean image and directions



> PCA finds the directions of most spread in the 784-dimensional space: each direction is itself an image (teal adds ink, red removes it).

PCA of the 60,000 MNIST training digits (eigenvectors of the 784 × 784 covariance). Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 3.3

PCA on 60,000 digits: a mean image and directions



$$\underbrace{z = W^\top (x - m)}_{\text{encode: } 784 \to d} \qquad \underbrace{\hat{x} = m + Wz = m + z_1 w_1 + \cdots + z_d w_d}_{\text{decode: } d \to 784}$$

> Encode: d numbers per image, z = $W^T(x - m)$. Decode: start from the mean and add z_1 times w_1, z_2 times w_2, and so on.

PCA of the 60,000 MNIST training digits (eigenvectors of the 784 × 784 covariance). Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 3.3

Rebuild our 3 from its first d PCA numbers



our 3: 784 numbers

→

10 numbers

$z_1 = 0.88, z_2 = 2.55, \ldots$

→



rebuilt from the first 10

26.4

squared error

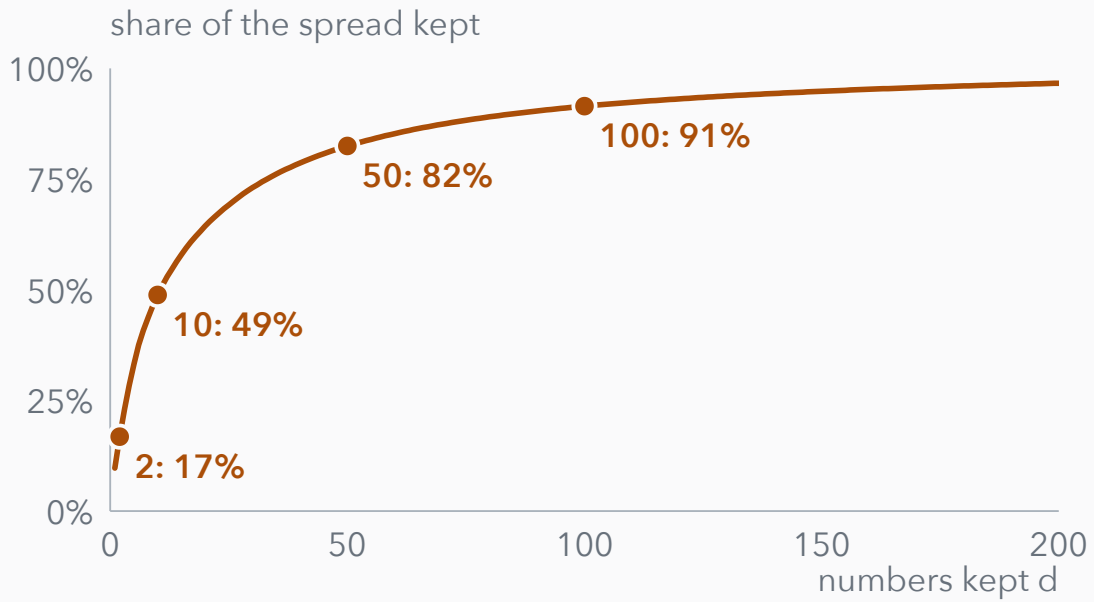
48.8%

of the variance of all 60,000 digits kept

Drag d: 2 numbers give a blur (error 41.3), 10 a rough 3 (26.4), 50 a clear 3 (7.2).

▶ Notebook 3.3

How many numbers do we need?

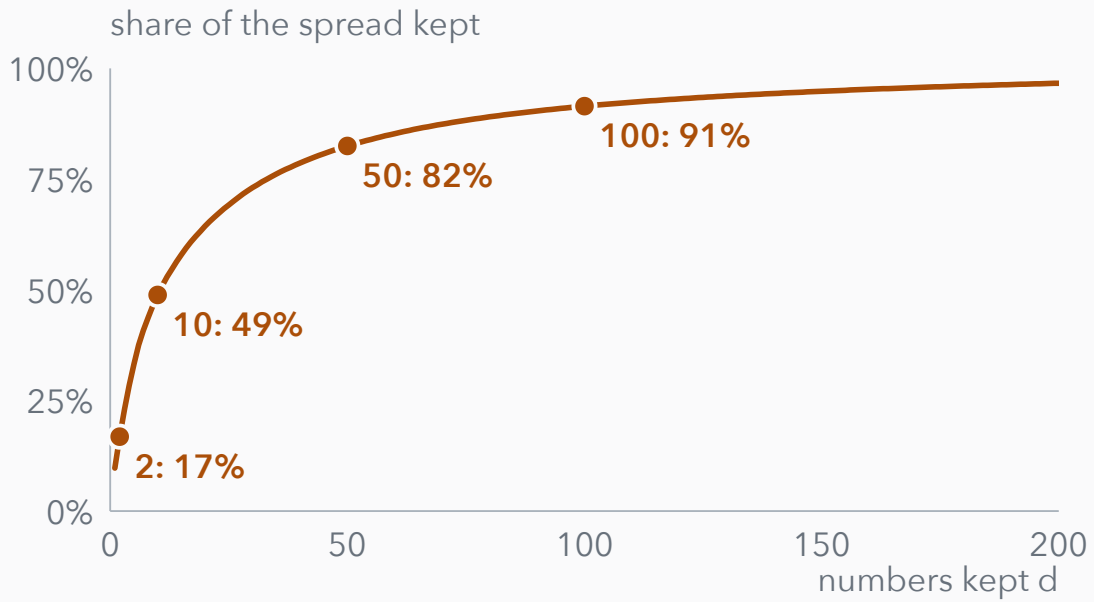


d = 10 keeps 49% of the spread; d = 50 keeps 82%.

> The first few directions carry the most spread; the curve flattens fast.

▶ Notebook 3.3

How many numbers do we need?



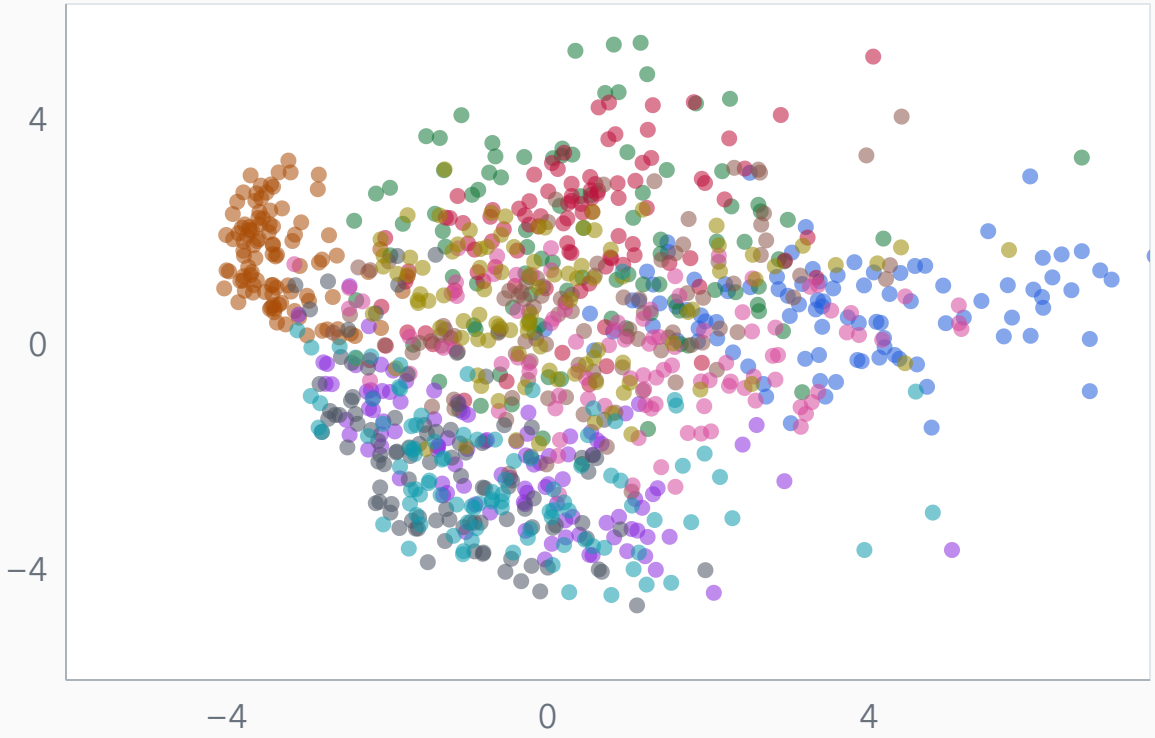
d = 10 keeps 49% of the spread; d = 50 keeps 82%.

784 numbers per digit, but about 50 carry most of the information.

> Digits really live near a much lower-dimensional surface. That is why compressing first is a good idea.

▶ Notebook 3.3

Each digit as 2 numbers: PCA's map of 1,000 test digits



0 1 2 3 4 5 6 7 8 9

z_1, z_2 from PCA

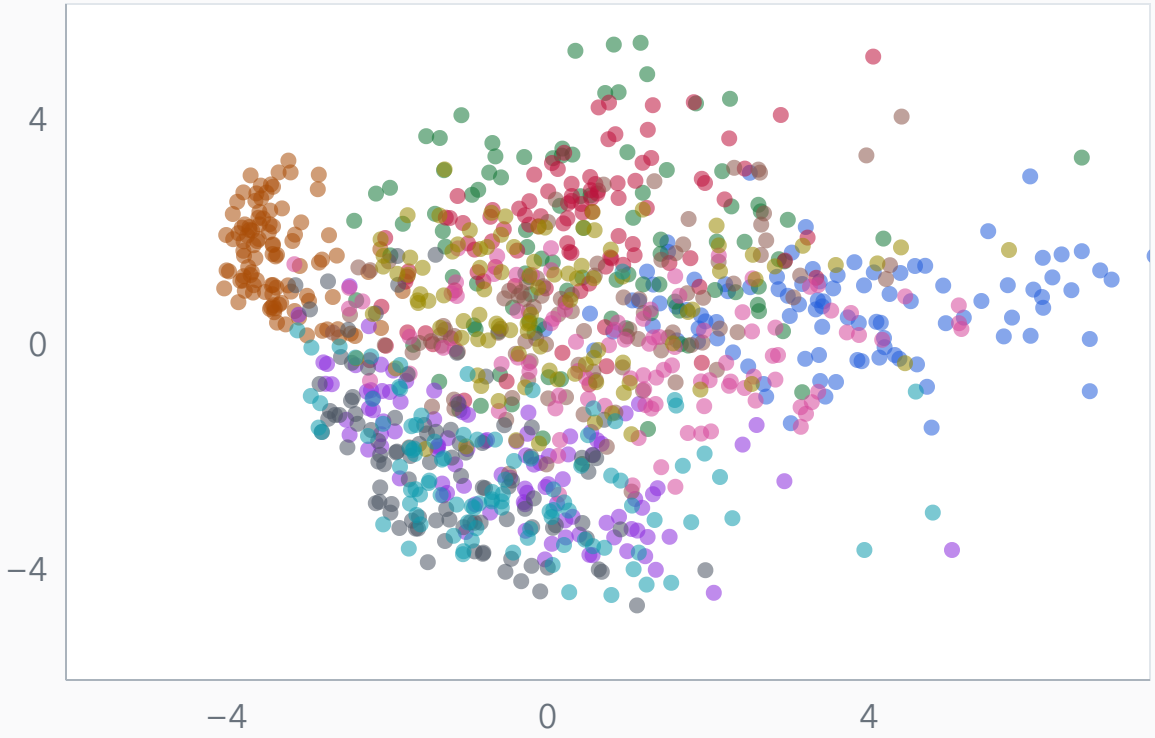
hover any point to see its digit

> Each point is one test digit, placed at its first two PCA numbers and coloured by its label. Hover to see the digit.

1,000 MNIST test digits, 100 per class (seed 2024). Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 3.3

Each digit as 2 numbers: PCA's map of 1,000 test digits



0 1 2 3 4 5 6 7 8 9

z_1, z_2 from PCA

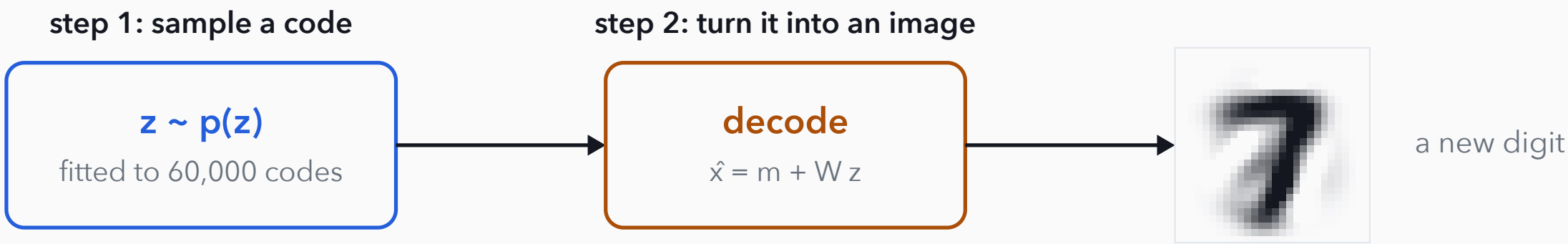
hover any point to see its digit

> 0s and 1s separate; 4, 7 and 9 overlap heavily. Two straight-line numbers cannot pull every digit apart.

1,000 MNIST test digits, 100 per class (seed 2024). Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 3.4

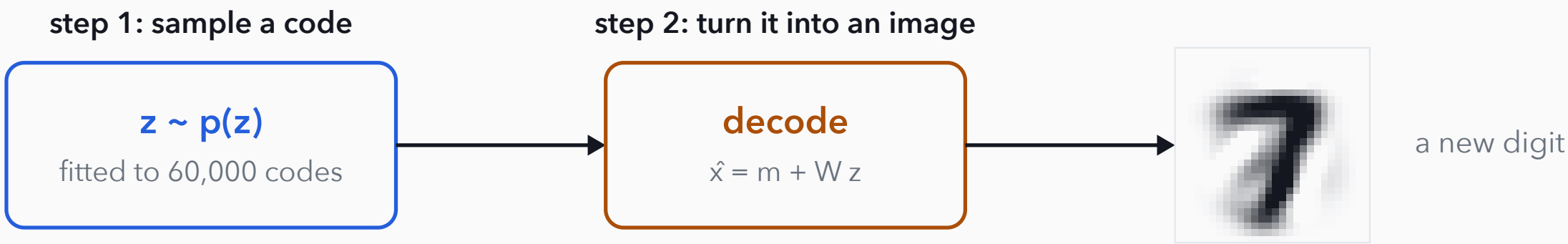
Generate in two steps: sample a few numbers, then rebuild an image



> Fit a distribution to the d PCA numbers of the 60,000 training digits, sample a new code, and decode it with $\hat{x} = m + W z$.

▶ Notebook 3.4

Generate in two steps: sample a few numbers, then rebuild an image



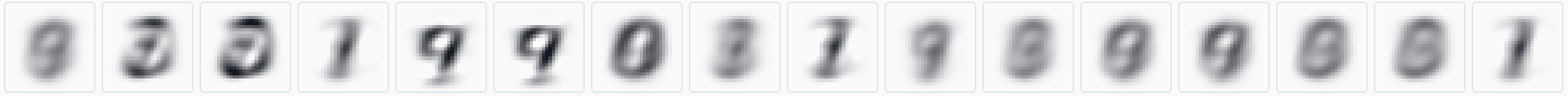
Part 2's recipe again: first a hidden thing (the code z), then the data. Here step 2 is PCA's decoder.

> The same shape as the mixture: pick a hidden value first, then produce x from it. The code z plays the role of the bump k.

▶ Notebook 3.4

One Gaussian over d PCA numbers

d = 2



d = 10



d = 50

> A Gaussian over 2 numbers gives blurry average-looking digits; over 10 numbers, rough digits appear.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 3.4

One Gaussian over d PCA numbers



> Over 50 numbers: more detail, but many samples are mixtures of two digits. One bump for ten kinds of digit.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 3.4

A mixture of K Gaussians over d numbers: try it



d = 20 PCA numbers · K = 10 bumps · 2,309 fitted numbers · held-out average log-likelihood −23.9

> Choose d and K, then press Sample: 16 codes from the mixture, each decoded with PCA. Compare K = 1 and K = 20 at d = 20.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 3.4

What limits the PCA recipe?



PCA's decoder is a straight-line map: $\hat{x} = m + W z$.

> With 2 numbers, PCA's rebuilt digits are blurs (test error 43.6 per image); with 16 they are recognisable but soft (21.1).

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 3.4

What limits the PCA recipe?



PCA's decoder is a straight-line map: $\hat{x} = m + W z$.

Bend it: replace both maps with small neural networks. That is an autoencoder.

> A flexible encoder and decoder should keep more from the same few numbers. Part 4.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 3.4

Part 3 in one slide

model	numbers	samples
independent pixel Gaussians	1,568	speckle
one full Gaussian	308,504	ghosts
Gaussian on 50 PCA numbers	1,325	blended digits
20 bumps on 20 PCA numbers	4,619	rough but real digits

> Raw pixels need huge Gaussians and still fail. Compressing to a few PCA numbers and fitting a mixture there works much better.

▶ Notebook 3.4

Part 3 in one slide

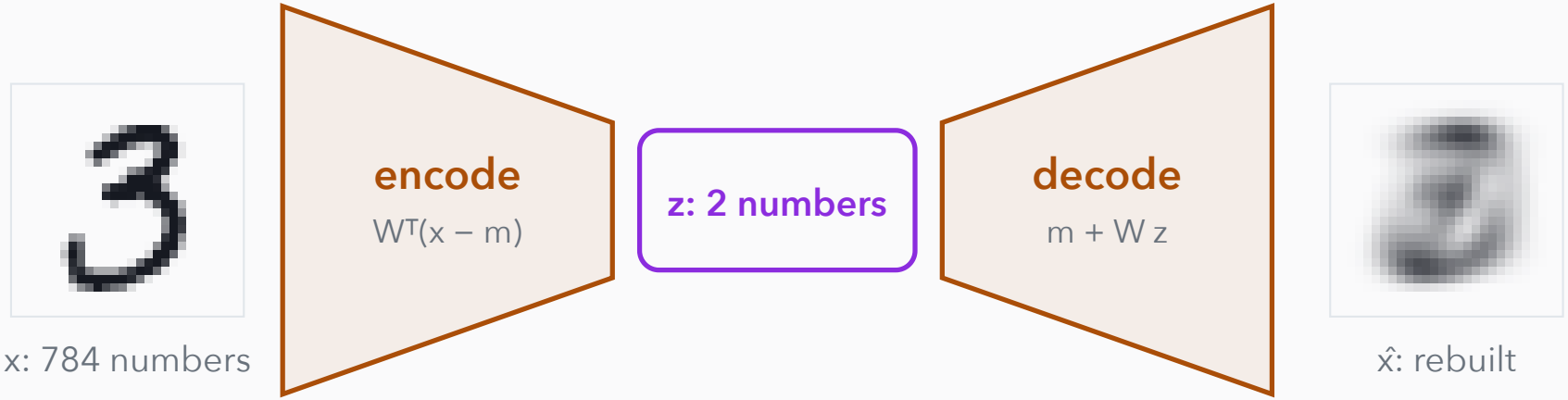
model	numbers	samples
independent pixel Gaussians	1,568	speckle
one full Gaussian	308,504	ghosts
Gaussian on 50 PCA numbers	1,325	blended digits
20 bumps on 20 PCA numbers	4,619	rough but real digits

Compress first, then fit: the code is the hidden variable. Next, learn the compression itself.

> The weak link is PCA's straight-line decoder. Part 4 learns a better one.

▶ Notebook 4.1

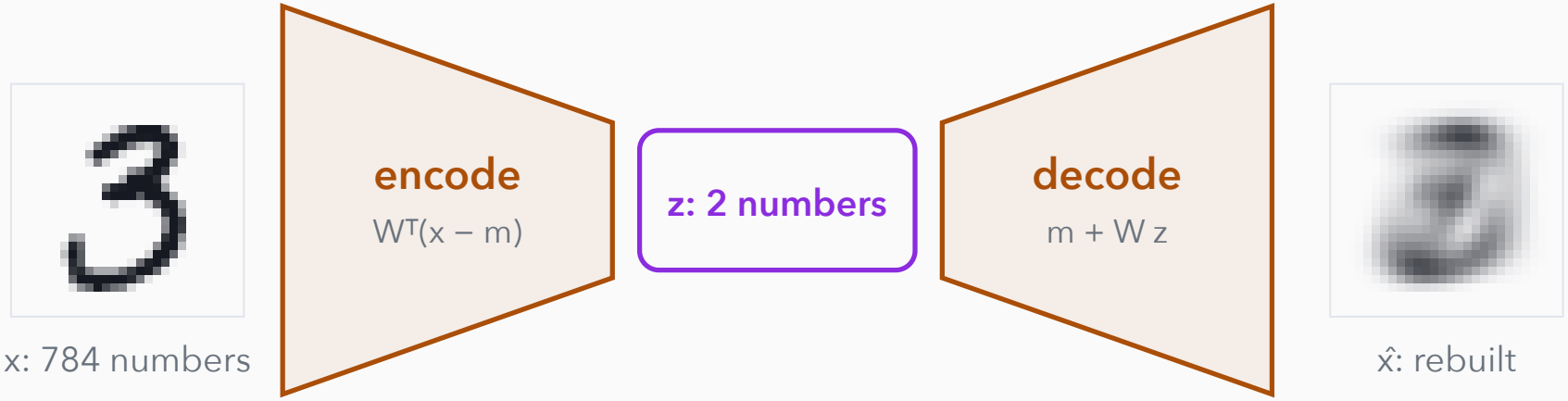
PCA is an encoder and a decoder



PCA already has the shape we want: squeeze 784 numbers into a short code, then expand back.

▶ Notebook 4.1

PCA is an encoder and a decoder

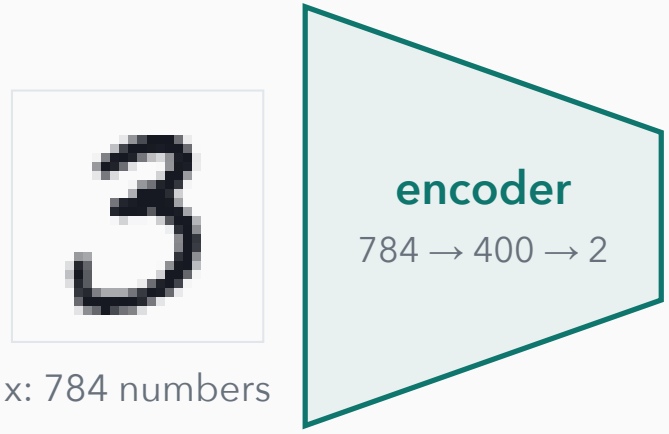


Both maps are straight lines (matrix products), chosen to make the rebuilt image close to the original.

> PCA's two maps are linear. For our 3 with 2 numbers the squared error is 41.3.

▶ Notebook 4.1

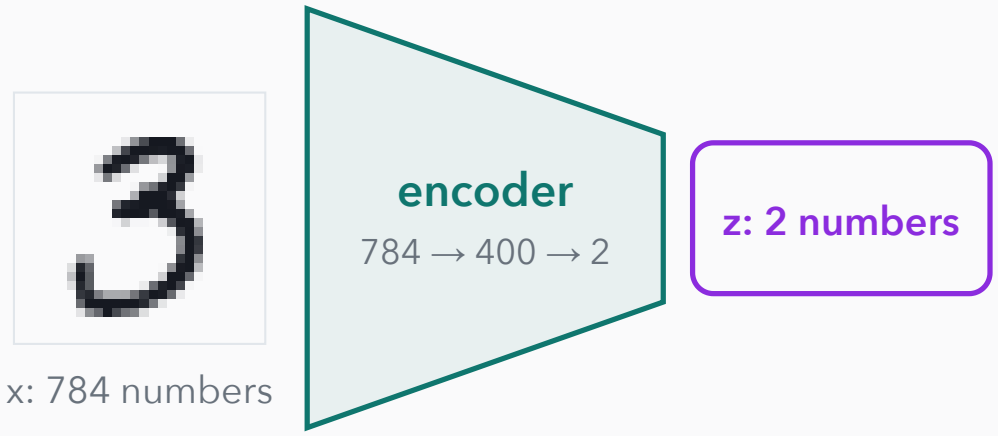
An autoencoder: replace both maps with small neural networks



The encoder is a small network: 784 inputs, 400 hidden units with ReLU, 2 outputs.

▶ Notebook 4.1

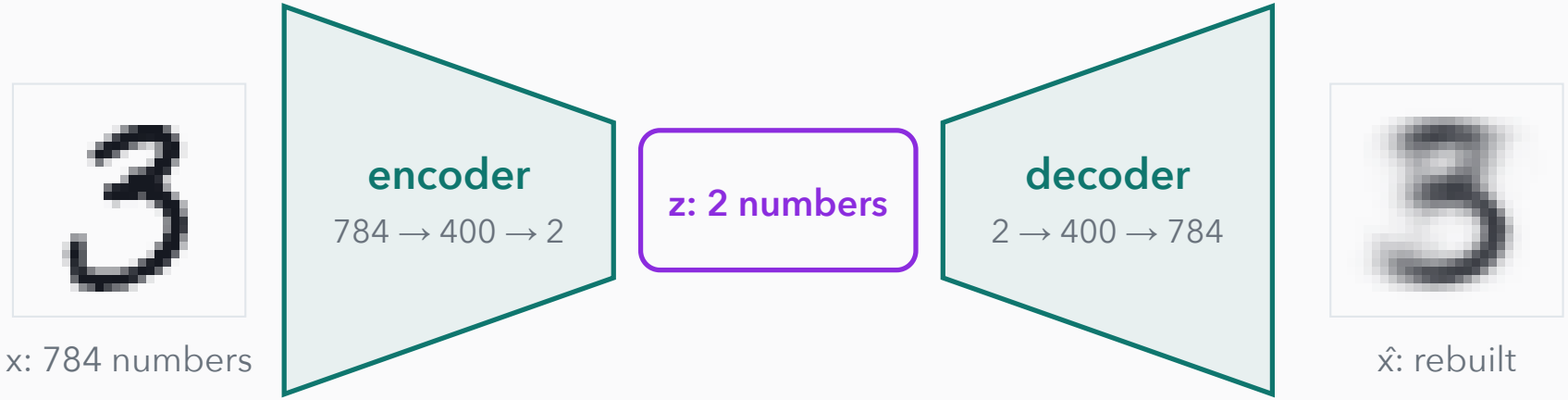
An autoencoder: replace both maps with small neural networks



> Its output is the code z, the bottleneck: like the neck of a bottle, everything must squeeze through 2 numbers.

▶ Notebook 4.1

An autoencoder: replace both maps with small neural networks



The decoder mirrors it: $2 \to 400 \to 784$, then a sigmoid. Train both so that $\hat{x} \approx x$.

▶ Notebook 4.1

Two small networks in PyTorch


```python
class AutoEncoder(nn.Module):
    def __init__(self, z=2):
        super().__init__()
        self.encoder = nn.Sequential(
            nn.Linear(784, 400), nn.ReLU(),
            nn.Linear(400, z))
        self.decoder = nn.Sequential(
            nn.Linear(z, 400), nn.ReLU(),
            nn.Linear(400, 784), nn.Sigmoid())

    def forward(self, x):
        z = self.encoder(x)     # 784 -> z numbers
        return self.decoder(z)  # z -> 784 numbers
```

Two nn.Sequential blocks. The forward pass encodes, then decodes.

▶ Notebook 4.1

Two small networks in PyTorch


```python
class AutoEncoder(nn.Module):
    def __init__(self, z=2):
        super().__init__()
        self.encoder = nn.Sequential(
            nn.Linear(784, 400), nn.ReLU(),
            nn.Linear(400, z))
        self.decoder = nn.Sequential(
            nn.Linear(z, 400), nn.ReLU(),
            nn.Linear(400, 784), nn.Sigmoid())

    def forward(self, x):
        z = self.encoder(x)     # 784 -> z numbers
        return self.decoder(z)  # z -> 784 numbers
```

Why a sigmoid at the end? Are we classifying?

No: it keeps every output pixel between 0 and 1, like the data.

> The final sigmoid squashes each of the 784 outputs into (0, 1), the range of our pixels.

▶ Notebook 4.1

The loss: squared error, summed over pixels, averaged over images

$$L(\phi, \theta) = \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{784} \left(x_{ij} - \hat{x}_{ij}\right)^2, \qquad \hat{x}_i = f_\theta\left(g_\phi(x_i)\right)$$

> g_φ is the encoder with weights φ; f_θ the decoder with weights θ. Both are trained together on one loss.

▶ Notebook 4.1

The loss: squared error, summed over pixels, averaged over images

$$L(\phi,\theta)=\frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{784}\left(x_{ij}-\hat{x}_{ij}\right)^2,\qquad \hat{x}_i=f_\theta\left(g_\phi(x_i)\right)$$

Our 3, through a trained 2-number autoencoder: squared error **26.3**.

> For our 3 the trained autoencoder leaves a squared error of 26.3, summed over the 784 pixels.

▶ Notebook 4.1

The loss: squared error, summed over pixels, averaged over images

$$L(\phi, \theta) = \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{784} \left(x_{ij} - \hat{x}_{ij} \right)^2, \qquad \hat{x}_i = f_\theta\left(g_\phi(x_i)\right)$$

Our 3, through a trained 2-number autoencoder: squared error **26.3**.

PCA on the same 3 with the same 2 numbers: **41.3**. Same budget, bent maps, less error.

> PCA, the best straight-line version, leaves 41.3. The bend buys a lot.

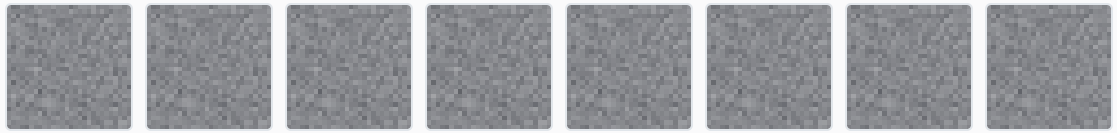
▶ Notebook 4.2

Before training: always look at what the network outputs

8 test digits



untrained output



Every output pixel is between 0.38 and 0.66: flat grey.

> With random weights the sigmoid outputs about 0.5 everywhere: grey squares, whatever the input.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

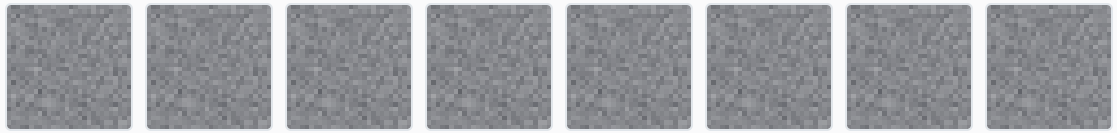
▶ Notebook 4.2

Before training: always look at what the network outputs

8 test digits



untrained output



Every output pixel is between 0.38 and 0.66: flat grey.

Test squared error: **185** per image.

> The starting loss is about 185 per image. This is the number training must bring down; a sanity check before any training.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 4.2

One training step: forward, loss, backward, update


```
model = AutoEncoder(z=2)
opt = torch.optim.Adam(model.parameters(), lr=1e-3)
for epoch in range(30):
    for x in loader:          # 128 images at a time
        x_hat = model(x)
        loss = ((x_hat - x)**2).sum(1).mean()
        opt.zero_grad()
        loss.backward()       # encoder AND decoder
        opt.step()
```

469 steps per pass over the 60,000 digits; 30 passes.

> Batches of 128 digits, the explicit loss from the last slide, Adam.

▶ Notebook 4.2

One training step: forward, loss, backward, update


```
model = AutoEncoder(z=2)
opt = torch.optim.Adam(model.parameters(), lr=1e-3)
for epoch in range(30):
    for x in loader:          # 128 images at a time
        x_hat = model(x)
        loss = ((x_hat - x)**2).sum(1).mean()
        opt.zero_grad()
        loss.backward()       # encoder AND decoder
        opt.step()
```

469 steps per pass over the 60,000 digits; 30 passes.

One loss, one backward pass: the gradient flows from the error back through the decoder, the code, and the encoder.

> loss.backward() sends gradients through the decoder, into the code, and on into the encoder: both are trained together.

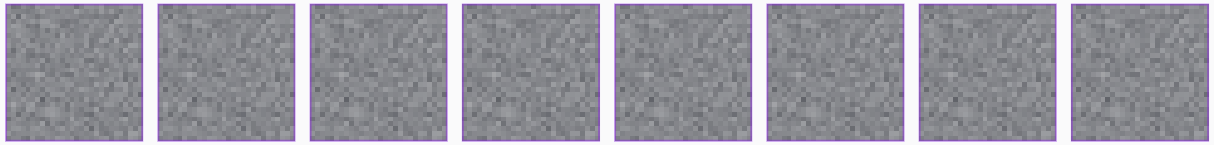
▶ Notebook 4.2

Press play: reconstructions sharpen as it trains

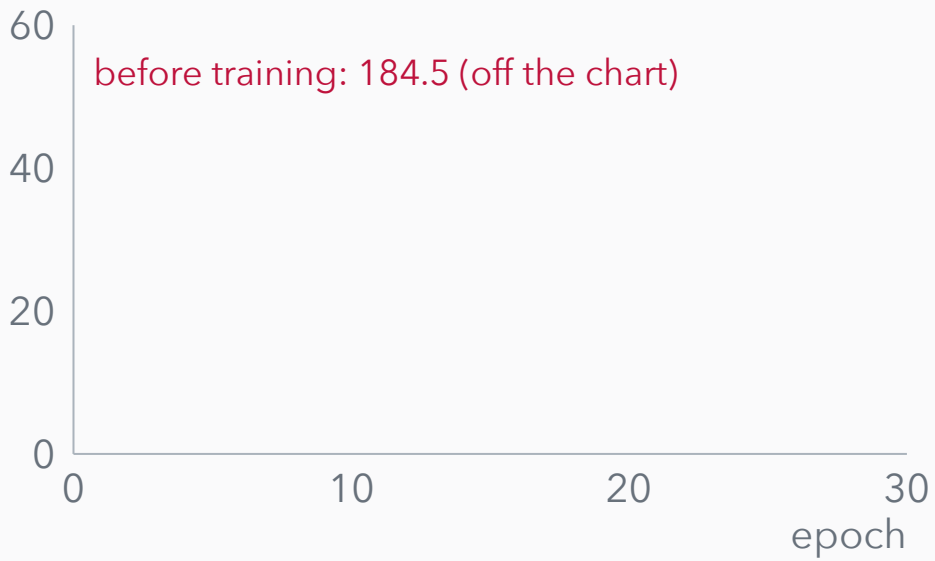
8 test digits



what the untrained network outputs



test squared error per image



Frame 0: grey. Within the first 100 steps, blurry average digits appear.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 4.2

Press play: reconstructions sharpen as it trains

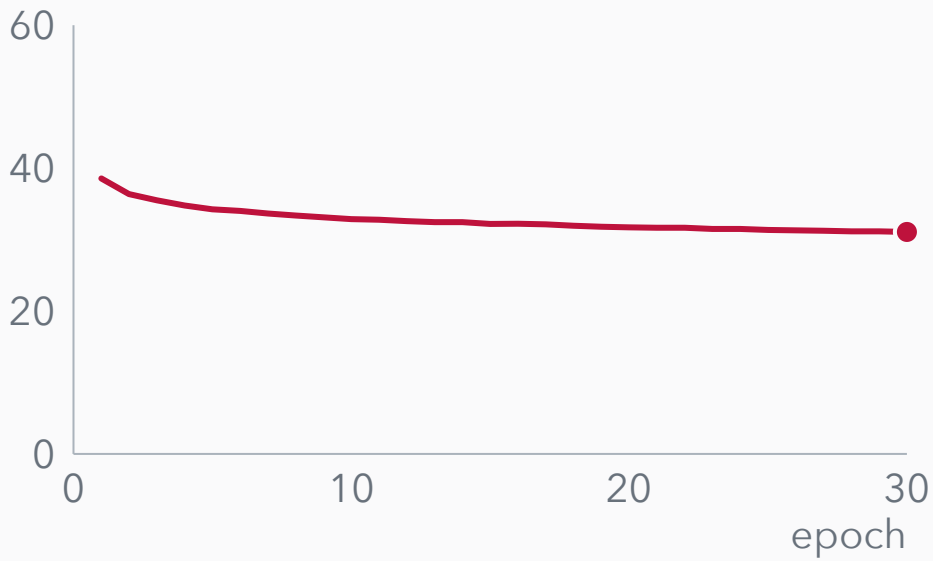
8 test digits



its reconstructions after 14,070 steps (30 epochs)



test squared error per image



By the end, each digit is recognisable from 2 numbers. Training took 25 s on a laptop.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 4.2

Did it memorise? Compare training and test error

2-number code	training error	test error
autoencoder	30.5	31.1
PCA		43.6

Test digits were never used for training.

> Average squared error per image on the training digits, and on the 10,000 test digits.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 4.2

Did it memorise? Compare training and test error

2-number code	training error	test error
autoencoder	30.5	31.1
PCA		43.6

Test digits were never used for training.

Training and test errors are close: it learned digits in general, not these particular 60,000.

> The gap is small (30.5 vs 31.1). And the autoencoder beats PCA by a wide margin on the same 2 numbers.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 4.2

Take out the ReLUs and the sigmoid: a linear autoencoder rediscovers PCA



Test error: linear autoencoder 43.8, PCA 43.6.

> Train the same way, but with straight-line encoder and decoder. The reconstructions match PCA's.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 4.2

Take out the ReLUs and the sigmoid: a linear autoencoder rediscovers PCA



Test error: linear autoencoder 43.8, PCA 43.6.

Its decoder spans almost the same plane as PCA's two directions (cosines of the angles between them: 1.000 and 0.92).

> Gradient descent found PCA's plane on its own. The autoencoder is PCA with bendable maps.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 4.3

Wider codes reconstruct better



Seven autoencoders, codes of 2 to 128 numbers, trained the same way.

The same autoencoder with codes of 2, 4, 8, … 128 numbers. The curves stack in order.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 4.3

Wider codes reconstruct better



Seven autoencoders, codes of 2 to 128 numbers, trained the same way.

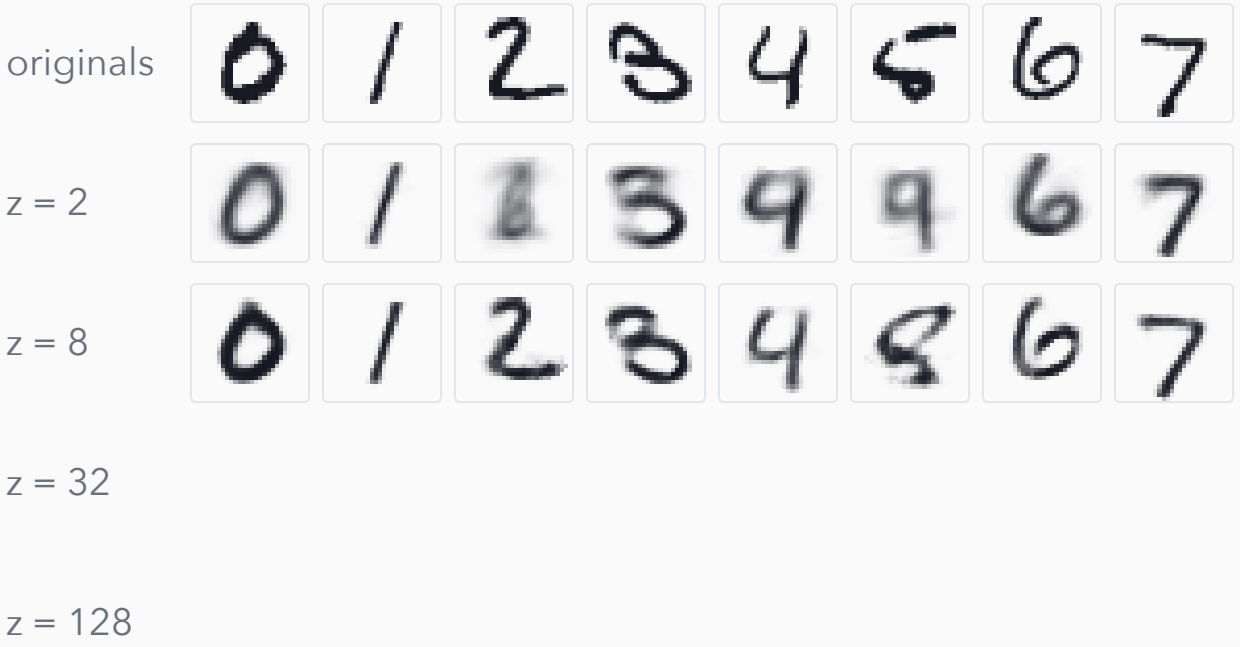
2 numbers: 31.1. 128 numbers: 1.2.

More numbers in the code, less error: with 128 numbers each digit is rebuilt almost exactly.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 4.3

The same 8 test digits through codes of 2, 8, 32 and 128 numbers



> 2 numbers keep the digit's identity and rough shape; 8 numbers add the style.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 4.3

The same 8 test digits through codes of 2, 8, 32 and 128 numbers

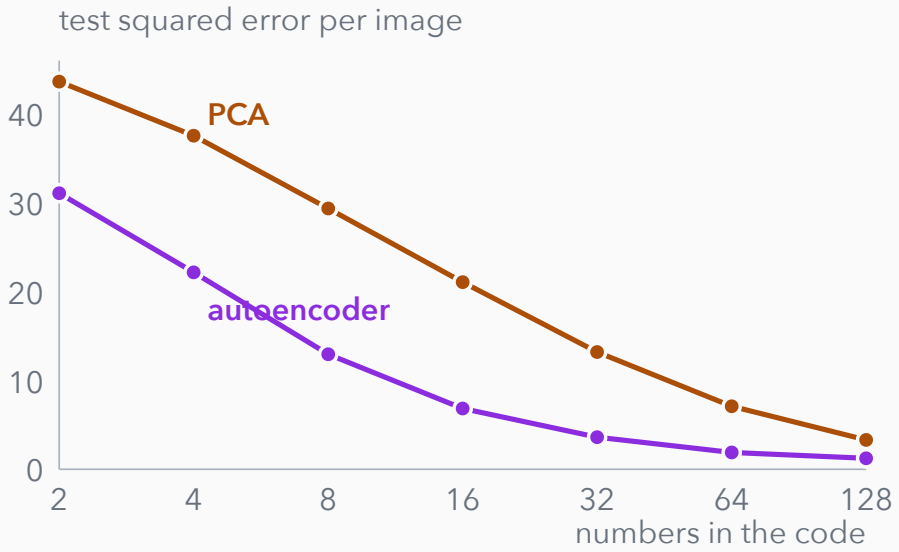


> 32 and 128 numbers reproduce strokes almost exactly.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 4.3

Autoencoder against PCA, at every code size



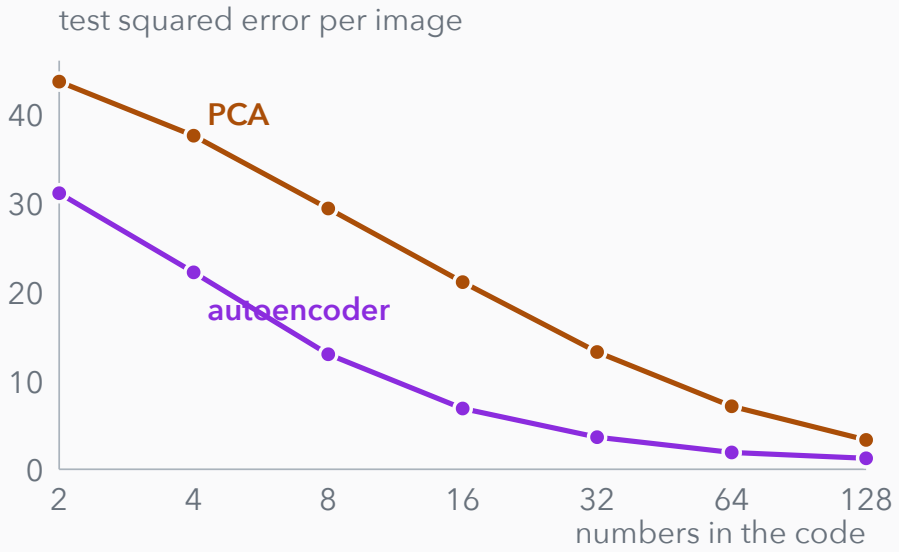
8 numbers: PCA 29.4, autoencoder 12.9.

At every code size the autoencoder rebuilds test digits with less error than PCA.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 4.3

Autoencoder against PCA, at every code size



8 numbers: PCA 29.4, autoencoder 12.9.

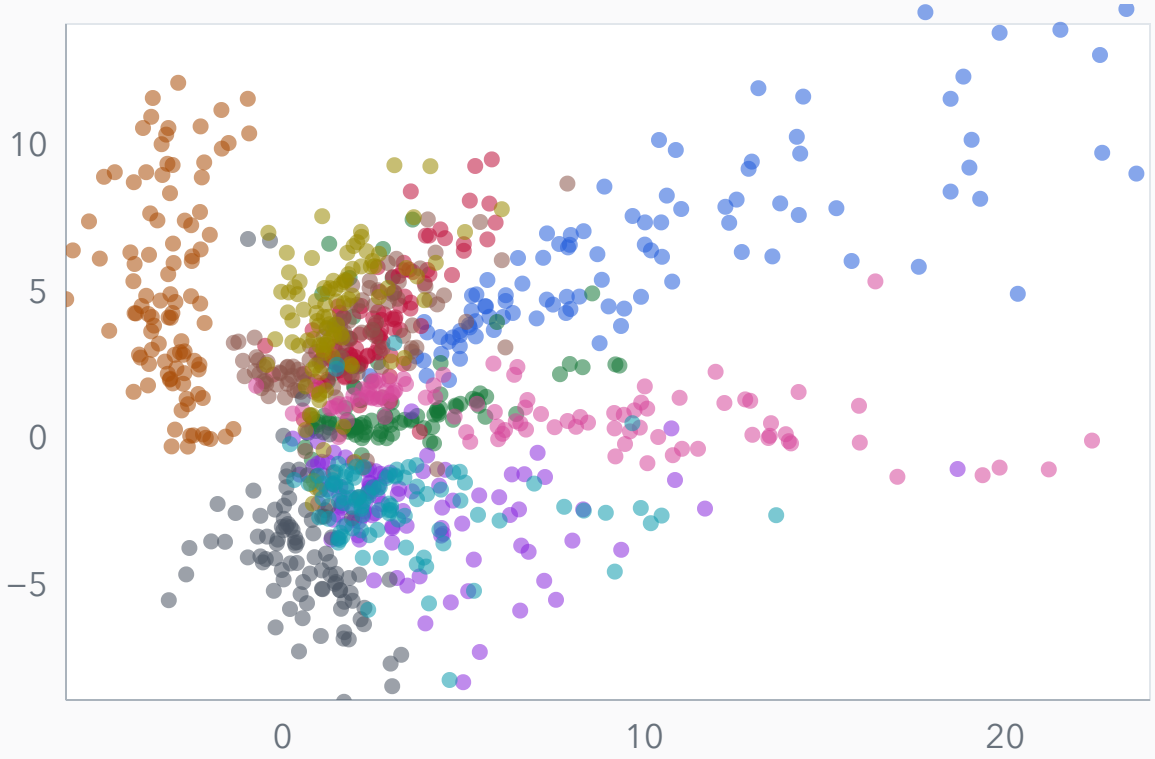
PCA needs about four times as many numbers for the same error.

The biggest gain is for short codes, where bending matters most.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 4.4

The 2-number codes of 1,000 test digits



0 1 2 3 4 5 6 7 8 9

z_1, z_2 from the autoencoder

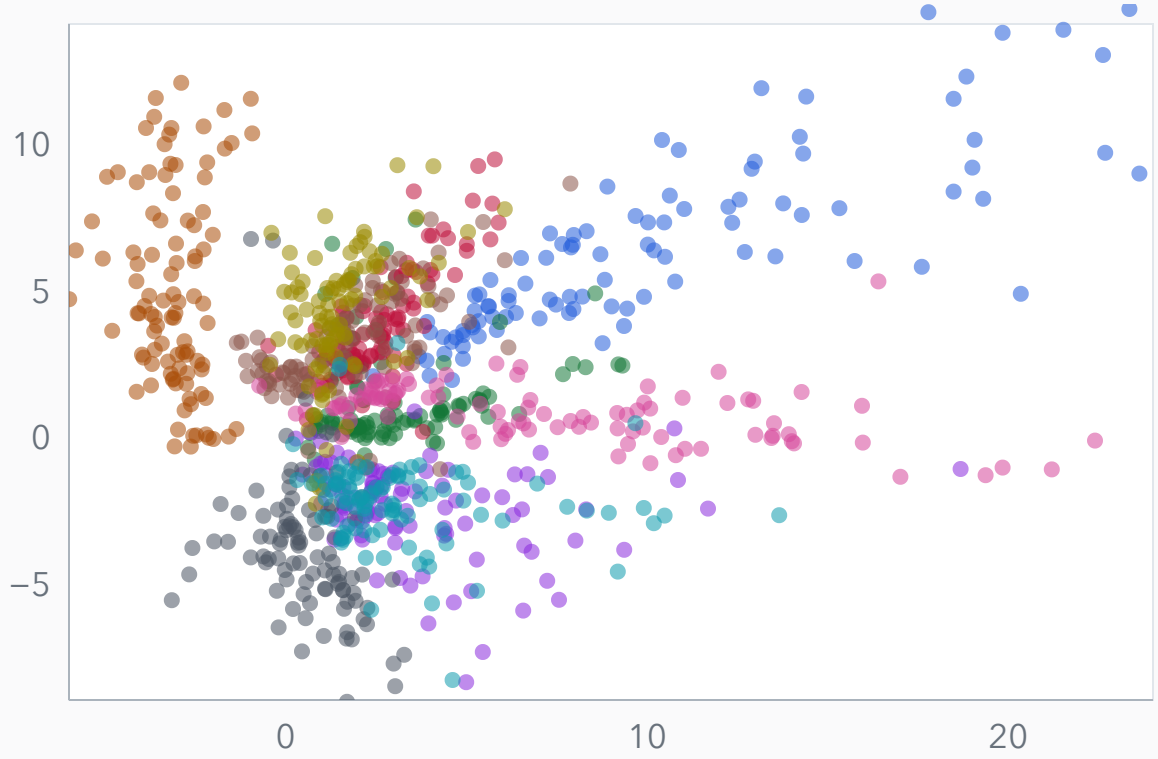
hover any point to see its digit

The same 1,000 test digits, now placed at their autoencoder codes. Hover to see digits.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 4.4

The 2-number codes of 1,000 test digits



0 1 2 3 4 5 6 7 8 9

z_1, z_2 from the autoencoder

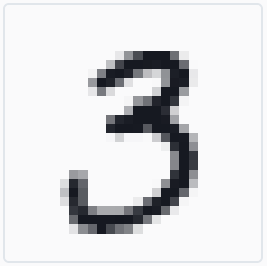
hover any point to see its digit

> Compared with PCA's map, the digits form tighter, better-separated groups. No labels were used in training.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 4.4

Search by code: the nearest digits to a query



query

Encode the query, encode all 60,000 training digits, return the 5 nearest codes.

Codes from a 16-number autoencoder. To search, compare 16 numbers instead of 784 pixels.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 4.4

Search by code: the nearest digits to a query



Encode the query, encode all 60,000 training digits, return the 5 nearest codes.

Music apps and "vector databases" do the same with codes of songs, images and text.

> The nearest codes are other 1s with the same slant: the code keeps what matters about the digit.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 4.4

Denoising: damage the input, ask for the clean image

clean



half the pixels removed



plain autoencoder

denoising autoencoder

> Remove a random half of the pixels of each test digit.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 4.4

Denoising: damage the input, ask for the clean image



Error against the clean digits: plain 34.2, denoising 11.9.

> A plain autoencoder rebuilds the damage faithfully. A denoising autoencoder, trained on damaged inputs, fills in the strokes.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 4.4

Denoising: damage the input, ask for the clean image



Error against the clean digits: plain 34.2, denoising 11.9.

$$L = \frac{1}{n} \sum_i \left\| x_i - f_\theta(g_\phi(\tilde{x}_i)) \right\|^2$$

Input: the damaged $\tilde{x}$. Target: the clean x.

> Its loss compares the output with the clean image, so it must learn what digits look like, not copy its input.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 4.4

Part 4 in one slide

	PCA	autoencoder
encoder, decoder	straight lines	small networks
trained by	an eigenvector formula	gradient descent on squared error
2-number test error	43.6	31.1
codes	overlapping	tighter groups

> An autoencoder is a learned, bent PCA: much better reconstructions from the same few numbers.

▶ Notebook 4.4

Part 4 in one slide

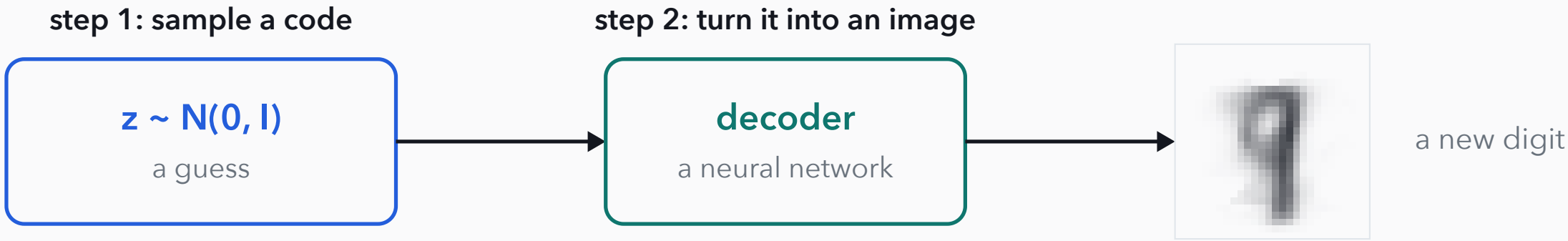
	PCA	autoencoder
encoder, decoder	straight lines	small networks
trained by	an eigenvector formula	gradient descent on squared error
2-number test error	43.6	31.1
codes	overlapping	tighter groups

PCA gave us generation by “sample a code, decode it”. Can the autoencoder’s decoder do the same?

> Part 5: keep only the decoder, feed it random codes, and see what it draws.

▶ Notebook 5.1

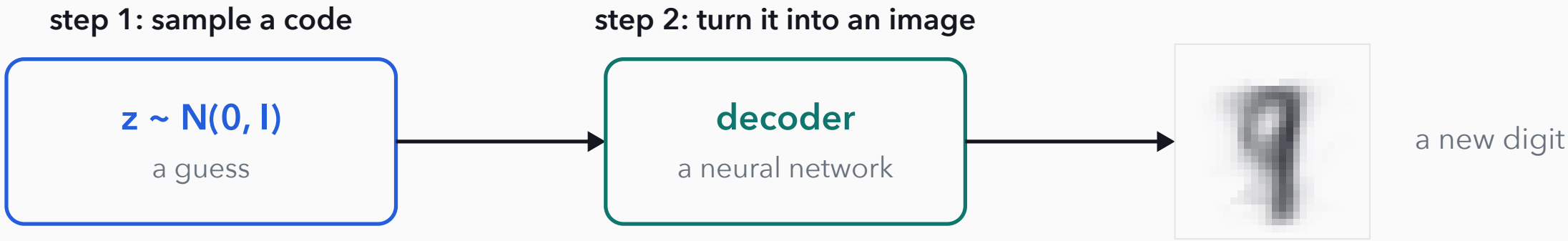
Keep the decoder, throw away the encoder, feed it random numbers



> The decoder turns 2 numbers into a digit. To generate, skip the encoder and invent the 2 numbers.

▶ Notebook 5.1

Keep the decoder, throw away the encoder, feed it random numbers

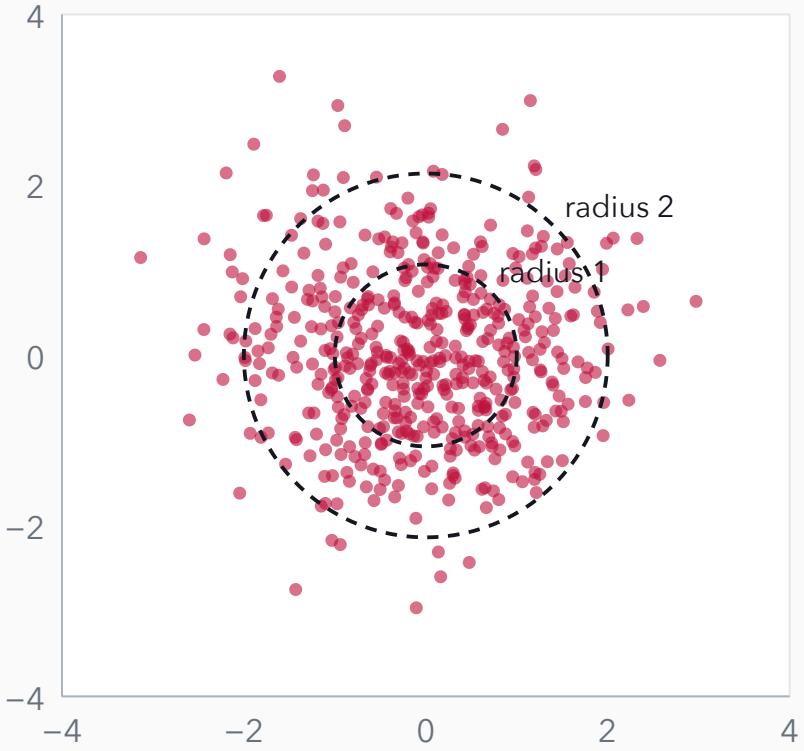


Which random numbers? The simplest choice: z ~ N(0, I).

> We need a distribution to draw codes from. Start with the simplest: the standard normal, N(0, I).

▶ Notebook 5.1

N(0, I) in two dimensions: two independent standard bells



$$z = (z_1, z_2)$$

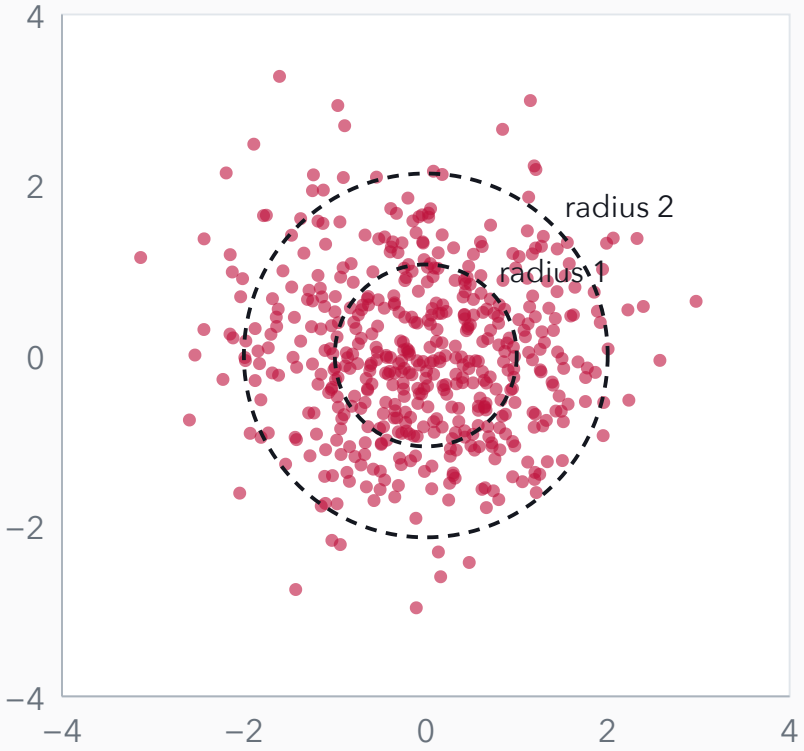
$$z_1 \sim \mathcal{N}(0,1), \quad z_2 \sim \mathcal{N}(0,1), \text{ independent}$$

N(0, I): each of the two numbers is a standard bell, drawn independently. “I” is the identity covariance.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 5.1

N(0, I) in two dimensions: two independent standard bells



$$z = (z_1, z_2)$$

$$z_1 \sim \mathcal{N}(0,1), \quad z_2 \sim \mathcal{N}(0,1), \text{ independent}$$

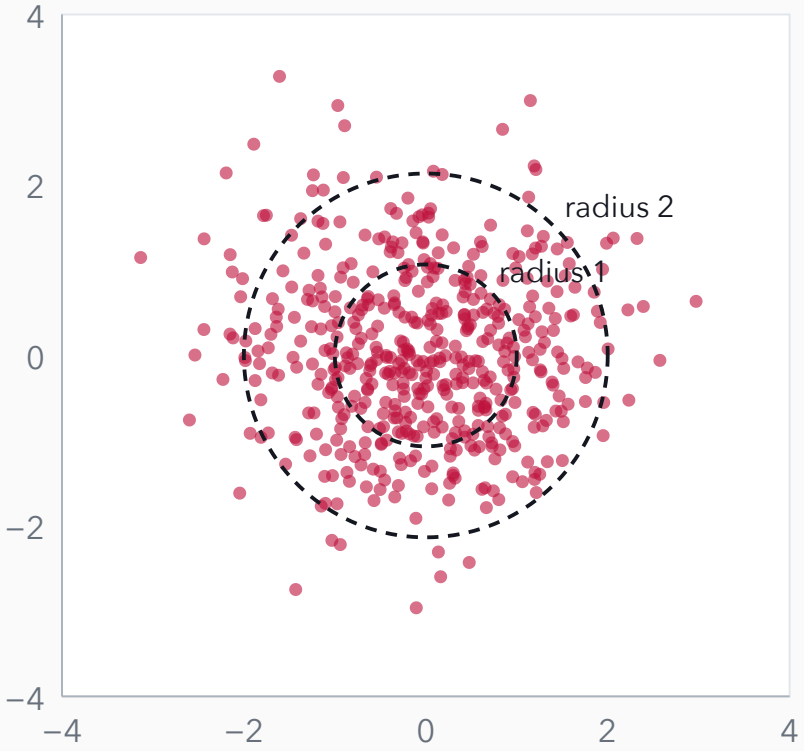
Of these 500 draws, 39% land within distance 1 of the origin and 88% within 2 (in theory 39% and 86%).

> Most draws land close to the origin: 39% within distance 1 and 88% within 2.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 5.1

N(0, I) in two dimensions: two independent standard bells



$$z = (z_1, z_2)$$

$$z_1 \sim \mathcal{N}(0,1), \quad z_2 \sim \mathcal{N}(0,1), \text{ independent}$$

Of these 500 draws, 39% land within distance 1 of the origin and 88% within 2 (in theory 39% and 86%).

In code: `torch.randn(500, 2)`. Draws pile up near (0, 0) and rarely go beyond 3.

That is all torch.randn does. Its draws live in a small disc around (0, 0).

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 5.1

Random codes through three autoencoders

z = 2



z = 8

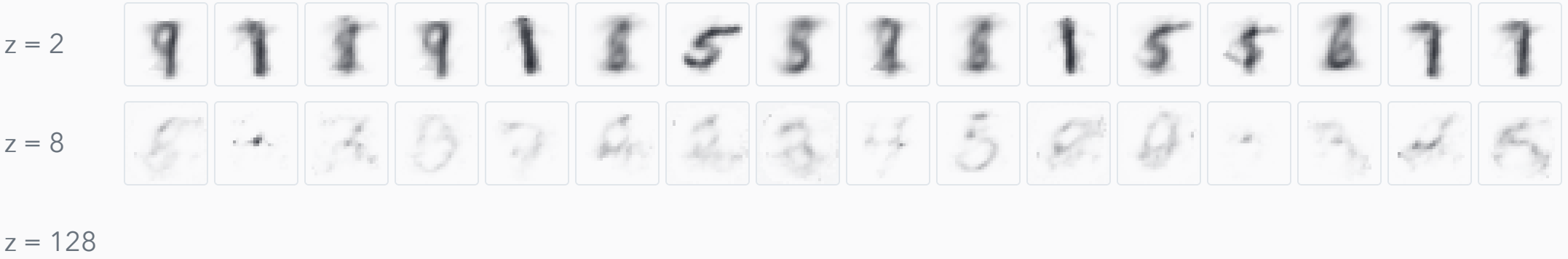
z = 128

2-number autoencoder: some digits, mostly the same few blurry shapes.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 5.1

Random codes through three autoencoders

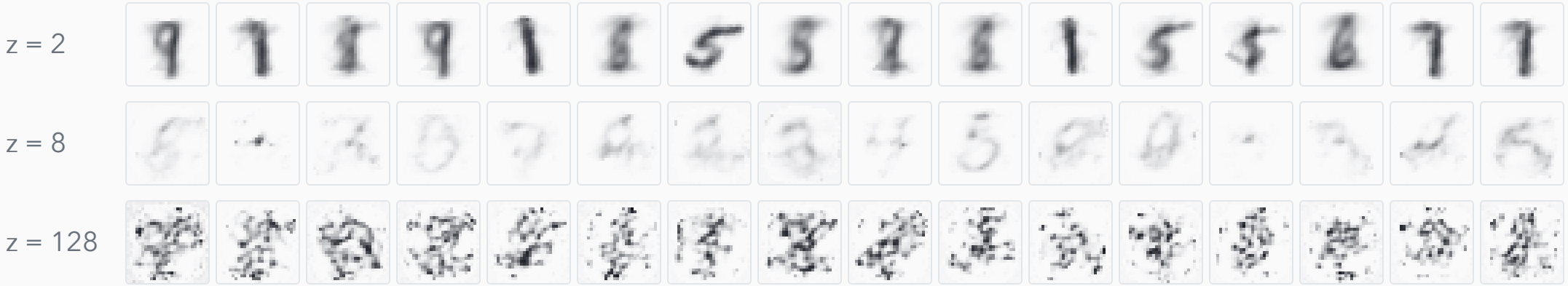


8 numbers: strokes that belong to no digit.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 5.1

Random codes through three autoencoders

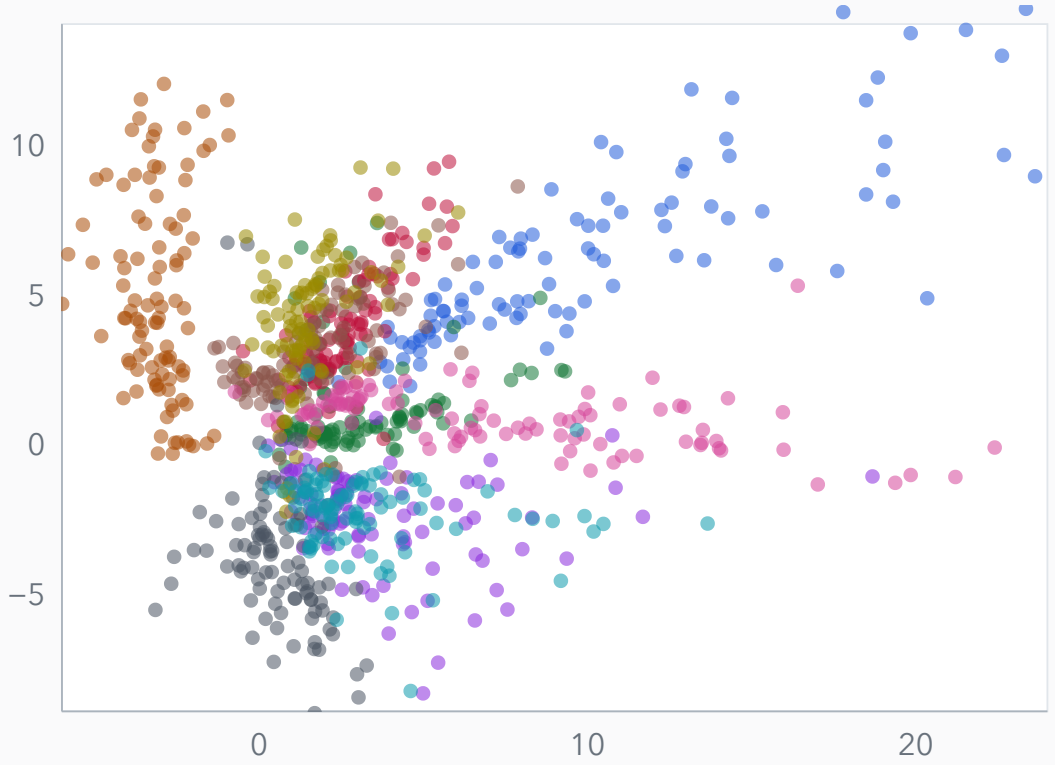


128 numbers, the best reconstructor of all: static. Better reconstruction, worse generation.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 5.2

Put the random codes on top of the real ones



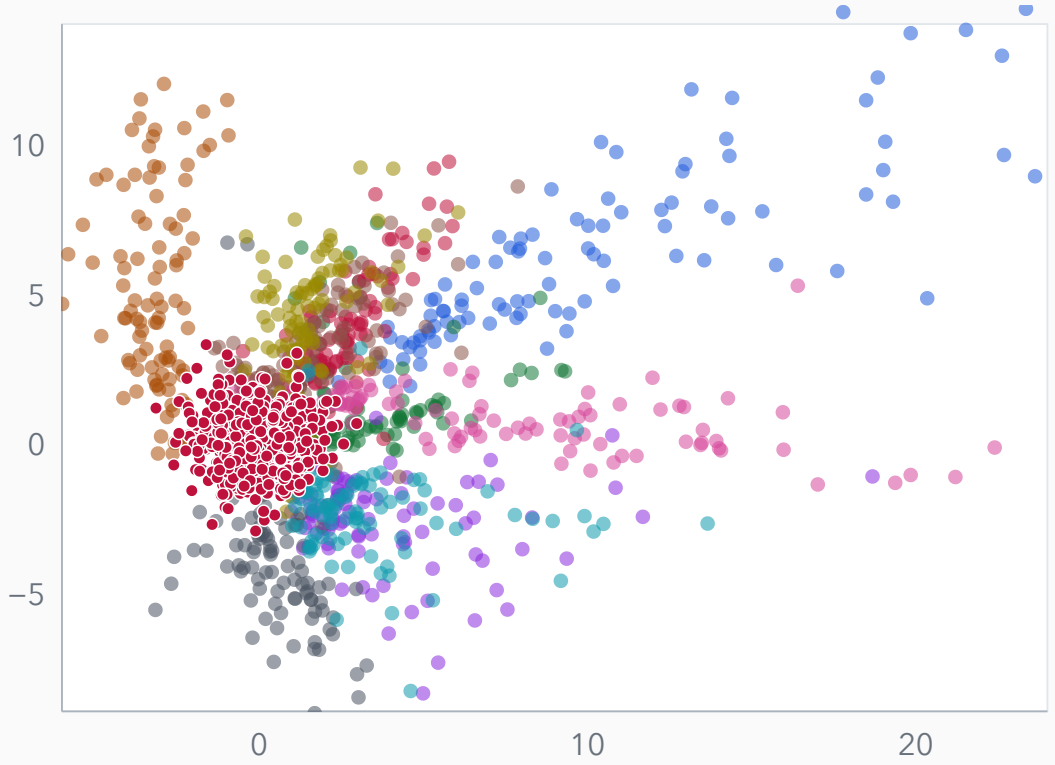
Real codes spread from about −4 to 22 and −7 to 12.

The 2-number codes of 1,000 test digits, coloured by digit.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 5.2

Put the random codes on top of the real ones



Real codes spread from about −4 to 22 and −7 to 12.

The 500 red draws from N(0, I) all land in one small corner.

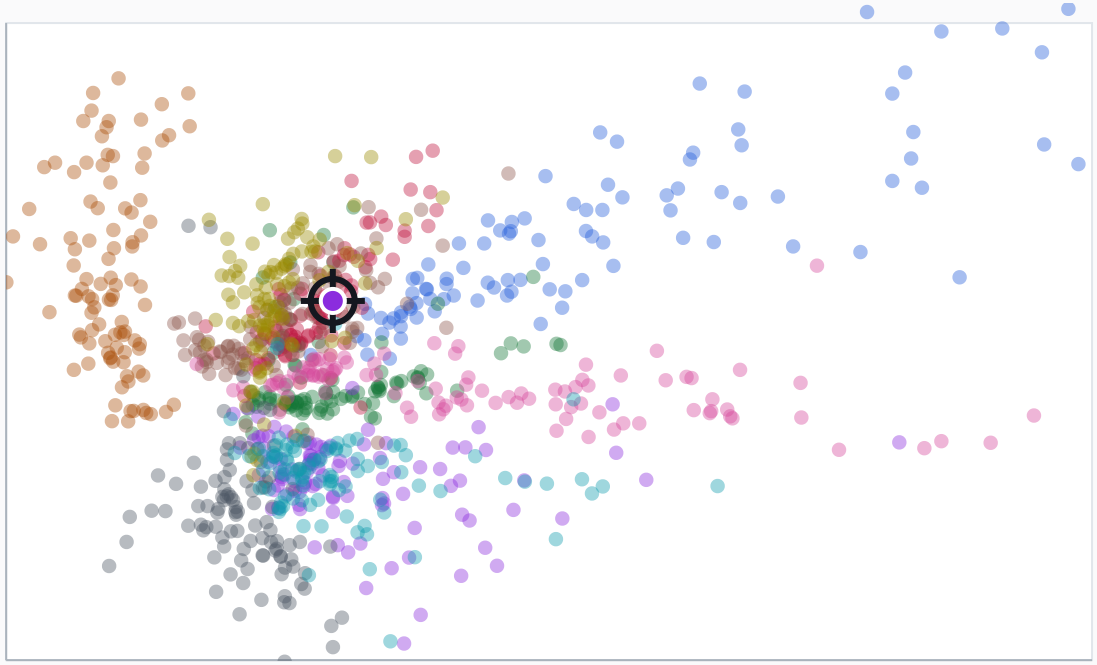
Nothing in the loss asked the codes to sit near N(0, I). Random codes miss most digits.

Add 500 draws from N(0, I) in red. They cover one corner: a few digit types, and nothing of the rest.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 5.2

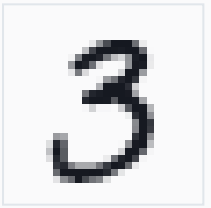
Your turn: three ways to choose a code



0 1 2 3 4 5 6 7 8 9

decoder of the autoencoder, run in your browser





the digit we encoded

z = (3.02, 3.97)

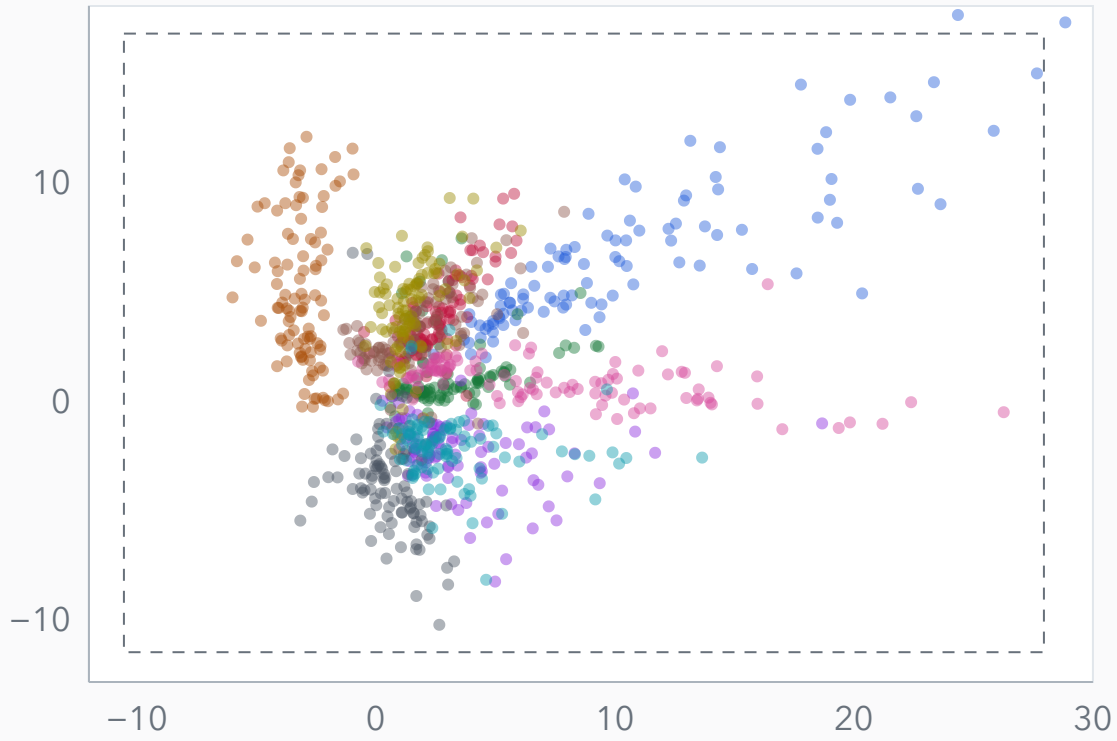
the code of our 3, decoded

(1) Draw a random code from N(0, I). (2) Encode a real test digit and decode its code. (3) Click anywhere on the plane. The decoder runs live in your browser.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 5.2

Decode a whole grid of codes



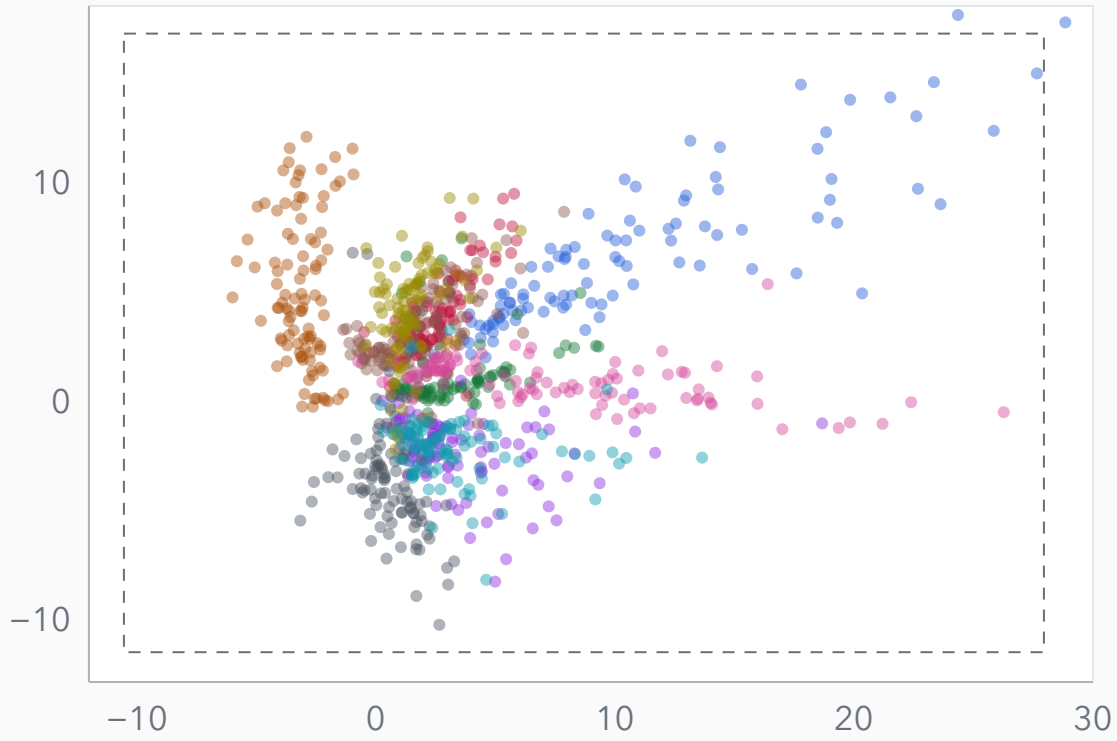


Decode a 12 × 12 grid of codes covering the dashed box: the codes on the left, the decoder's output at each grid point on the right.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 5.2

Decode a whole grid of codes





> Over the real codes the grid shows digits blending into each other. Far from them the decoder only extrapolates: thick 1s and 0s, and broken strokes in the bottom-left corner.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 5.2

Wider codes spread out

code size	two 0s apart	a 0 and a 1 apart	ratio
2	8.0	14.7	0.55
8	11.7	24.6	0.47
32	18.8	24.1	0.78
128	21.2	23.5	0.90

Average distance between codes of two 0s, and between a 0 and a 1.

> As the code grows, codes of the same digit drift apart.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 5.2

Wider codes spread out

code size	two 0s apart	a 0 and a 1 apart	ratio
2	8.0	14.7	0.55
8	11.7	24.6	0.47
32	18.8	24.1	0.78
128	21.2	23.5	0.90

Average distance between codes of two 0s, and between a 0 and a 1.

With 128 numbers two 0s are almost as far apart as a 0 and a 1 (ratio 0.90).

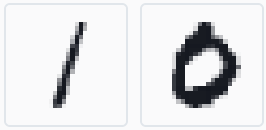
> In the widest codes, similarity is nearly gone: every digit gets its own private spot. Great for reconstruction, useless for sampling.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 5.2

Walk in a straight line from a 1 to a 0

ends



z = 2



z = 16

> Encode a 1 and a 0, take 9 evenly spaced codes on the line between them, decode each. With 2 numbers the path crosses the clusters of 8, 3 and 5 on its way.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 5.2

Walk in a straight line from a 1 to a 0

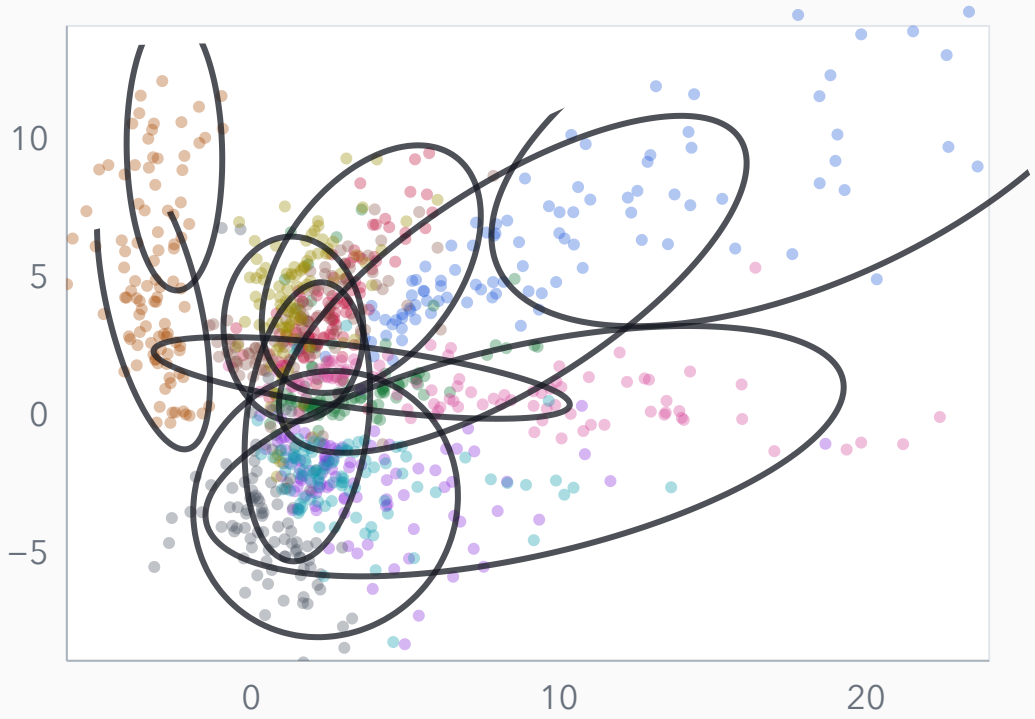


> With 16 numbers the endpoints are sharp, but the middle passes through blobs that are no digit: the straight line leaves the region where codes live.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 5.3

A fix we already know: fit a mixture to where the codes are



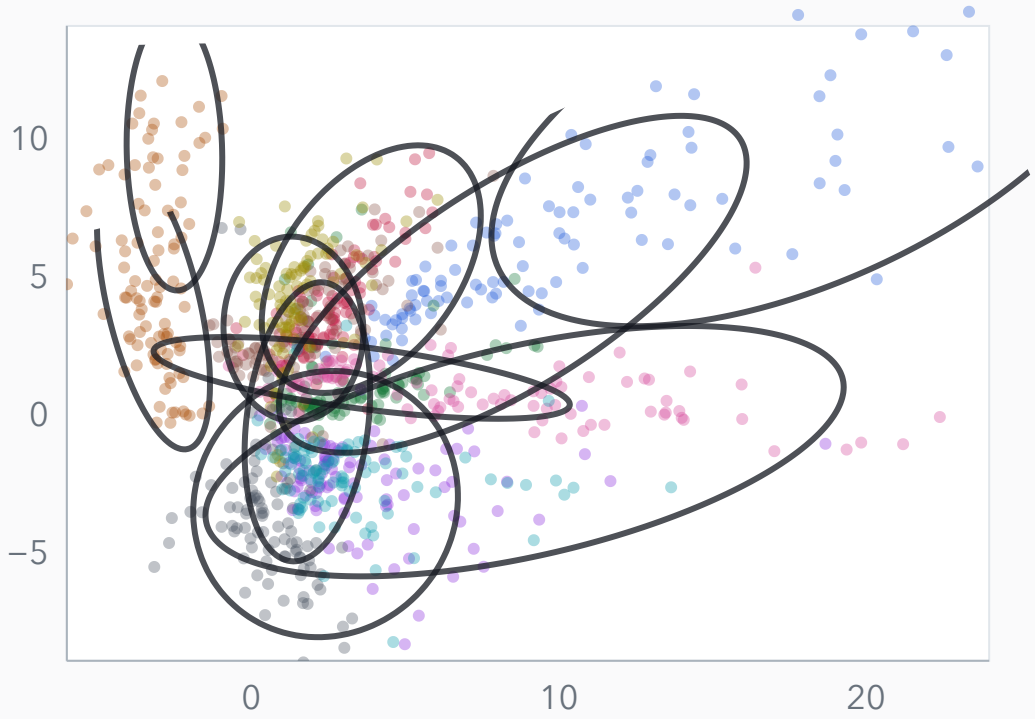
Part 3's recipe: fit 10 Gaussians to the codes of the 60,000 training digits, sample a code, decode.

> Fit a mixture of 10 Gaussians to the training codes: the ellipses follow the clusters.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 5.3

A fix we already know: fit a mixture to where the codes are



Part 3's recipe: fit 10 Gaussians to the codes of the 60,000 training digits, sample a code, decode.

2-number autoencoder + mixture

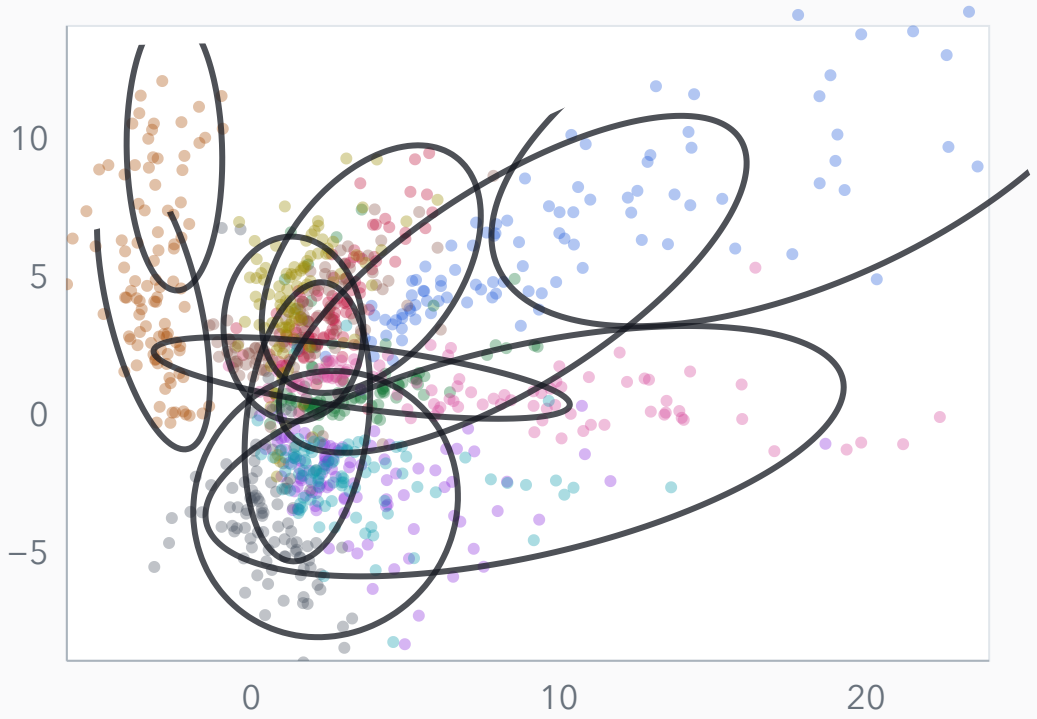


Sample codes from it, decode: real-looking digits from the 2-number autoencoder.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 5.3

A fix we already know: fit a mixture to where the codes are



Part 3's recipe: fit 10 Gaussians to the codes of the 60,000 training digits, sample a code, decode.

2-number autoencoder + mixture



16-number autoencoder + mixture



The same with 16-number codes: sharper digits. A density fitted after training works.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 5.3

What we would like instead

1 · Codes we can sample

Codes that fill a simple, known shape, such as N(0, I), so that torch.randn gives good codes.

> First wish: decide the code distribution in advance, N(0, I), and make the codes fit it.

▶ Notebook 5.3

What we would like instead

1 · Codes we can sample

Codes that fill a simple, known shape, such as N(0, I), so that torch.randn gives good codes.

2 · Smooth codes

Nearby codes decode to similar images, so the space between digits holds plausible digits too.

> Second: each image should own a small region of codes, not a single point, so the space between is filled.

▶ Notebook 5.3

What we would like instead

1 · Codes we can sample

Codes that fill a simple, known shape, such as N(0, I), so that torch.randn gives good codes.

2 · Smooth codes

Nearby codes decode to similar images, so the space between digits holds plausible digits too.

3 · One probability model

One model, trained by maximum likelihood as in Part 2, that can both score and sample.

> Third: write it as one probability model and train by maximum likelihood. The variational autoencoder does all three.

▶ Notebook 6.1

Write the recipe first: pick a code, then draw an image

$$z \sim \mathcal{N}(0, I) \qquad 2 \text{ numbers, decided in advance}$$

$$x \mid z \sim \mathcal{N}\big(f_\theta(z),\ \sigma^2 I\big) \qquad 784 \text{ numbers, centred on the decoder's output}$$

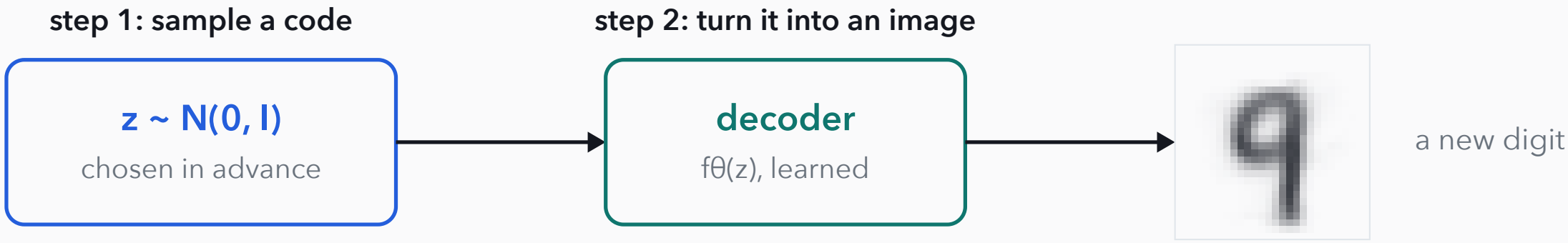
> Step 1: draw a 2-number code from N(0, I). Step 2: the decoder f_θ turns it into a mean image, and each pixel is that mean plus a little Gaussian noise of size σ.

▶ Notebook 6.1

Write the recipe first: pick a code, then draw an image

$$z \sim \mathcal{N}(0, I) \qquad 2 \text{ numbers, decided in advance}$$

$$x \mid z \sim \mathcal{N}\big(f_\theta(z),\ \sigma^2 I\big) \qquad 784 \text{ numbers, centred on the decoder's output}$$



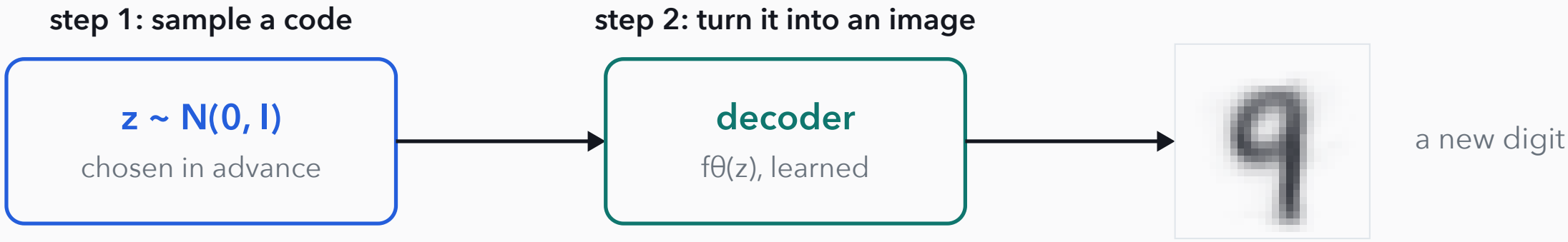
To generate, run the two steps. No encoder needed: the prior N(0, I) is fixed in advance.

▶ Notebook 6.1

Write the recipe first: pick a code, then draw an image

$$z \sim \mathcal{N}(0, I) \qquad 2 \text{ numbers, decided in advance}$$

$$x \mid z \sim \mathcal{N}\big(f_\theta(z),\ \sigma^2 I\big) \qquad 784 \text{ numbers, centred on the decoder's output}$$



> It is Part 2's recipe with a neural network in step 2.

▶ Notebook 6.1

The likelihood of an image, given its code, is a squared error

$$\log p_\theta(x \mid z) = \sum_{j=1}^{784} \log \mathcal{N}\left(x_j;\ f_\theta(z)_j, \sigma^2\right)$$

Each pixel is an independent Gaussian around the decoder's output, so the log-likelihood is a sum over 784 pixels.

▶ Notebook 6.1

The likelihood of an image, given its code, is a squared error

$$\log p_\theta(x \mid z) = \sum_{j=1}^{784} \log \mathcal{N}\left(x_j;\ f_\theta(z)_j, \sigma^2\right)$$

$$= -\frac{1}{2\sigma^2} \left\| x - f_\theta(z) \right\|^2 \ - \ 784 \log\left(\sigma\sqrt{2\pi}\right)$$

Expand the Gaussian: minus the squared error over 2σ², plus a constant. The autoencoder's loss was a likelihood all along.

▶ Notebook 6.1

The likelihood of an image, given its code, is a squared error

$$\log p_\theta(x \mid z) = \sum_{j=1}^{784} \log \mathcal{N}\left(x_j;\ f_\theta(z)_j, \sigma^2\right)$$

$$= -\frac{1}{2\sigma^2} \left\| x - f_\theta(z) \right\|^2 \ - \ 784 \log\left(\sigma\sqrt{2\pi}\right)$$

We use σ = 0.2: about the typical pixel error of a 2-number code, √(31 / 784) ≈ 0.2. The constant term, −784 log(σ√2π), is +541.4.

> σ = 0.2 matches the size of a typical pixel error; it is a modelling choice, and Section 6.7 shows what changing its weight does.

▶ Notebook 6.1

Train like Part 2: maximise log p(x) for every image

$$p_\theta(x) = \int p_\theta(x \mid z)\, p(z)\, dz$$

> The likelihood of an image adds up all the ways it could have been made: every code z, weighted by its prior probability.

▶ Notebook 6.1

Train like Part 2: maximise log p(x) for every image

$$p_\theta(x) = \int p_\theta(x \mid z)\, p(z)\, dz$$

Part 2: a sum over two kinds

$$p(x) = \sum_{k=1}^{2} \pi_k \mathcal{N}(x; \mu_k, \sigma_k^2)$$

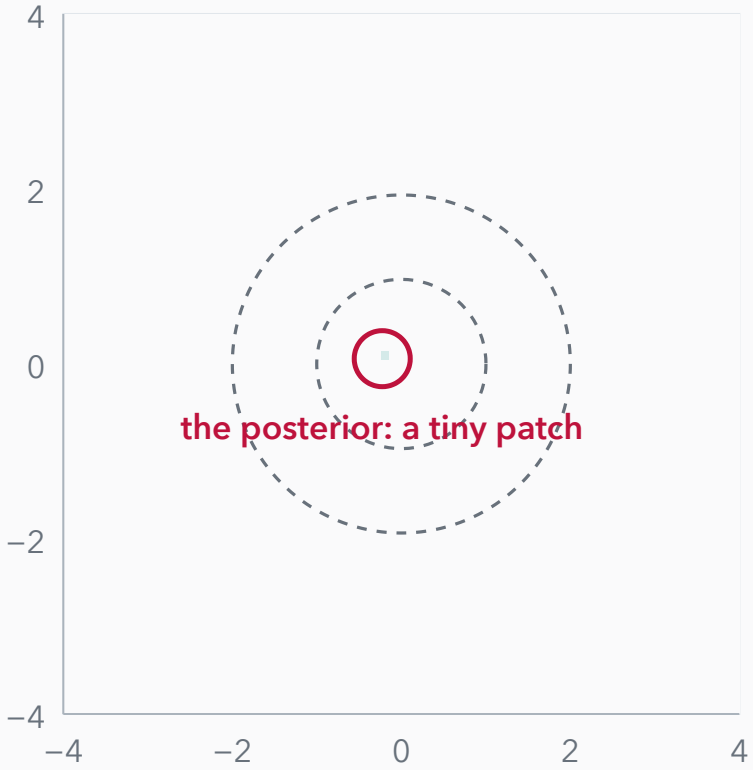
Now: an integral over every possible code

$$p_\theta(x) = \int \mathcal{N}(x; f_\theta(z), \sigma^2 I)\, \mathcal{N}(z; 0, I)\, dz$$

> For the geyser the sum had two terms. Here it is an integral over all codes, through a neural network. No formula.

▶ Notebook 6.2

For our 3: which codes could have made it?

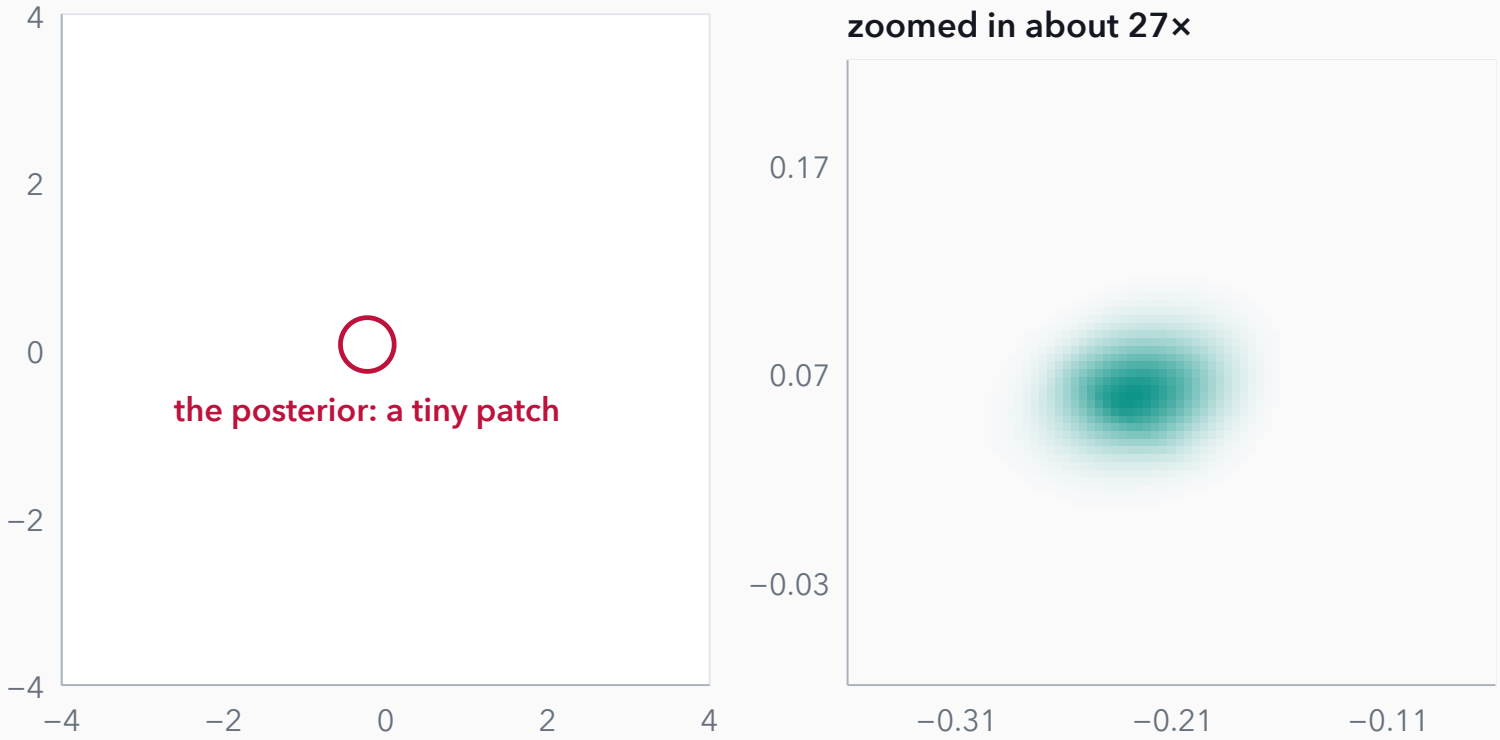


The trained VAE's decoder, our 3, and every code on a fine grid: shade = how well that code explains the 3. Dashed circles: the prior N(0, I).

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 6.2

For our 3: which codes could have made it?

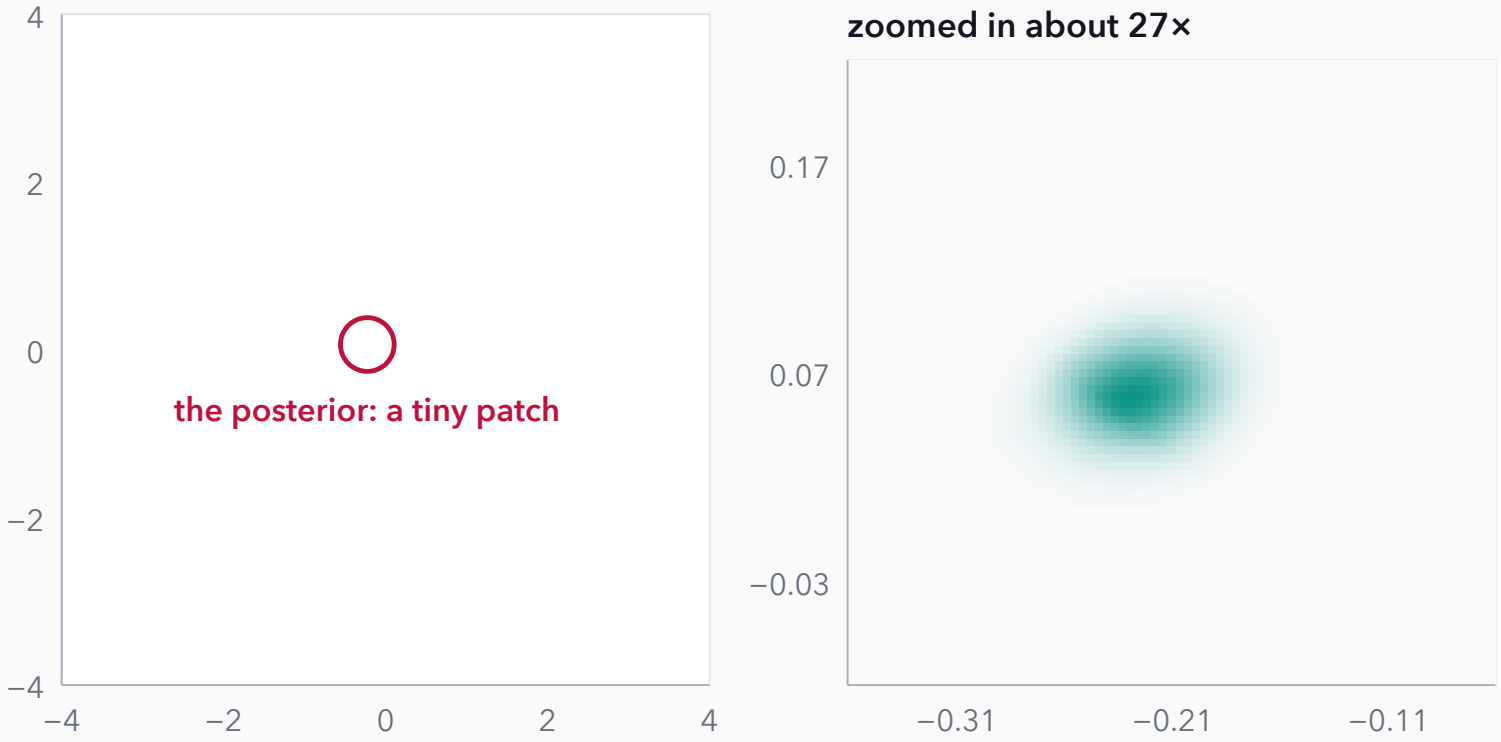


> Zoomed in: the codes that explain this 3 form a tiny patch, about 0.018 wide, near (−0.23, 0.06).

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 6.2

For our 3: which codes could have made it?



$$p(z \mid x) = \frac{p(x \mid z)\, p(z)}{p(x)}$$

> This is the posterior over codes, by Bayes' rule. Its denominator is the p(x) we want.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 6.2

With 2 numbers we can add up rectangles; with 16 we cannot

$$\log p(x) \approx \log \sum_{\text{cells}} p(x \mid z_c)\, p(z_c)\, \Delta z = 244.07$$

Checked by importance sampling with 100,000 draws: 244.07.

> For our 3 the grid gives log p(x) = 244.07. (A density over 784 numbers can exceed 1, so its log can be positive.)

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 6.2

With 2 numbers we can add up rectangles; with 16 we cannot

$$\log p(x) \approx \log \sum_{\text{cells}} p(x \mid z_c)\, p(z_c)\, \Delta z = 244.07$$

Checked by importance sampling with 100,000 draws: 244.07.

A grid with 161 points per axis (from −4 to 4 in steps of 0.05) has $161^2 \approx 26{,}000$ cells for 2 numbers, and $161^{16} \approx 10^{35}$ cells for 16. Lecture 1's intractable evidence, again.

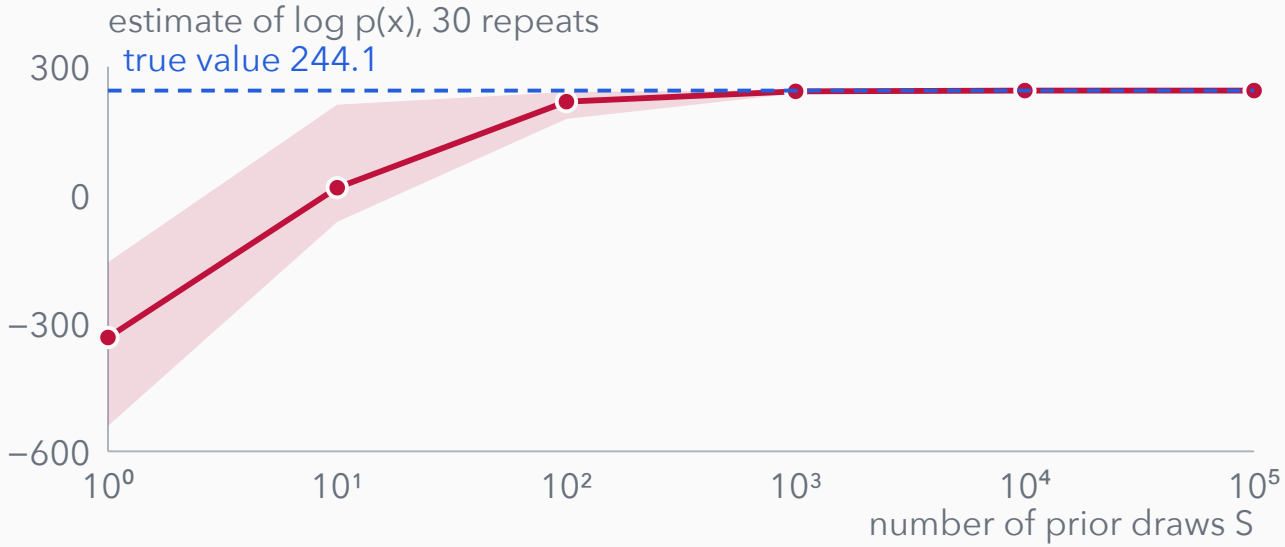
> Grids grow exponentially with the number of code dimensions. We need another way.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 6.2

Average p(x | z) over random codes? Most draws are wasted

$$p(x) = \mathbb{E}_{z\sim p(z)}\big[p(x \mid z)\big] \approx \frac{1}{S}\sum_{s=1}^{S} p(x \mid z_s)$$



With 100 draws the estimate is still off by about 26.

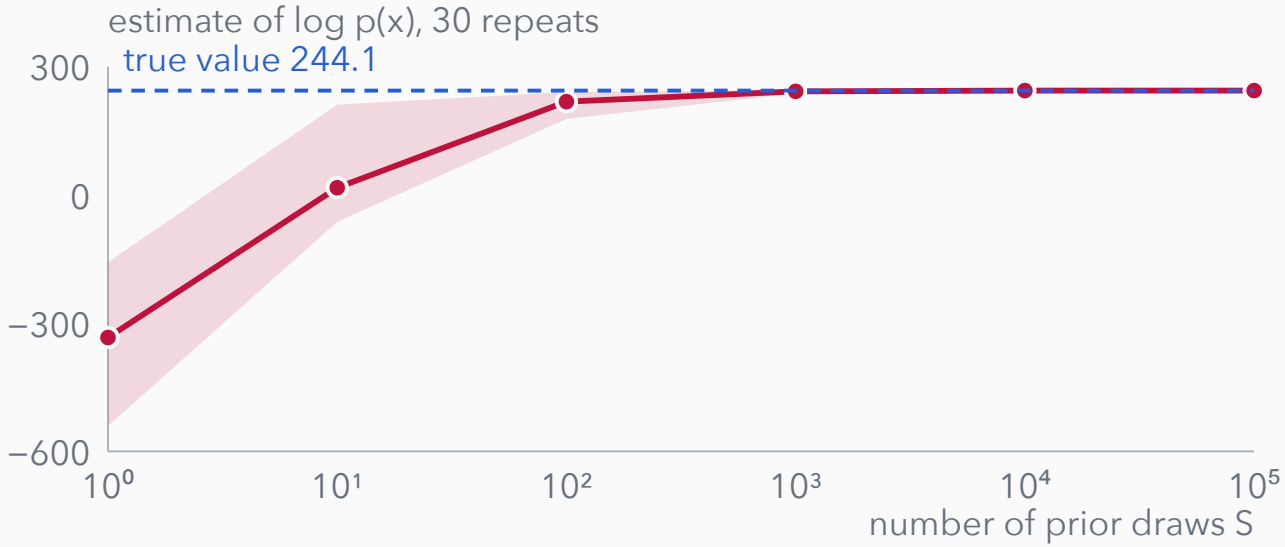
Lecture 1's Monte Carlo: p(x) is an average of p(x | z) over prior codes. It converges, but only after tens of thousands of draws.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 6.2

Average p(x | z) over random codes? Most draws are wasted

$$p(x) = \mathbb{E}_{z\sim p(z)}\big[p(x \mid z)\big] \approx \frac{1}{S}\sum_{s=1}^{S} p(x \mid z_s)$$



With 100 draws the estimate is still off by about 26.

Of 100,000 prior draws, the best 10 carry 38% of the total.

Almost every code from the prior explains this 3 terribly. We need codes from the posterior's tiny patch.

A handful of lucky draws carry the whole sum. We should draw codes where the posterior is, not from the prior.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 6.3

Lecture 1's tool: fit a simple q to the posterior

$$q(z) = \mathcal{N}\left(m,\ \mathrm{diag}(s^2)\right)$$

> Approximate the posterior over codes with a Gaussian q, with its own mean m and widths s.

▶ Notebook 6.3

Lecture 1's tool: fit a simple q to the posterior

$$q(z) = \mathcal{N}\big(m,\ \mathrm{diag}(s^2)\big)$$

$$\log p(x) = \underbrace{\mathbb{E}_q\big[\log p(x \mid z)\big] - \mathrm{KL}\big(q(z) \,\|\, p(z)\big)}_{\mathrm{ELBO}(q)} + \mathrm{KL}\big(q(z) \,\|\, p(z \mid x)\big)$$

Lecture 1's identity: log p(x) = ELBO + KL(q ‖ posterior). Raising the ELBO pushes q towards the posterior, without ever computing p(x).

▶ Notebook 6.3

Lecture 1's tool: fit a simple q to the posterior

$$q(z) = \mathcal{N}\big(m,\ \mathrm{diag}(s^2)\big)$$

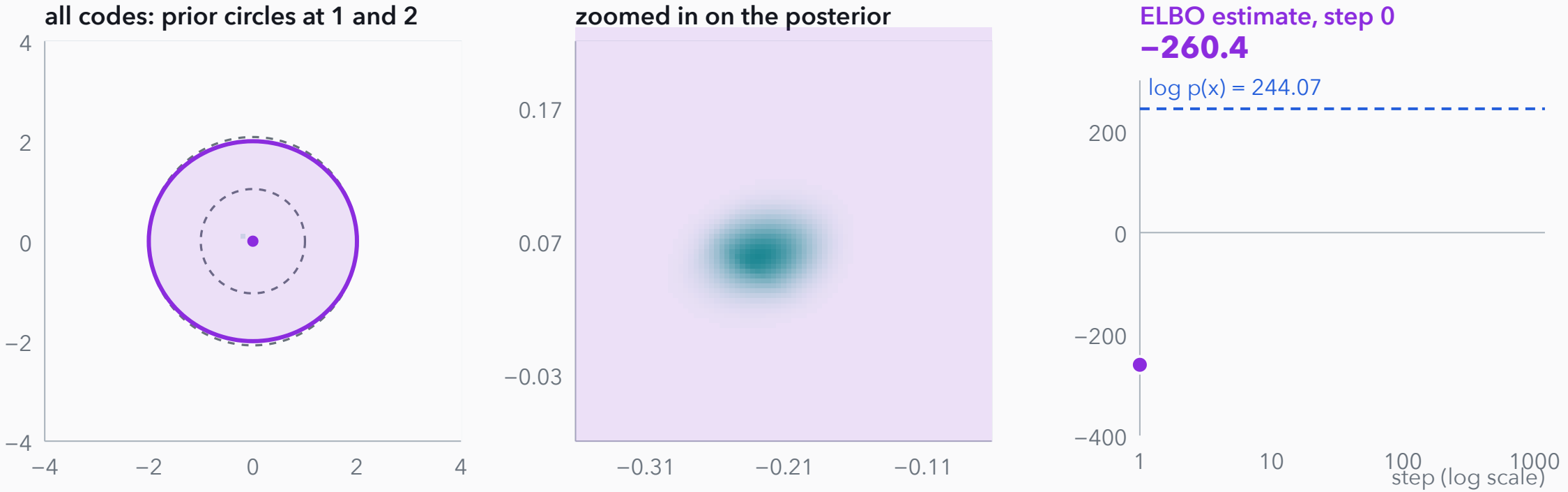
$$\log p(x) = \underbrace{\mathbb{E}_q\big[\log p(x \mid z)\big] - \mathrm{KL}\big(q(z) \,\|\, p(z)\big)}_{\mathrm{ELBO}(q)} + \mathrm{KL}\big(q(z) \,\|\, p(z \mid x)\big)$$

Lecture 1	today
unknown: network weights θ	unknown: the code z of one image
data $\mathcal{D}$: 17 points	data: one image x
prior N(0, 1) on each weight	prior N(0, I) on the code

> Same maths, new roles: the unknown is now the code of one image, not the weights of a network.

▶ Notebook 6.3

Fit q for our 3: press play

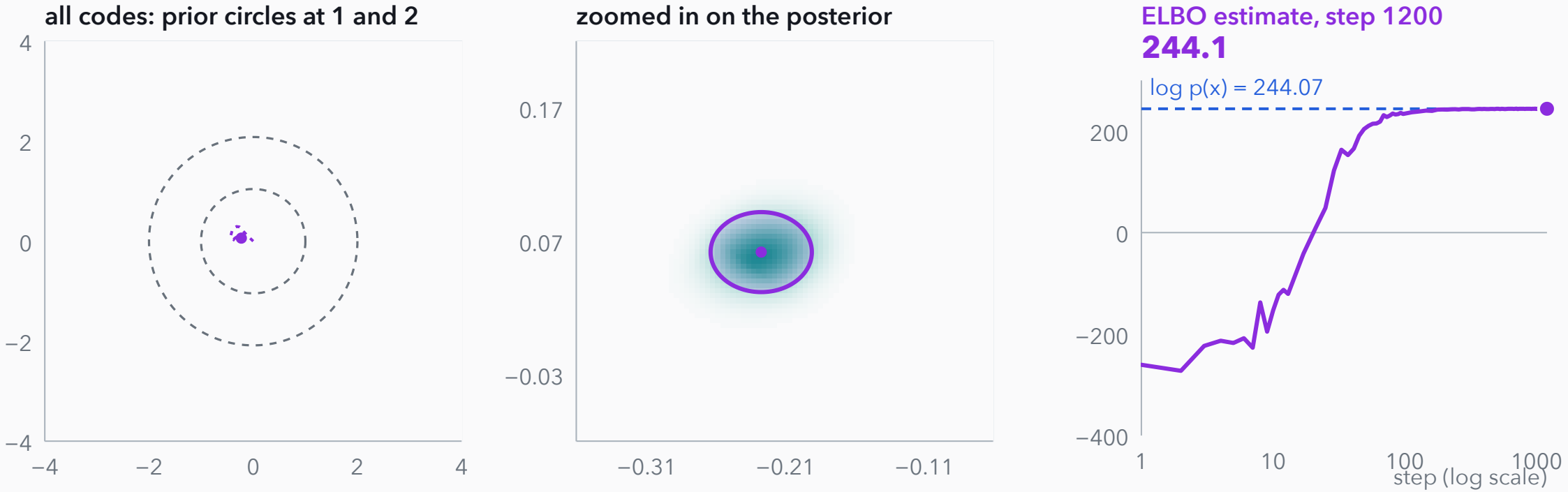


> Start q at the prior N(0, I) (a circle of radius 2 in the left panel) and climb the ELBO with reparameterised draws.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 6.3

Fit q for our 3: press play



> q moves into the posterior's patch and shrinks to fit it. The ELBO ends at 244.05, just below log p(x) = 244.07.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 6.4

60,000 images would mean 60,000 separate fits

Each image has its own posterior, so its own m and s: 1,200 steps each, and again for every new image.

> Per-image VI works, but it is slow: 60,000 fits during training, and a fresh fit for every new image.

▶ Notebook 6.4

60,000 images would mean 60,000 separate fits

Each image has its own posterior, so its own m and s: 1,200 steps each, and again for every new image.

Instead, train a network that reads an image and outputs its m and s in one pass: the **encoder**.

> Learn the fit instead: an encoder network outputs the mean and spread of q for any image.

▶ Notebook 6.4

60,000 images would mean 60,000 separate fits

Each image has its own posterior, so its own m and s: 1,200 steps each, and again for every new image.

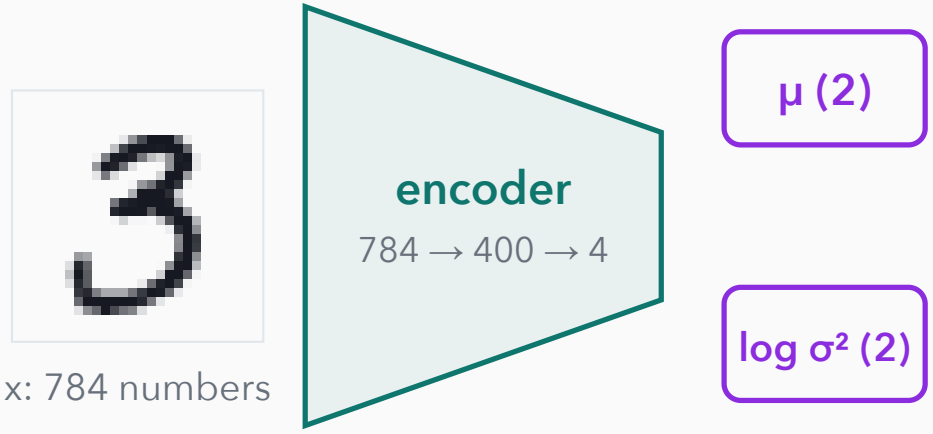
Instead, train a network that reads an image and outputs its m and s in one pass: the **encoder**.

$$q_\phi(z \mid x) = \mathcal{N}\big(\mu_\phi(x),\ \operatorname{diag}\sigma_\phi(x)^2\big)$$

> One network with weights φ, shared by all images. This is called amortised inference: pay once in training, then every image is one forward pass.

▶ Notebook 6.4

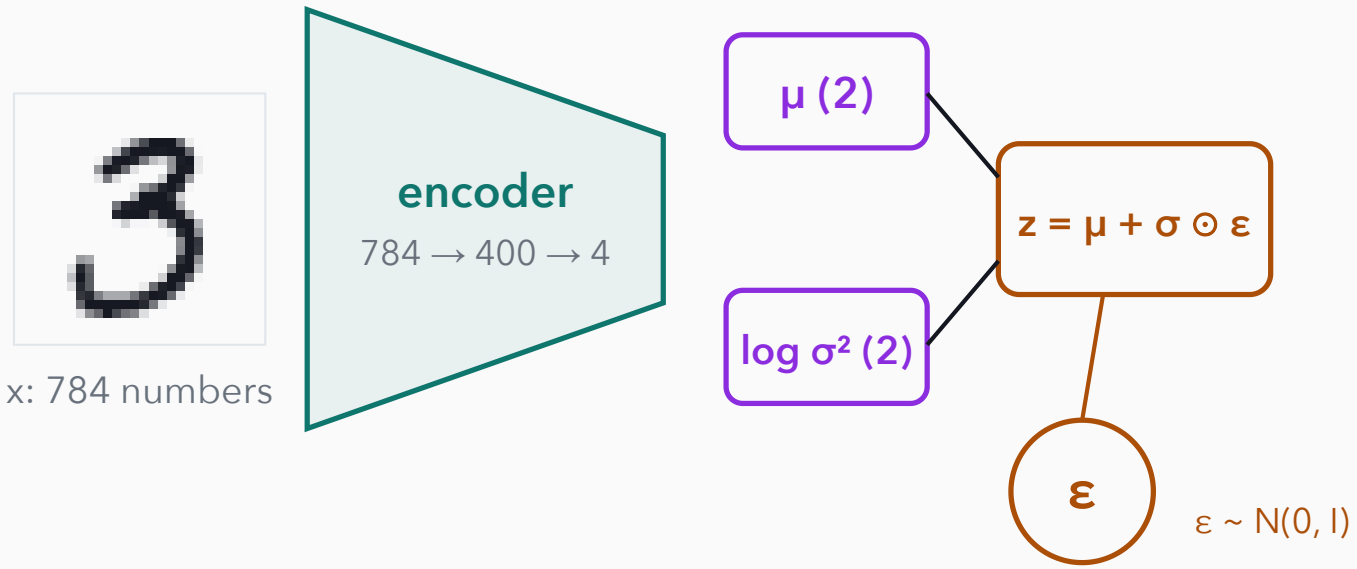
The VAE: an encoder that outputs q, a code drawn from it, and the decoder



> The encoder now ends in 4 numbers, not 2. A bug? No: a mean and a log variance for each code number.

▶ Notebook 6.4

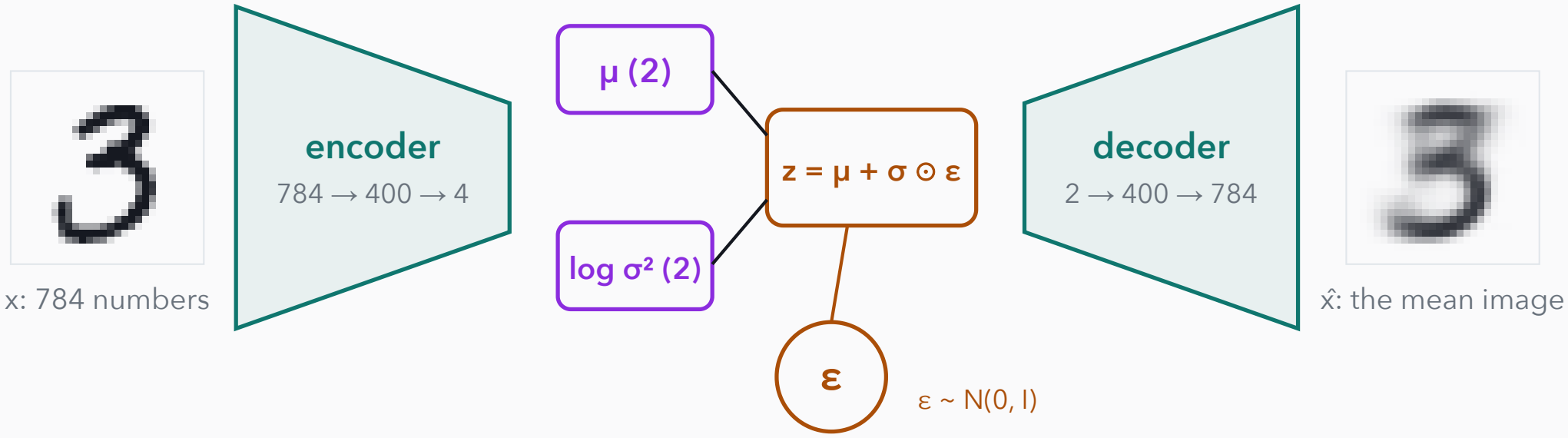
The VAE: an encoder that outputs q, a code drawn from it, and the decoder



> Draw the code by reparameterisation (Lecture 1, section 3.5): $z = \mu + \sigma \odot \varepsilon$ with $\varepsilon \sim N(0, I)$. Sampling z directly would block the gradient; written this way, backpropagation reaches μ and σ.

▶ Notebook 6.4

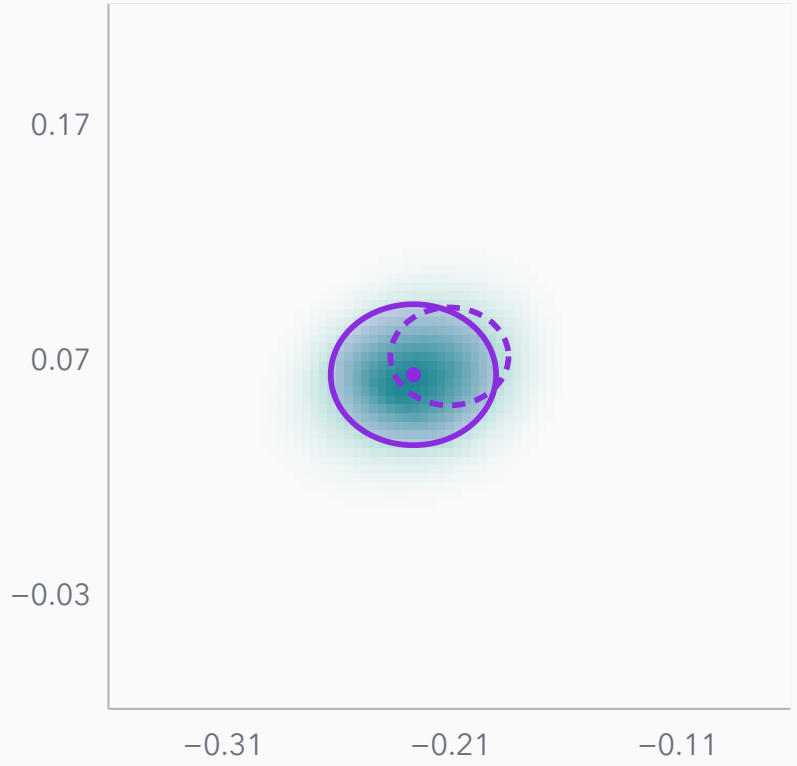
The VAE: an encoder that outputs q, a code drawn from it, and the decoder



> The decoder is unchanged: from z to a mean image.

▶ Notebook 6.4

The encoder on our 3: one pass, almost as good as 1,200 steps



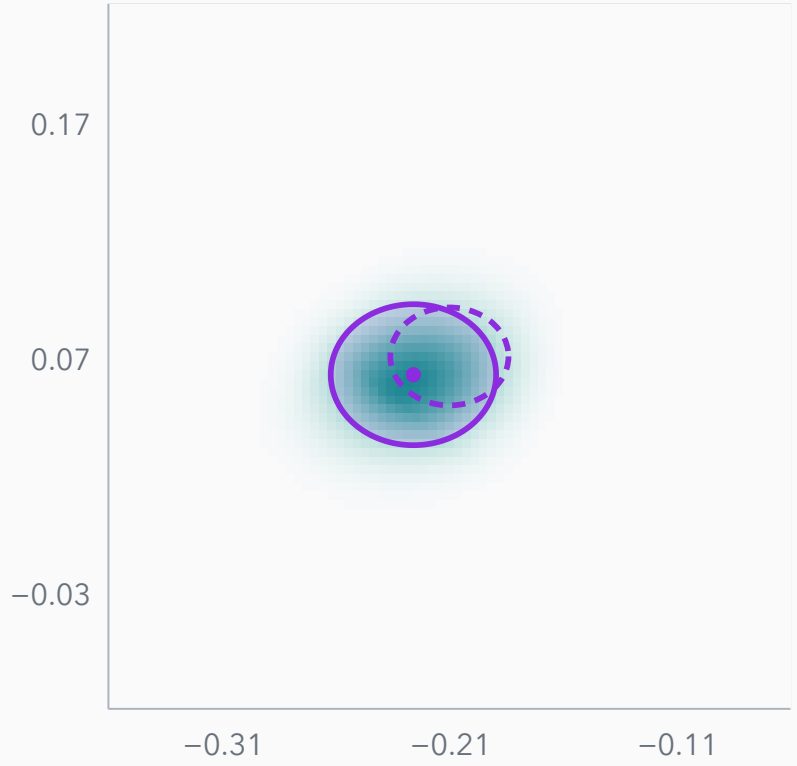
	ELBO
log p(x), the exact target	244.07
q fitted for this image (1,200 steps)	244.05
q from the encoder (one pass)	243.37

> The per-image fit gets within 0.02 of log p(x); the encoder, in one pass, within 0.70.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 6.4

The encoder on our 3: one pass, almost as good as 1,200 steps



	ELBO
log p(x), the exact target	244.07
q fitted for this image (1,200 steps)	244.05
q from the encoder (one pass)	243.37

Encoder: $\mu = (-0.211, 0.070)$, $\sigma = (0.013, 0.010)$.

Dashed: the encoder's q. Shaded: the per-image fit. Both 2 standard deviations.

> The encoder's q sits slightly off-centre and a little too narrow. That small loss is the price of amortisation.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 6.5

The loss for one image is the negative ELBO

$$L(\phi, \theta; x) = \underbrace{\mathbb{E}_{q_\phi(z|x)}\left[\frac{1}{2\sigma^2}\|x - f_\theta(z)\|^2\right]}_{\text{reconstruction}} + \underbrace{\mathrm{KL}\big(q_\phi(z \mid x) \,\|\, \mathcal{N}(0, I)\big)}_{\text{stay near the prior}}$$

Minus the ELBO, up to the constant 541.4 from section 6.1: an expected scaled squared error, plus a KL from q to the prior.

▶ Notebook 6.5

The loss for one image is the negative ELBO

$$L(\phi, \theta; x) = \underbrace{\mathbb{E}_{q_\phi(z|x)}\left[\frac{1}{2\sigma^2}\|x - f_\theta(z)\|^2\right]}_{\text{reconstruction}} + \underbrace{\mathrm{KL}\big(q_\phi(z \mid x) \,\|\, \mathcal{N}(0, I)\big)}_{\text{stay near the prior}}$$

$$\text{reconstruction} \approx \frac{1}{2\sigma^2}\|x - f_\theta(\mu + \sigma \odot \epsilon)\|^2, \qquad \epsilon \sim \mathcal{N}(0, I) \text{ (one draw)}$$

The expectation is estimated with one reparameterised draw per image per step.

▶ Notebook 6.5

The loss for one image is the negative ELBO

$$L(\phi, \theta; x) = \underbrace{\mathbb{E}_{q_\phi(z|x)}\left[\frac{1}{2\sigma^2}\|x - f_\theta(z)\|^2\right]}_{\text{reconstruction}} + \underbrace{\mathrm{KL}\big(q_\phi(z \mid x) \,\|\, \mathcal{N}(0, I)\big)}_{\text{stay near the prior}}$$

$$\text{reconstruction} \approx \frac{1}{2\sigma^2}\|x - f_\theta(\mu + \sigma \odot \epsilon)\|^2, \qquad \epsilon \sim \mathcal{N}(0, I) \text{ (one draw)}$$

$$\mathrm{KL} = \frac{1}{2}\sum_{k=1}^{2}\left(\mu_k^2 + \sigma_k^2 - \log \sigma_k^2 - 1\right) \qquad \text{(Lecture 1, section 2.7)}$$

> The KL between two diagonal Gaussians has a closed form: no sampling needed.

▶ Notebook 6.5

One step for our 3, with numbers

$$\mu = (-0.211,\ 0.070), \quad \sigma = (0.013,\ 0.010), \quad \epsilon = (0.5,\ -1.0)$$

$$z = \mu + \sigma \odot \epsilon = (-0.204,\ 0.060)$$

The encoder gives μ and σ; draw ε; the code is z = μ + σ ⊙ ε.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 6.5

One step for our 3, with numbers

$$\mu = (-0.211,\ 0.070), \quad \sigma = (0.013,\ 0.010), \quad \epsilon = (0.5,\ -1.0)$$

$$z = \mu + \sigma \odot \epsilon = (-0.204,\ 0.060)$$



vs



$$\frac{1}{2(0.2)^2} \times 23.18 = 289.8$$

Decode z and compare with our 3: squared error 23.2, scaled by 1/(2σ²) = 12.5.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 6.5

One step for our 3, with numbers

$$\mu = (-0.211,\ 0.070), \quad \sigma = (0.013,\ 0.010), \quad \epsilon = (0.5,\ -1.0)$$

$$z = \mu + \sigma \odot \epsilon = (-0.204,\ 0.060)$$



vs



$$\frac{1}{2(0.2)^2} \times 23.18 = 289.8$$

$$\mathrm{KL} = 7.94 \qquad L = 289.8 + 7.94 = 297.7$$

Check: L − 541.4 = −243.7, close to −ELBO = −243.4. The loss drops the constant, and one draw is noisy.

> Add the KL to the prior: the loss for this image on this step.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 6.5

The VAE in PyTorch: three changes to the autoencoder


```python
class VAE(AutoEncoder):           # same decoder
    def __init__(self, z=2):
        super().__init__(z)
        self.encoder[-1] = nn.Linear(400, 2 * z)  # (1)

    def forward(self, x):
        mu, logvar = self.encoder(x).chunk(2, dim=1)
        eps = torch.randn_like(mu)
        z = mu + torch.exp(0.5 * logvar) * eps       # (2)
        return self.decoder(z), mu, logvar

x_hat, mu, logvar = model(x)
recon = ((x_hat - x)**2).sum(1) / (2 * 0.2**2)
kl = 0.5 * (mu**2 + logvar.exp() - logvar - 1).sum(1)  # (3)
loss = (recon + kl).mean()
```

> Same encoder and decoder as Part 4, with three changes.

▶ Notebook 6.5

The VAE in PyTorch: three changes to the autoencoder


```python
class VAE(AutoEncoder):           # same decoder
    def __init__(self, z=2):
        super().__init__(z)
        self.encoder[-1] = nn.Linear(400, 2 * z)  # (1)

    def forward(self, x):
        mu, logvar = self.encoder(x).chunk(2, dim=1)
        eps = torch.randn_like(mu)
        z = mu + torch.exp(0.5 * logvar) * eps       # (2)
        return self.decoder(z), mu, logvar

x_hat, mu, logvar = model(x)
recon = ((x_hat - x)**2).sum(1) / (2 * 0.2**2)
kl = 0.5 * (mu**2 + logvar.exp() - logvar - 1).sum(1)  # (3)
loss = (recon + kl).mean()
```

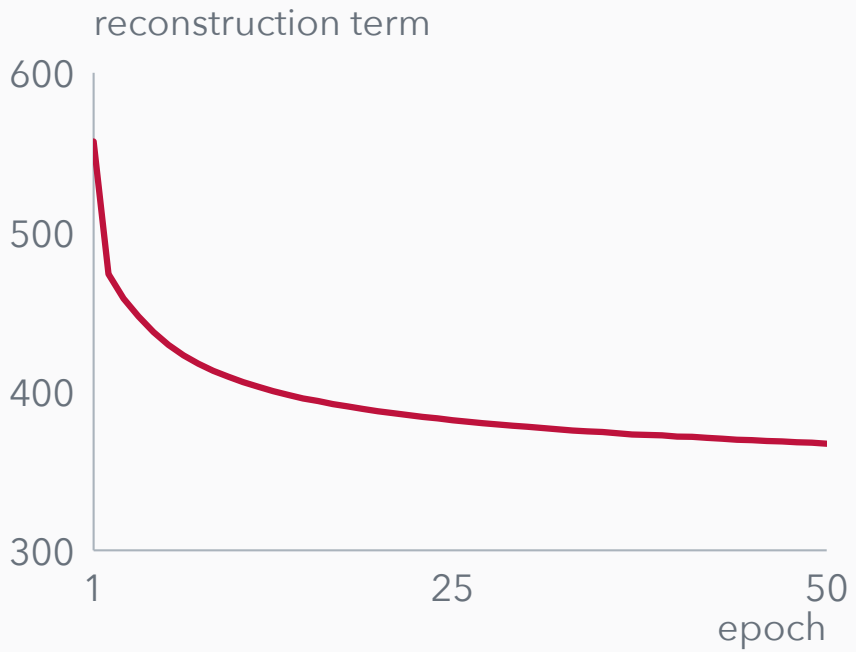
(1) the encoder outputs 2z numbers, (2) the code is a reparameterised draw, (3) the loss adds the KL.

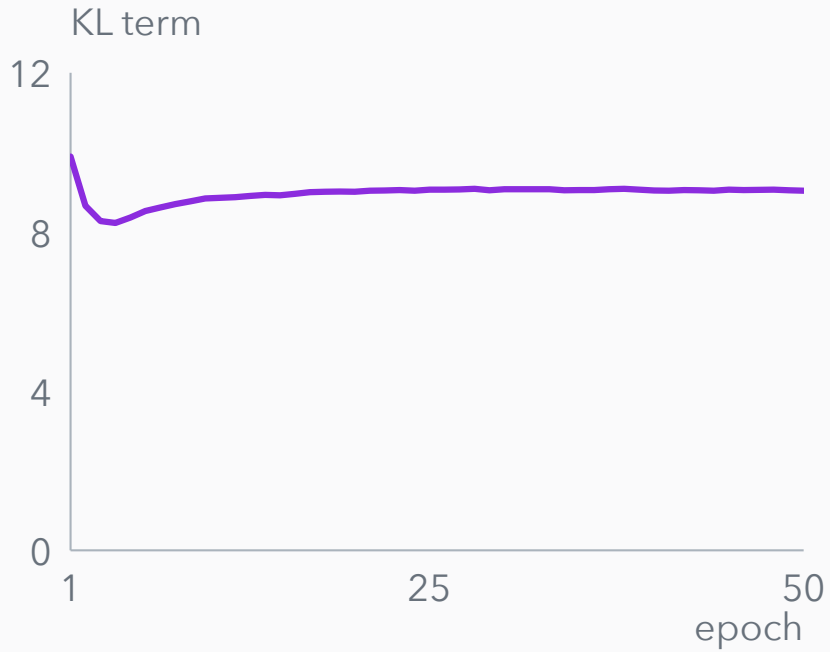
Training loop: identical to the autoencoder's.

> Everything else, the batches, Adam and 50 passes over the data, is the autoencoder's loop.

▶ Notebook 6.5

While training: reconstruction falls, the KL settles



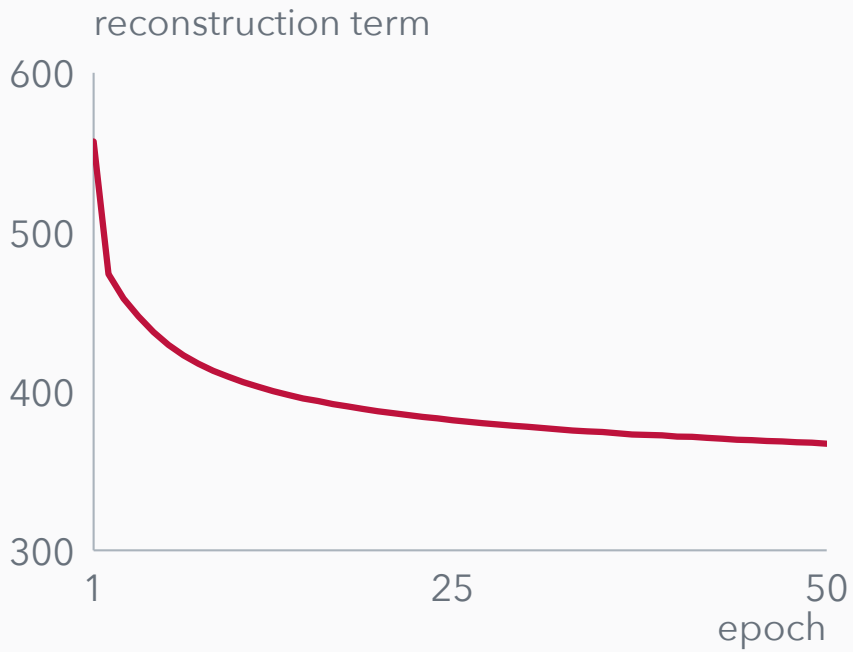


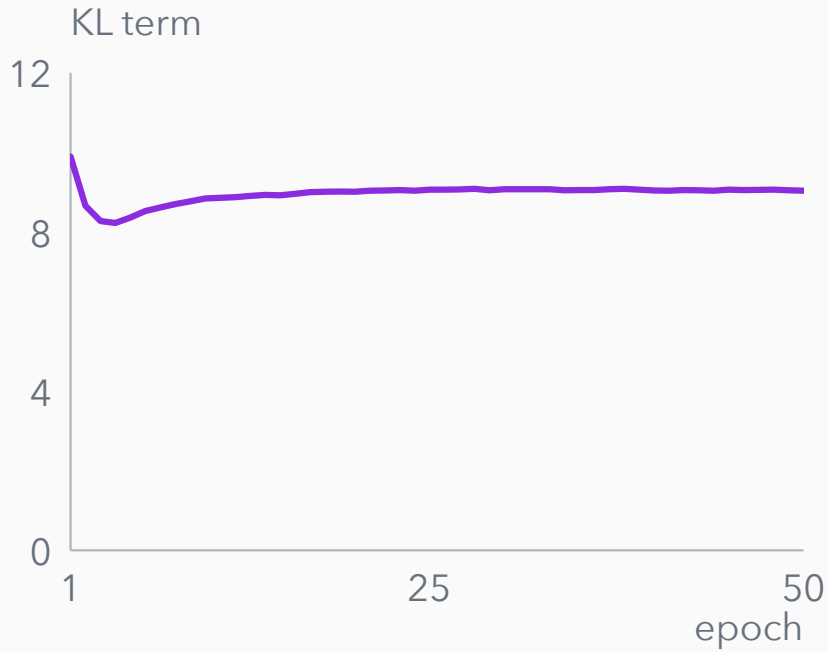
The reconstruction term falls steadily, as for the autoencoder.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 6.5

While training: reconstruction falls, the KL settles





After 50 epochs: reconstruction 367, KL 9.0 per image.

> The KL stays near 9 nats per image: each code is allowed to say about 9 nats more than the prior, no more.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 6.6

Now sample: z ~ N(0, I), then decode

autoencoder, 2



VAE, 2

VAE, 16

> The same 16 draws from N(0, I) as before, through the autoencoder's decoder.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 6.6

Now sample: z ~ N(0, I), then decode

autoencoder, 2



VAE, 2



VAE, 16

> Through the 2-number VAE: almost every draw is a digit, and all ten digits appear.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 6.6

Now sample: z ~ N(0, I), then decode

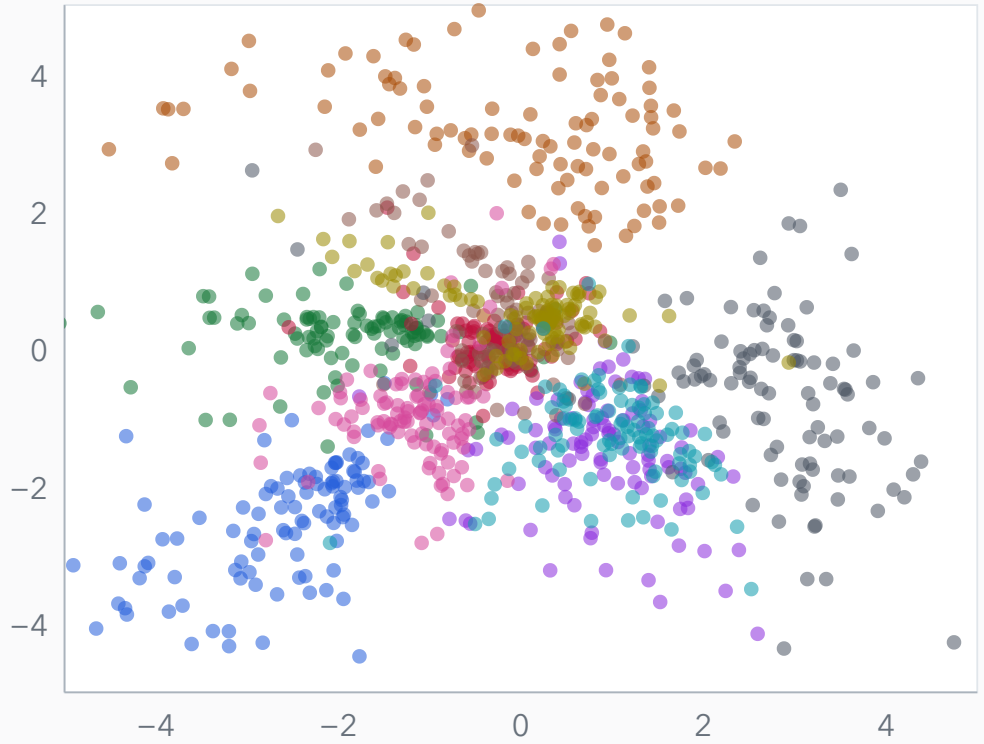


> Through the 16-number VAE: sharper and more varied. These include the fakes from the start of the lecture.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 6.6

The codes now fill N(0, I)



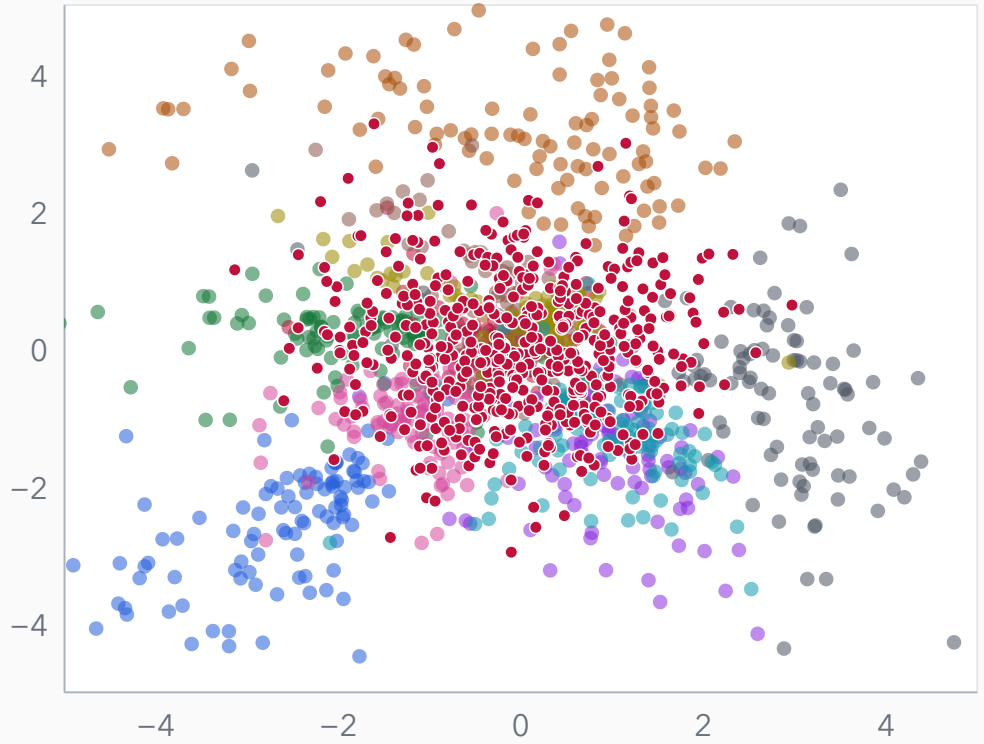
Codes of all 10,000 test digits: mean (−0.2, −0.2), spread (1.8, 1.7).

The VAE's codes (the means μ of q) for the same 1,000 test digits: one round cloud around the origin, digits side by side.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 6.6

The codes now fill N(0, I)



Codes of all 10,000 test digits: mean (−0.2, −0.2), spread (1.8, 1.7).

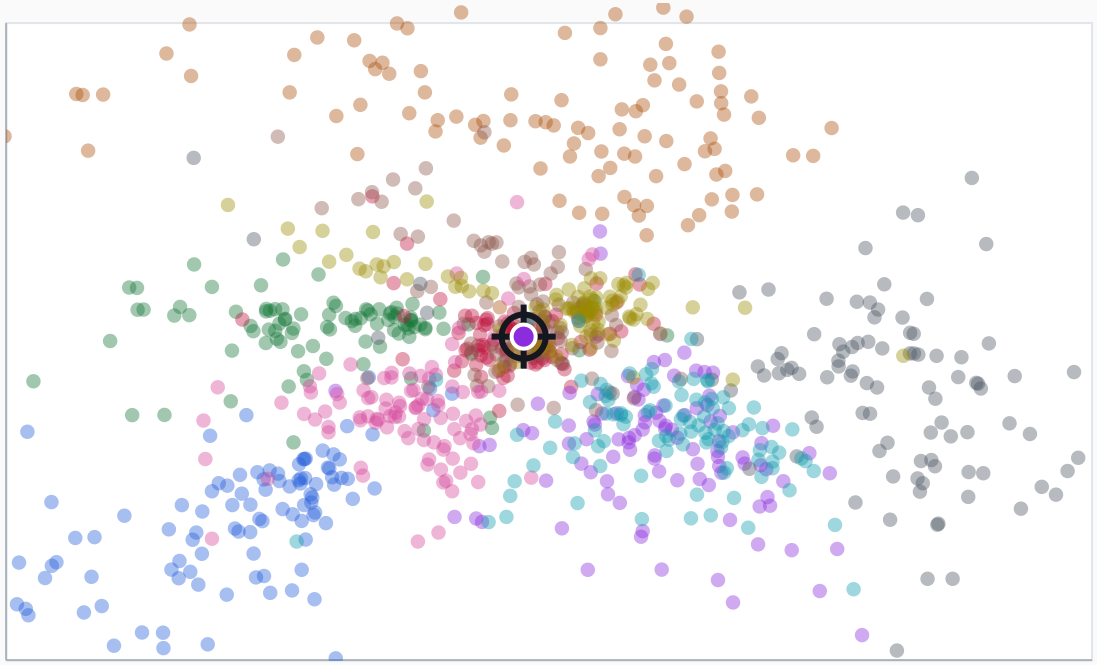
Random draws (red) now land among real codes, on every digit.

> Overlay 500 draws from N(0, I): they fall on real codes of every digit. That is why sampling works.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 6.6

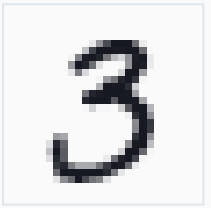
Your turn again: the same three ways, now with the VAE



0 1 2 3 4 5 6 7 8 9

decoder of the VAE, run in your browser





the digit we encoded

z = (−0.21, 0.07)

the code of our 3, decoded

> Random codes, encoded test digits, or click anywhere. Switch between the VAE and the autoencoder to compare.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 6.6

Decode a grid of codes: digits everywhere

VAE: codes from −3 to 3 in each direction



> Every point of a 12 × 12 grid over $[-3, 3]^2$ decodes to a digit, morphing smoothly from one to the next.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 6.6

Decode a grid of codes: digits everywhere

VAE: codes from −3 to 3 in each direction



VAE: walk from the same 1 to the same 0



16-number VAE



Walk from the same 1 to the same 0: every step is a plausible digit.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 6.7

Put a weight on the KL: β

$$L_\beta = \mathbb{E}_q\left[\frac{1}{2\sigma^2}\|x - f_\theta(z)\|^2\right] + \beta\,\mathrm{KL}\big(q(z \mid x) \,\|\, \mathcal{N}(0, I)\big)$$

β = 1 is the true ELBO. Changing β is the same as assuming noisier or cleaner pixels (a different σ).

> Weight the KL by β, as in β-VAE. Like ridge regression: data fit plus β times a penalty.

▶ Notebook 6.7

Put a weight on the KL: β

$$L_\beta = \mathbb{E}_q\left[\frac{1}{2\sigma^2}\|x - f_\theta(z)\|^2\right] + \beta\,\mathrm{KL}\big(q(z \mid x)\,\|\,\mathcal{N}(0, I)\big)$$

β = 1 is the true ELBO. Changing β is the same as assuming noisier or cleaner pixels (a different σ).

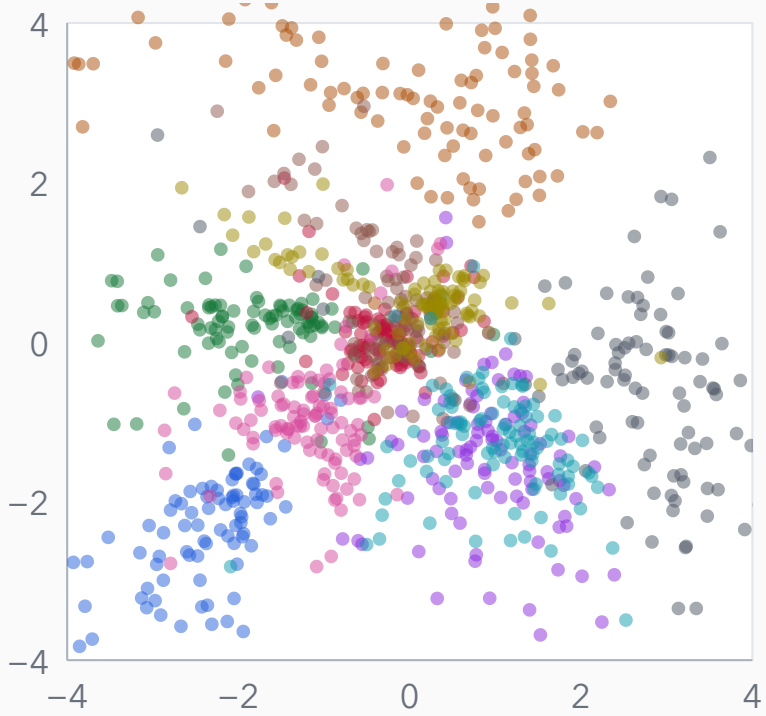
Small β
codes free to spread: close to an autoencoder

Large β
codes squeezed towards the prior: closer to N(0, I), less detail

Predict before we look: what happens at β = 0, and at very large β?

▶ Notebook 6.7

Seven VAEs, β from 0 to 256: try them

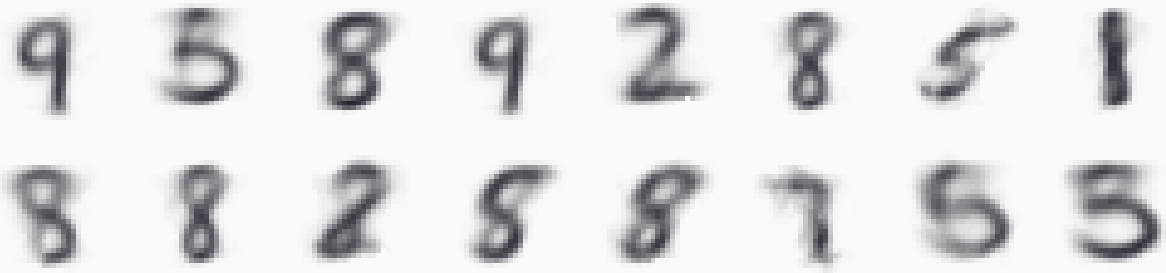


β = 1 KL 9.0 · squared error 30.8 · average σ of q 0.034

reconstructions of 8 test digits



16 samples: z ~ N(0, I), then decode



distance between two 0s ÷ distance from a 0 to a 1: 0.23

> Click through β. Watch three things: how spread the codes are, the reconstructions, and the samples.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 6.7

β = 0: is it just an autoencoder? (An exam question)

β	KL per image	average σ of q	spread of codes
0	68.3	0.0001	5.36
1	9.0	0.0343	1.68
256	0.0	1.0001	0.00

With β = 0 nothing stops q from shrinking.

> β = 0 keeps the random draw $z = \mu + \sigma\varepsilon$, so it is not quite the same model as an autoencoder.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 6.7

β = 0: is it just an autoencoder? (An exam question)

β	KL per image	average σ of q	spread of codes
0	68.3	0.0001	5.36
1	9.0	0.0343	1.68
256	0.0	1.0001	0.00

With β = 0 nothing stops q from shrinking.

Structurally it still draws $z = \mu + \sigma\varepsilon$, unlike an autoencoder. But training drives σ to 0.0001, so in practice it behaves like one: spread-out codes, poor samples.

> But training drives σ almost to zero, and then it behaves like one: spread-out codes and poor samples.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 6.7

β = 256: every code decodes to the same blur

β = 256, recon.



β = 256, samples



KL = 0.00: q(z | x) equals the prior for every image.

With a huge weight on the KL, the cheapest solution is q = N(0, I) for every image.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 6.7

β = 256: every code decodes to the same blur

β = 256, recon.



β = 256, samples



KL = 0.00: q(z | x) equals the prior for every image.

The decoder learns to ignore z and outputs the average digit. This is called **posterior collapse**.

> Then z carries no information, so the decoder ignores it and draws the average digit every time.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

▶ Notebook 6.8

The whole lecture in one table

model	how it generates	how it is fitted	what goes wrong
Gaussian or mixture (Part 2)	pick a bump, draw a number	maximum likelihood	fine for a few numbers
Gaussians on PCA numbers (3)	sample a code, decode linearly	maximum likelihood, then EM	straight-line decoder: blurry
autoencoder (4–5)	no rule for choosing codes	squared error	random codes miss the data
autoencoder + mixture (5)	sample a fitted code, decode	two separate fits	no single p(x)
VAE (6)	$z \sim N(0, I)$, decode	the ELBO: reconstruction + KL	soft samples; collapse if β is large

> From one number per example to 784, from formulas to networks: always a hidden code first, then the data.

▶ Notebook 6.8

Next: can we drop the likelihood altogether?

VAE samples are soft because each pixel is an independent Gaussian around a mean image.

> The pixel-wise Gaussian likelihood is the VAE's weak point: it rewards the safe average stroke.

▶ Notebook 6.8

Next: can we drop the likelihood altogether?

VAE samples are soft because each pixel is an independent Gaussian around a mean image.

Next: a generator trained by a second network that judges real against fake (GANs). Then: many small denoising steps (diffusion).

> Next lectures replace it: a learned critic (GANs) and repeated denoising (diffusion).

A sparse autoencoder: a KL between two coins

Wide hidden layer, but each unit should be on only rarely, say for ρ = 5% of images.

$$L = \text{reconstruction} + \lambda \sum_j \mathrm{KL}\big(\text{Bernoulli}(\rho) \,\|\, \text{Bernoulli}(\hat{\rho}_j)\big)$$

> A sparse autoencoder penalises hidden units that are on too often.

Ng, “Sparse autoencoder”, CS294A lecture notes; Lilian Weng, “From Autoencoder to Beta-VAE”.

A sparse autoencoder: a KL between two coins

Wide hidden layer, but each unit should be on only rarely, say for ρ = 5% of images.

$$L = \text{reconstruction} + \lambda \sum_j \mathrm{KL}\big(\text{Bernoulli}(\rho) \,\|\, \text{Bernoulli}(\hat{\rho}_j)\big)$$

$\hat{\rho}_j$ is unit j's average activation over the data. Lecture 1's Bernoulli KL, used as a penalty.

> The penalty is the KL between two coins: the target rate ρ and the unit's actual rate.

Ng, "Sparse autoencoder", CS294A lecture notes; Lilian Weng, "From Autoencoder to Beta-VAE".

Held-out scores of the PCA mixtures

d	K = 1	K = 5	K = 10	K = 20
2	−4.4	−4.1	−4.1	−4.1
10	−18.5	−16.0	−15.3	−14.8
20	−31.6	−25.6	−23.9	−21.9
50	−55.9	−40.4	−32.1	−26.0

Average log-likelihood of the 10,000 test codes. Compare along a row only: different d are densities in different spaces.

Computed for this lecture: `scripts/science.py` from the networks trained by `scripts/train.py` (MNIST, fixed seeds).

EM in one slide: the classic way to fit a mixture

$$r_{ik} = \frac{\pi_k \mathcal{N}(x_i; \mu_k, \Sigma_k)}{\sum_j \pi_j \mathcal{N}(x_i; \mu_j, \Sigma_j)} \qquad \text{(E: who made each point?)}$$

E step: for each point, the posterior probability that each bump made it (the eruption “kind”, given the wait).

EM in one slide: the classic way to fit a mixture

$$r_{ik} = \frac{\pi_k \mathcal{N}(x_i; \mu_k, \Sigma_k)}{\sum_j \pi_j \mathcal{N}(x_i; \mu_j, \Sigma_j)} \qquad \text{(E: who made each point?)}$$

$$\pi_k = \tfrac{1}{n}\textstyle\sum_i r_{ik}, \quad \mu_k = \frac{\sum_i r_{ik} x_i}{\sum_i r_{ik}}, \quad \Sigma_k = \frac{\sum_i r_{ik}(x_i - \mu_k)(x_i - \mu_k)^\top}{\sum_i r_{ik}} \qquad \text{(M)}$$

> M step: refit each bump to the points, weighted by those probabilities. Repeat: the likelihood never decreases.